PLAN STRONG™

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2ND EDITION

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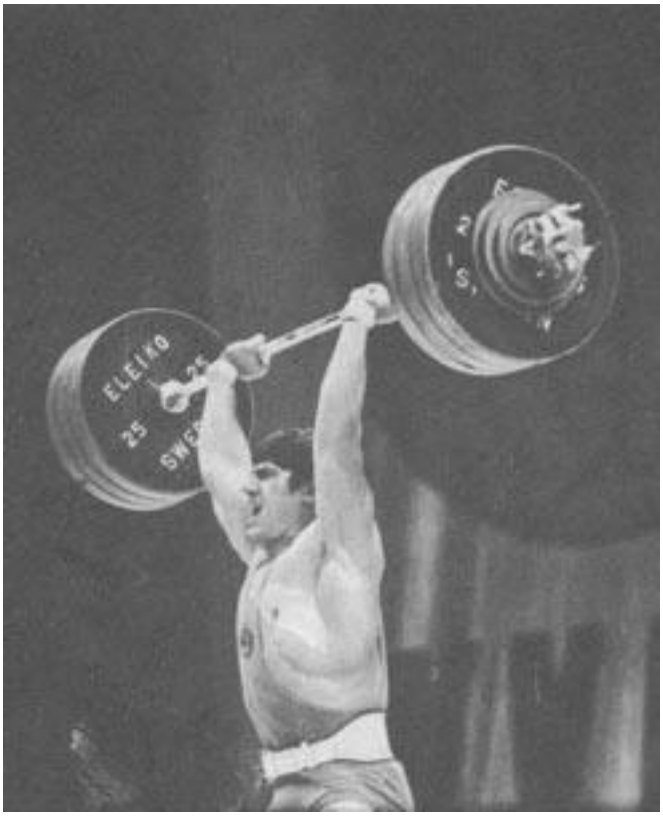
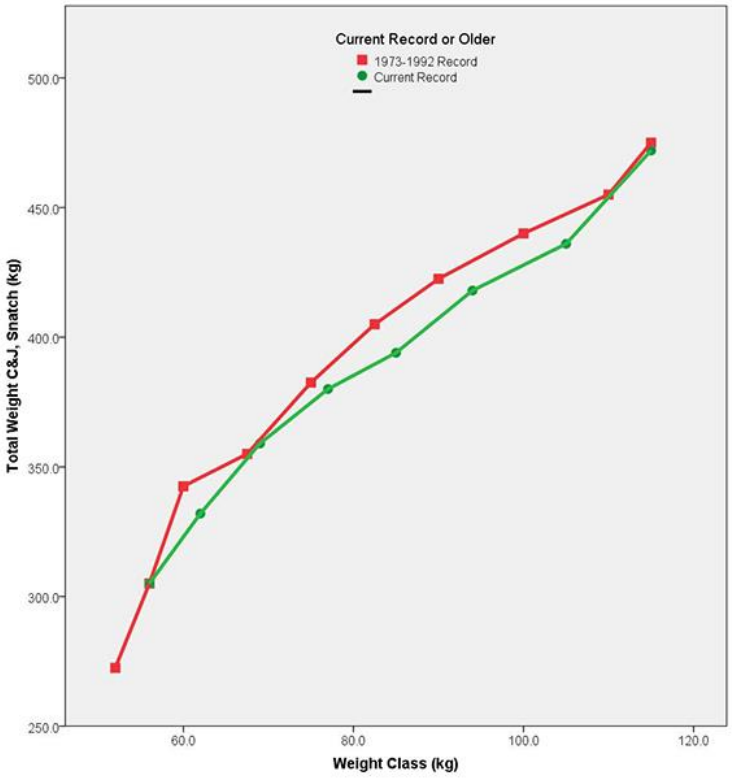


Plan Strong™ is a general strength training system faithfully adapted from the Soviet weightlifting methodology that produced records that still stand.



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PLAN STRONG™ 70 SUMMARY

BIG CYCLE

Preparatory period (1-2 months)

+

Competition period (1 month)

+

Transition period (1 week to 1 month)

=

Big cycle

The preparatory period has higher volume and lower intensity than the competition period.

The transition period is typically a week of "random acts of variety".

PREPARATORY PERIOD

If there are two months in the preparatory cycle, either month can be higher in volume. Increase or decrease by 20-50%. (Up to a 100% increase for well-trained lifters)

A minimum NL change, increase or decrease, between two adjacent training units (months, weeks, days) is 20%.

Either preparatory month can be higher in intensity. Change the HARI by at least 1% 1RM between two preparatory months. Also change the shares of individual zones.

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Step I: Monthly volume

Monthly NL per class of lift in the preparatory period, ≥70% 1RM

- Squats and/or deadlifts—**150-250**
- Presses, pullups—**200-300** and higher

Step II: Volume dynamics within a month

Classic monthly volume distributions by week

Variant	Percentage of the monthly total			
	Week 1	**Week 2**	**Week 3**	**Week 4**
1	35	28	22	15
2a	15	35	28	22
2b	28	35	22	15
2c	22	35	28	15
3a	15	22	35	28
3b	22	28	35	15
3c	15	28	35	22
1-3a	35	15	28	22
1-3b	35	22	28	15
3-1a	28	15	35	22
3-1b	28	22	35	15
4	15	22	28	35
2-4a	15	35	22	28
2-4b	22	35	15	28
4-2a	22	28	15	35
4-2b	15	28	22	35
The following combinations are rarely used for total volume but are useful for varying intensity:				
1-2	35	28	15	22
1-4a	35	15	22	28
1-4b	35	22	15	28
2-1	28	35	15	22
3-4	22	15	35	28
4-1a	28	15	22	35
4-1b	28	22	15	35
4-3	22	15	28	35

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Guidelines for customizing volume distribution among weeks

	Week's volume			
	1 (low)	**2 (med)**	**3 (med)**	**4 (high)**
Standard	14-16%	22-23%	27-28%	34-36%
More variable	11-13%	20-21%	29-30%	37-46%
Less variable	17-19%	24-25%	25-26%	31-33%
When the highest volume week is in the maximal range, one of the medium weeks may become small. Or both medium weeks may stay in the bottom half of the medium range, 20-25%.				
May combine standard, more variable, and less variable percentages.				
A minimum of 20% difference between two adjacent weeks.				

Special planning considerations for ladies

- In the, ovulating, premenstrual, and menstrual phases reduce the volume in exercises loading the pelvic floor, especially squats.
- Highest strength and speed-strength performance, maximal efficiency and speed of recovery is in the post-menstrual and post-ovulating phases. These are the phases to plan the highest volume and intensity loads.
- Performance is not at its highest in the ovulatory phase.

Step III: Monthly intensity (NL in the 91-100% zone, distribution of the volume by intensity zones, HARI)

≥90% NL is individual and lift specific. Starting point for experimentation, per lift per week:

- NL 0-3 in the preparatory period and the last two weeks of the competition period;
- NL 2-5 in the first two weeks of the competition period; 0-3 in the third.

Having established the 90-100% NL, convert the reps into percentage of the monthly total.

Guidelines for monthly volume distribution in the high intensity zones

% 1RM	70-80%	81-90%	91-100%	HARI
Average	55	35	10	**80%**
Primary	**50-60**	**30-40**	**0-10**	**79-81%**
Expanded	20-90	10-60	0-20	**75-85%**

Select 1-2 weights per zone in the two bottom zones. In the 91-100% zone select 0-3 weights.

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High intensity zones' kettlebell training weights as % of 1RM

1RM ↓	12 kg	14 kg	16 kg	18 kg	20 kg	22 kg	24 kg	28 kg	32 kg	36 kg	40 kg	44 kg
48kg									67%	75%	83%	92%
44kg									73%	82%	91%	
40kg								70%	80%	90%		
36kg							67%	78%	89%			
32kg						69%	75%	88%				
28kg					71%	79%	86%					
24kg			67%	75%	83%	92%						
22kg			73%	82%	91%							
20kg		70%	80%	90%								
18kg	67%	78%	89%									

Step IV: Intensity dynamics within a month

There is no trend in intensity dynamics from week to week.

Generally change the intensity from week to week and month to month by 1-5% 1RM.

Unless the ratio of 90% and heavier lifts is unusually high, the distribution of the 81-90% lifts is likely to determine the intensity variant for the month.

Intensity is varied by assigning lifts in the 91-100% and 81-90% zones to weeks according to the same procedure as for assigning the total volume but using different variant(s) and, as an option, different percentages.

1. Assign the 91-100% zone—select the variant and the percentages. If the number of these lifts is low, calculating the percentages is unnecessary. Assigning all lifts to two or three weeks only is an option.

Distribution of volume between 2 or 3 training units, from the least to the most variable, %

2	3
40, 60	25, 33, 42
35, 65	20, 35, 45
30, 70	22, 28, 50
25, 75	20, 30, 50
20, 80	15, 35, 50
	15, 30, 55
	10, 40, 50

2. Assign the ≥95% lifts out of the 91-100% NL total to given weeks without any consideration for the chosen variant but guided by the athlete's presumed ability to perform these lifts without excessive excitation at this point in a cycle.

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3. Apply step one to the 81-90% weights. The variants and percentages for 81-90% and 91-100% zones may or may not match.

   When assigning 81-90% lifts, generally do not place their highest number in either of the two lowest total volume weeks.

   If you have two weights in the 81-90% zone, generally assign them according to the same variant (although it is advisable to change the percentages).

   An alternative tactic for planning two weights in the 81-90% zone is to first lump them together and assign the total according to the same variant and percentages. Then asymmetrically distribute the two weights between the weeks.

4. Calculate the 70-80% NL by subtracting the middle and top zone lifts from the total in each week. If you have two weights in the 70-80% zone, assign the weight representing the majority of that zone according to the total volume variant, then determine the NL of the remaining weight by the subtraction procedure described above.

5. Calculate the HARI in every week, identify the overall intensity dynamics, and make changes to the plan if necessary.

Step V: Volume dynamics within a week

Stagger schedules for multiple lifts to train them together on some days and separately on others. Plan at least one day off lifting a week.

Single lift weekly schedule "R", 4 days a week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
X	Add on high volume weeks.	**X**		**X**	**X** Remove on low volume weeks.	

Single lift weekly schedule "V", 4 days a week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
X	**X**		**X**	**X**	Add on high volume weeks.	
	Remove either Tuesday or Thursday session on low volume weeks.					

Single lift weekly schedule, 3 days a week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
X		**X** Remove on low volume weeks.		**X**	Add on high volume weeks.	

Single lift weekly schedule, 2 days a week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
X				**X**		
On higher volume weeks add a session on any other day but, preferably, Wednesday.						

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Considerations when a given lift is trained two days in a row:

- If day one is low in volume and low to moderate in intensity, it improves performance on day two.

- If day one is low in volume and high in intensity, it improves performance on day two in a session of any volume and low to medium intensity.

- If day one is high in volume, a low volume/low intensity day two amplifies the anabolic effect.

- If on the first day one of the load variables, volume or intensity, is high and the other is medium, the second day's training will be conducted in a state of fatigue. The training effects of both sessions will be summated. It is an effective advanced tactic but it should not be used too frequently.

NL per class of exercise per day, ≥70% 1RM

- **10-20**—minimal
- **20-30** —optimal
- **30-50**—maximal

If a lift's daily volume exceeds 30, separate it into 2-3 series separated by unrelated exercises.

Option: separate the daily load into 2-3 sessions with at least 30min of rest between them. The ratio of volume between two daily sessions ranges between 1:1 and 1:3 with either the earlier or the later session being higher in volume. Do not plan two days with twice-a-day training back to back. Ideally, take a day off after a day with two training sessions or at the most follow it up with a day with only one easy session.

Assign the week's volume to separate days using one of the following ratios. Stay with it for a month, then change it:

Distribution of a weekly volume by days, from least to most variable, %

Training days per week			
2	**3**	**4**	**5**
40, 60 35, 65 30, 70 25, 75 20, 80	25, 33, 42 20, 35, 45 22, 28, 50 20, 30, 50 15, 35, 50 15, 30, 55	15, 22, 28, 35 10, 20, 30, 40 Or any other combination used in planning weeks	10, 15, 20, 25, 30

The above numbers do not imply that you must use them in that order (even though increasing the volume towards the end of the week is a common prep period tactic.)

Maintain the same volume dynamics for one month. If you have a second prep month, you may change the dynamics and/or the training schedule or not.

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Step VI: Intensity dynamics within a week

Intensity is varied independently from volume.

A standard preparatory period tactic, used together with an increase of volume towards the end of the week, is placing the heaviest lifts to the middle of the week after a low to medium volume and intensity session one or two days earlier. Stagger different lifts' highest intensity days.

Possible schedules for facilitating a high intensity mid-week session for a lift

<table>
<tr><th>Monday</th><th>Tuesday</th><th>Wednesday</th><th>Thursday</th><th>Friday</th><th>Saturday</th><th>Sunday</th></tr>
<tr><td>L</td><td></td><td>H</td><td colspan="3" rowspan="3">Increase the volume
towards the end of the week</td><td rowspan="5">Off</td></tr>
<tr><td></td><td>L</td><td>H</td></tr>
<tr><td>L</td><td>H</td><td></td></tr>
<tr><td></td><td>L</td><td></td><td>H</td><td></td><td></td></tr>
<tr><td></td><td></td><td>L</td><td>H</td><td></td><td></td></tr>
</table>

Do not keep the same ARI in a given exercise from session to session; vary it by selecting different main weights.

An effective tactic is selecting different main weights for each session in a week, except one. In that one session use variable weights, typically with weight ladders.

When distributing top zone lifts between days, generally do not exceed:

- Presses and pullups: NL 7 @ 90% or NL 3-4 @ 95-100%
- Squats and deadlifts: NL 3 @ 90% or NL 2 @ 95-100%

You may maintain the same intensity dynamics for a month or change it from week to week.

Step VII: Load in a session

Organize the load as a pyramid or a half-pyramid with flexible rep and weight ladders built in.

Use the repetition ranges recommended below with a bias towards the optimal number in the middle. The same lifter may use one table for some lifts and the other for others.

Vary the reps from set to set. When your rep choices do not allow you to exactly fill the day's volume with a given weight, use lower or higher reps then specified in the tables.

Individualize the reps per set keeping the same volume and intensity. You may choose to use the first table for some lifts and the second for others.

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Reps per set for lifters with average to high endurance with a given RM

% 1RM	70%	75%	80%	85%	90%	95%
RM, ≥	12RM	10RM	8RM	6RM	4RM	≥2RM
Optimal	6	5	4	3	2	1
Range	**4-8**	**3-6**	**3-5**	**2-4**	**1-2**	**1**
Warm-ups; comp week	2-3	1-2		1		

Reps per set for lifters with low endurance with a given RM

% 1RM	70%	75%	80%	85%	90%	95%
RM	10RM	8RM	6RM	5RM	3RM	2RM
Optimal	5	4	3	2	1-2	1
Range	**3-6**	**3-5**	**2-4**	**2-3**	**1-2**	**1**
Warm-ups; comp week	1-3	1-2		1		

Before the planned 70% and up lifts perform joint mobility exercises and warm-up lifts according to individual needs.

Once you have worked up to the top weight of the day you can:

a) Stay with this weight until you have completed all the reps with it and then perform the remaining lighter sets if they are listed (the most common tactic);

b) Sandwich some lighter sets between heavier sets. You can alternate sets of different weights, alternate series of several sets, or use a combination of both.

Use option A at least twice as often as option B.

c) An option to exercise infrequently is to do the heaviest lifts after a fatiguing series of lighter lifts. Back-off sets afterwards are optional.

Rest as long as needed between sets. Walk and perform "fast and loose" drills, then sit down.

Step VIII: Interaction of different lifts within a week and within a session

You are training 2 or 3 lift categories:

1. Squat/deadlift
2. Press
3. Pullup (optional)

The volume and intensity in all classes of lifts are independent of each other. Use different volume and intensity dynamics within a week and within a session for different lifts.

All combinations of the 2-3 main lifts are possible in a session—1, 2, or 3 of them. Vary them.

Vary the order in which lifts are performed. Alternating between harder and easier exercises is advisable.

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A tactic: alternate one or two sets of an exercise done for a low volume on the given day with a series of several sets of a high volume exercise.

Emphasize different intensity zones for different lifts in the same session.

Regarding interaction of lower and upper body exercises on two consecutive days:

- A low to medium volume with medium to high intensity on day one for the upper body improves performance for the lower body on day two and vice versa.

- A high volume lower body session on day one will have a negative effect on upper body performance on day two. However, if the upper body session is high in volume and hypertrophy oriented, it will benefit from the anabolic hormones released by the lower body session.

Step IX: Programming assistance exercises. Training more than one main lift prioritizing the same muscle groups.

Share of the main lift as a percentage of the category (main lift plus assistance):

- Beginner—100%
- Intermediate—70-80%
- Advanced—up to 50%

Change the ratio of the main lift to assistance every month, e.g., 70:30, 80:20, 75:25. Reduce the number and share of assistance exercises in the comp month, especially the last two weeks.

Change the assistance exercise(s) every month. Mostly—or even totally—restrict them to the 70-80% intensity zone. Plan your assistance as back-off sets and/or on separate session(s). Exercises that promote skill development of the main lift may also be inserted into the ramp-up to heavy weights.

If the given SV exercise's 1RM is within 5% of that of the main lift, use the main lift's max as 100%; otherwise measure the intensity as a percentage of the assistance exercise's own max.

If you feel that you have gotten stronger in an assistance exercise, go for a 1-3RM to recalibrate the loads. Select a week and a day when you are relatively fresh. To stay with the planned distribution of the NL by intensity zones, max on a day when top zone NL are planned for the main lift and do it at the expense of these lifts.

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COMPETITION PERIOD

Step I: Monthly volume

Reduce the volume in the competition month by 10-50% (typically around 30%) compared to the last preparatory month.

Step II: Volume dynamics within a competition month

Taper according to the last preparatory month's volume

Variant	Last month's volume	Deload from week #
1	Very high	1
2	High	2
1-3	Moderate	3
3, 3-1	Low (insufficient)	4

Classic competition month variants

Variant	Week 1	Week 2	Week 3	Week 4
1	35	28	22	15
2b	28	35	22	15
2c	22	35	28	15
3b	22	28	35	15
1-3b	35	22	28	15
3-1b	28	22	35	15

To develop your own variant or tweak an existing one refer to the table entitled "Guidelines for customizing volume distribution among weeks" in the preparatory period section.

Step III: Monthly intensity (91-100% NL, distribution of the volume by intensity zones, HARI)

Compared to the last prep month, increase the HARI by at least 1% 1RM.

Change the share each weight contributes to the total.

Increase the NL in the 91-100% zone. A 2:3 ratio between the preparatory and the competition months is standard (NL; not %). Assign at least 2/3 of the zone's NL to 90-92.5% weights

The number of 91-100% lifts varies greatly between lifts and between lifters. Starting point:

- NL 2-5 in the first two weeks of the comp period; 0-3 in the third.

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Step IV: Intensity dynamics within a competition month

Match an increase or decrease in volume by same in intensity.

Use intensity variants 1, 2, 1-3, 3-1.

Assign the desired number of 91-100% lifts NL to the competition week (usually NL 3-4), then distribute the remaining lifts between the first three weeks or two out of the three. Refer to the table "Distribution of volume between 2 or 3 training units" in the prep period section.

Match the dynamics of the main variant's first three weeks or use a similar variant:

Main variant	Similar variants
1	2b, 2c, 1-3b
2b	1, 1-3b, 2c
2c	1, 1-3b, 2b
3b	2b, 2c, 3-1b
1-3b	1, 2b, 2c
3-1b	2b, 2c, 3b

Some peaking tactics:

- MP, BP, PU: week 1—90%, week 2—95-100%, week 3—90%
- SQ, DL: week 0 (last week of the prep period)—95-100%, week 2—90%
- Any lift: week 2—90%, week 3—95%

For 81-90% assign 10% (or an exact NL demanded by your competition week design) to the competition week and distribute the remaining lifts between the first three weeks using one of the sets of three percentages. Match the variant used for the total volume. If you have more than one weight choice in the 81-90% zone, use the same variant but preferably different percentages.

If you have more than one weight in the 70-80% zone, assign the heavier one(s) to all four weeks according to the main variant, then calculate the number of the lifts with the lightest weight by subtraction.

Finish by calculating the HARI for each week. Evaluate the resulting ARI, variability, and dynamics. Adjust the plan if necessary.

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Step V: Volume, intensity, and their dynamics within a week and a session

Start with the competition week. If the existing NL by zone do not suit you, go back to the previous step and make trades with other weeks. When done, recalculate the HARI for each week to check the variability and the dynamics.

Competition week

<table>
<tr><th colspan="2">Day 1</th><th colspan="2">Day 3 (if applies)</th><th colspan="2">Day 5</th></tr>
<tr><td colspan="2"></td><td colspan="2">Approximately 50% of the first day's volume</td><td colspan="2"></td></tr>
<tr><td colspan="2">MP/BP up to 85-90%
SQ up to 75-80%
DL up to 70-75%</td><td colspan="2">MP/BP up to 70-75%
SQ, DL up to 70%</td><td colspan="2">Competition or test</td></tr>
<tr><td colspan="6">All repetitions are in the warm-up range:</td></tr>
<tr><td>70%</td><td>75%</td><td>80%</td><td>85%</td><td>90%</td><td>95%</td></tr>
<tr><td>1-3</td><td colspan="2">1-2</td><td colspan="3">1</td></tr>
<tr><td colspan="3" rowspan="2"></td><td colspan="2">Day 4 (Optional)</td><td rowspan="2"></td></tr>
<tr><td colspan="2">A specific to the contested lift warm-up to 50% 1RM exactly 24 hours before the meet.</td></tr>
</table>

Weekly loads in competition month

Day 1	Day 3 (if applies)	Day 5
Tuesday	Thursday	Saturday
Or		
Monday	Wednesday	Friday
High volume, moderate to high intensity	**Low volume, low to moderate intensity**	**Competition simulation**

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PLAN STRONG™ 50 SUMMARY

Most of the recommendations from high intensity zones planning apply, unless specified otherwise.

VOLUME

Monthly NL per class of lift in the preparatory period, ≥50% 1RM

- Squats and/or deadlifts—**200-400**
- Presses, pullups—**300-500** and higher

NL per class of exercise per day, ≥50% 1RM

- **10-25**—minimal
- **25-50**—optimal
- **50-100**—maximal

Volume in all main exercises in a session, ≥50% 1RM

- ≤ **50**—small
- **51-100**—medium
- **>100**—large

ARI AND ITS VARIABILITY

ARI = 67-77% 1RM	
Preparatory period	Competition period
<70%	**>70%**
Change from week to week and month to month by 1-5% 1RM	

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INTENSITY ZONES

Kettlebell training weights in five zones as % of 1RM

1RM ↓	8 kg	10 kg	12 kg	14 kg	16 kg	18 kg	20 kg	22 kg	24 kg	28 kg	32 kg	36 kg	40 kg	44 kg
48kg									50%	58%	67%	75%	83%	92%
44kg								50%	55%	64%	73%	82%	91%	
40kg							50%	55%	60%	70%	80%	90%		
36kg						50%	56%	61%	67%	78%	89%			
32kg					50%	56%	63%	69%	75%	88%				
28kg				50%	57%	64%	71%	79%	86%					
24kg			50%	58%	67%	75%	83%	92%						
22kg			55%	64%	73%	82%	91%							
20kg		50%	60%	70%	80%	90%								
18kg		56%	67%	78%	89%									
16kg	50%	63%	75%	88%										

How to assign a limited number of weights by five zones

1. Distribute the 91-100% and 81-90% lifts according to your chosen variants.
2. Distribute the 50-60% and 61-70% weights according to the main variant.
3. Calculate the NL in the 71-80% zone by subtraction of the above from the total.

The classic distribution

50-60%	61-70%	71-80%	81-90%	91-100%	ARI
10	25	**35**	25	5	**73.9%**

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How to distribute a monthly volume across five zones with central weights

<table>
<tr><th>50-60%</th><th>61-70%</th><th>71-80%</th><th>81-90%</th><th>91-100%</th></tr>
<tr><td colspan="5">First establish the monthly ARI in the 67-77% range.
Generally, stay below 70% in the prep period and above 70% in the comp period.</td></tr>
<tr><td colspan="5">Calculate the ARI with the central weights of each zone—55, 65, 75, 85, 92.5%:</td></tr>
<tr><th>55%</th><th>65%</th><th>75%</th><th>85%</th><th>92.5%</th></tr>
<tr><td colspan="2">Assign the shares of these zones last after establishing the contribution of the top three zones.

Roughly 30-60% (usually 40-50%) of your volume should be in these two zones. For heavyweights focusing on hypertrophy the number can be as high as 75%

Adjust the ratio between the 50-60% and 61-70% zones depending on the target average intensity: more lifts in the first zone to lower it; more in the second to raise.

The share of 50-70% lifts usually decreases in the competition period.</td><td colspan="3">Increase a share of these zones in the competition period.</td></tr>
<tr><td rowspan="2"></td><td colspan="3">Once you have established your ARI, the next step is to choose the predominant zone: one of these three.</td><td rowspan="2"></td></tr>
<tr><td>For beginners, heavyweights, and during hypertrophy emphasis, this is the primary zone—35-40% of the total.</td><td>For most lifters this is the dominant zone—35-50% of the total (and at times as high as 60%).

If you are focusing on hypertrophy, reduce to 25-35% and increase the share of the lighter zones.</td><td>Typically assign 5-10% of your NBL to this zone. If you respond well to high volume heavy lifts, gradually up this number to 20-40%. Adjust the volume in the lower three zones to keep the ARI the same. Migrate some reps from the 61-70% and 71-80% zones down to the 50-60% zone.</td><td>Do not plan lifts in this zone by % but decide on an appropriate for you NL—and then convert this number into percentage of the total.

Many lifters can do well without lifts in this zone in the preparatory period.</td></tr>
<tr><td></td><td colspan="3">Aim for a smooth curve but at times flatten the peak (a relatively even distribution across zones 2-4) or make two peaks (decrease the volume in zone 3 and increase in zones 2 and 4).</td><td></td></tr>
</table>

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MONTHLY INTENSITY DYNAMICS

Algorithm for arriving at the desired intensity dynamics within a month

Zones	**50-60%**	**61-70%**	**71-80%**	**81-90%**	**91-100%**
Central weights	**55%**	**65%**	**75%**	**85%**	**92.5%**
Dynamics	Main variant	Main variant	By subtraction of all other zones' NL from the total volume	The variant chosen to define the month's intensity dynamics	Try to abide by the chosen monthly intensity variant—but let your ability to perform heavy lifts on each given week prevail. Do not automatically assign ≥92.5% lifts to low volume weeks; Monday-Wednesday of a high volume week are also options.
				These zones' variant(s) tend to determine the monthly intensity dynamics.	
After you have completed all the calculations you may need to move some lifts around to achieve the desired intensity dynamics and variability. Generally trade between the 81-90% zone (above average) and the 61-70% zone (below average). If the 81-90% zone has too few lifts, trade the 71-80% zone.					

WEEKLY PLANNING

When making weekly planning, push up the intensity in some reps by 5% (or 2.5%, or any other percentage less than five) while decreasing it by as much in the same number of reps.

Limit the use of this tactic in the 81-90% zone as this would introduce unplanned 90% lifts. One option is to stick with 2.5% increases and decreases in this zone.

Reps per set for lifters with average to high endurance with a given RM

% 1RM	**50%**	**55%**	**60%**	**65%**	**70%**	**75%**	**80%**	**85%**	**90%**	**95%**
RM, ≥					12RM	10RM	8RM	6RM	4RM	2RM
Optimal					6	5	4	3	2	1
Range	**4-10**				**4-8**	**3-6**	**3-5**	**2-4**	**1-2**	**1**
Warm-ups	4-5		2-3			1-2		1		
Hypertrophy	5-10 (preparatory period only)									

Reps per set for lifters with low endurance with a given RM

% 1RM	**50%**	**55%**	**60%**	**65%**	**70%**	**75%**	**80%**	**85%**	**90%**	**95%**
RM					10RM	8RM	6RM	5RM	3RM	2RM
Optimal					5	4	3	2	1-2	1
Range	**4-6**				**3-6**	**3-5**	**2-4**	**2-3**	**1-2**	**1**
Warm-ups	3-5		1-3			1-2		1		
Hypertrophy	5-10*		5-8*							
	* Preparatory period only									

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PLAN STRONG™ 50 TEMPLATE

% 1RM→	55%	65%	75%	85%	92.5%	ARI
NL	**PREP**					
ARI						
NL	**COMP**					
ARI						

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
NL					
ARI					
	92.5%				
NL					
	Out of the above number ≥**95%**				
NL					
	85%				
NL					
	65%				
NL					
	55%				
NL					
	Subtract for **75%**				

PREP		**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**					
92.5%						
Out of that ≥95%:						
85%						
75%						
65%						
55%						
ARI						

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COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
NL					
ARI					
	92.5%				
NL					
	Out of the above number ≥**95%**				
NL					
	85%				
NL					
	65%				
NL					
	55%				
NL					
	Subtract for **75%** in the first three weeks				

COMP		**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Month**
% 1RM	**NL**					
92.5%						
Out of that ≥95%:						
85%						
75%						
65%						
55%						
ARI						

Week #

	55%	**65%**	**75%**	**85%**	**92.5%**	**ARI**
NL						

Mon	**Tue**	**Wed**	**Thur**	**Fri**	**Sat**	**Sun**

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THE BIG PICTURE

- In science and engineering the solution that meets all the set requirements at a minimal "cost" is called optimal." (Vorobyev, 1981)

- "The optimal load and the optimal interval between loads are determined empirically." (Luchkin, 1962)

- Hill, Nobel laureate, stressed that the highest number of concentrated physiological data is not in physiology textbooks but in world records. (Matveev, 1991)

- "One must take into an account that the human organism is a probabilistic system[1]. In other words, even when the conditions and the training program appear optimal, one cannot always count on the expected result." (Vorobyev, 1981)

- "The existence of the regulation of hormonal signal on two levels—on the level of cells producing signal molecules, and on the level of receptors receiving the signal—makes variability of tactics of adaptive responses possible." (Viru, 1991)

- The term "load" "partly corresponds with such terms as "exercise", "work", etc. but is not synonymous to them. It primarily characterizes the magnitude of the demands imposed on the organism by the exercise—that is, how large they are and to what degree they are within the capabilities of the [athlete]." (Matveev, 1991)

- "The term "load" reflects the obvious fact that performance of any physical exercise is associated with a shift of the organism's functional state to a level of activity higher than at rest. Thus it is something extra that loads functional systems and causes fatigue, if it is large enough. *Load*, in this sense*, is an additional, compared to that at rest, degree of the organism's functional activity brought on by performance of exercise or exercises and the degree of the difficulties being overcome."* (Matveev, 1991)

- "Internal" and "external" aspects of the load. On one hand, purely physical-mechanical measurements like weight, speed, distance, etc. ("external"). On the other, internal changes in the body in response to this external work: higher heart rate, lung ventilation, etc. Internal load describes the organism's reaction to the external load. "[Both internal and external load criteria] are important for evaluating and directing loads… External load criteria must be factored in because they provide guidance for setting achievable and necessary load values. The organism's responses are then compared to them and loads are rationed accordingly." (Matveev, 1991)

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TERMINOLOGY & CALCULATIONS

Soviet weightlifting science had some fascinating markers, some of them applicable to general strength training and others not. The Soviets started attempting to quantify training early on. Until the end of the 1950s the number of sets was a standard measure. Luchkin (1940) suggested adding up the tonnage. This practice is not perfect, but still useful and still in use.

Roman & Falameev (1969) proposed tracking volume in the number of barbell lifts, or NBL. It turned out to be a very informative measure.

Matveev (1959) proposed measuring the intensity as the average weight lifted in a workout, a week, etc. It is still measured that way and referred to as the "absolute intensity".

While the average weight lifted is very telling when you are looking at the progress of one lifter and helpful in comparing lifters from the same weight class, it is not informative when comparing lifters of different weight classes or for writing a training prescription. Chernyak proposed measuring the "relative intensity"—compared to the lifter's 1RM, e.g. 75.1% 1RM. This was measured for individual exercises, groups of exercises, and all exercises. In every training unit, from a session to a year, the "average relative intensity" (ARI) was calculated. Here is how, using a bench press and deadlift workout as an example.

Say, your bench max is 300 is your deadlift max is 500. You benched this:

135x5/2, 185x3, 205x2, 225x4/6 (reps/sets)

First, add up the NBL. Since 135 is below 50% of your max, start with 185.

3+2+(4x6) = 29

NBL=29

Second, add up the tonnage, again not counting the weights lighter than 50%:

185x3=555
205x2=410
225x24=5400

555+410+5400=6,365

Tonnage=6,365

(Or "poundage", to be exact.)

Third, calculate the average weight by dividing the tonnage by the number of lifts and round up:

6,365 : 29 = 219.48

Average weight=219

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Fourth, calculate the average relative intensity by dividing the average weight by your 1RM and multiplying the result by 100:

(219 : 300) x 100 = 73%

ARI=73%

Fifth, calculate the "volume coefficient", VC, that characterizes the "general load" (Chernyak) of the given exercise. It is done by multiplying the NBL by the ARI. Although not usually written in the training journal, the VC is a useful intermediate step for some calculations, as you will see shortly.

29 x 73 = 2,117

VC=2,117

Do the same for your deadlift. You pulled:

225x3/3, 315x2/2, 365x1, 405x1, 445x1, 2, 1, 2, 1, 2, 1

Add up the NBL starting with 315:

(2x2)+1+1+1+2+1+2+1+2+1 = 16

NBL=16

Add up the tonnage from 315 and up:

315x4=1,260
365
405
445x10=4,450

1, 260+365+405+4,450=6,480

Tonnage=6,480

Calculate the average weight by dividing the tonnage by the NBL:

6,480 : 16 = 405

Average weight=405

Calculate the average relative intensity by dividing the average weight by your 1RM and multiplying the result by 100:

(405 : 500) x 100 = 81%

ARI=81%

Calculate the volume coefficient by multiplying the NBL by the ARI.

16 x 81= 1,296

VC=1,296

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Now combine the numbers for both lifts:

	NBL	**Tonnage, Lbs.**	**Average weight, Lbs.**	**VC**	**ARI, % 1RM**
BP	29	6,365	219	2,117	73
DL	16	6,480	405	1,296	81
	Add up the NBL for both lifts	*Add up the tonnage for both lifts*	*Divide the workout tonnage by the workout NBL*	*Add up the VC for both lifts*	*Divide the workout VC by the workout NBL*
	29+16 = 45	*6,365 + 6,480 = 12,846*	*12,846 : 45 = 285*	*2,117 + 1,296 = 3,413*	*3,413 : 45 = 75*
Session	**45**	**12,846**	**285**	**3,413**	**75**

I have darkened the cells with the most informative data. For obvious reasons, the weight averaged out between the BP and the DL is not going to tell you much and neither will the total tonnage.

The total of reps is more telling and so is the ARI—useful measures of volume and intensity respectively. The VC is mostly used as an intermediate step to calculate the workout's ARI.

When it is all said and done, all you have to write down in your log is this:

	NBL	**ARI, %**
BP	29	73
DL	16	81
Session	**45**	**75**

These numbers give you a very good idea of the volume and intensity of each exercise and the session at large.

Carl (1967) proposed intensity zones using 20% and 10% jumps and Roman (1969) proposed 5% wide intensity zones. Chernyak & Medvedev (1972) suggested 10% and it stuck. Carl's zones were too broad to accurately describe the training load; Roman's were too many and thus cumbersome. 10% hit the spot.

Intensity zones in weightlifting
(Chernyak & Medvedev, 1972)

0. 40-50%—Minimal
1. 51-60%—Low
2. 61-70%—Small
3. 71-80%—Medium
4. 81-90%—Large
5. 91-100%—Nearmaximal and maximal
6. 101-110%—Supramaximal I
7. Over 110%—Supramaximal II

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The supramaximal zones exist because snatch pulls and clean pulls are prescribed in percentages of the competition snatch and C&J maxes. First, for ease of planning. Second, because Soviet scientists determined very heavy pulls to be detrimental to weightlifting performance, hence lifters were not encouraged to find our their maxes.[2]

For general strength purposes we use five zones, as does Boris Sheyko for powerlifting:

Zone #	% 1RM
1	50-60%
2	61-70%
3	71-80%
4	81-90%
5	91-100%

Obviously, there is a long way between the top and the bottom weights of each 10% intensity zone—especially for stronger athletes. Thus came about the so-called "central weights" that split each zone in the middle: 65% for the 61-70% zone, 75% for the 71-80% zone, etc. Establishment of this term greatly simplified the calculations. To calculate the average relative intensity in each column multiply the central weight in % by the NBL. (Nechepurenko)

The average relative and absolute intensity—% 1RM and the weight—are determined for all training units: a session, a month, a training period, a training cycle, a year. It must be noted that a training month equals four weeks, rather than a calendar month.

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VOLUME

- "Accumulation of quantity leads to a jump in quality." (Ozolin, 2006) A possibility of delayed transformation. (Matveev)

- "Chronically maintained high demands are necessary for a stable adaptation." (Viru, 1995)

- "Mastering large loads is connected with long term reconstruction of the organism." (Roman, 1986)

- "Up to a certain volume of training (called the saturation threshold) there exists a direct dependence of increase [in results] on the volume." (Viru, 1995)

- It is believed that the plastic reserve of the organism (adaptivity) is not only exhausted but also is supercompensated. (Viru, 1980)

- In the first years of training results linearly grow along with volume and intensity. (Medvedev, 1965)

- [Later] there is no correlation between annual volume and results. (Vorobyev, 1981)

- There is a low correlation between volume and gains. (Prilepin, 1975)

- "For experienced athletes, the number of lifts with 70% 1RM and higher influences the growth of results in the snatch [and C&J]: the higher this number, the more the results grow. It is natural for athletes to attempt to increase the number of such lifts, but it should be done within reason and individually for each athlete. The most important condition—the athlete's organism must recover from such loads." (Roman, 1986)

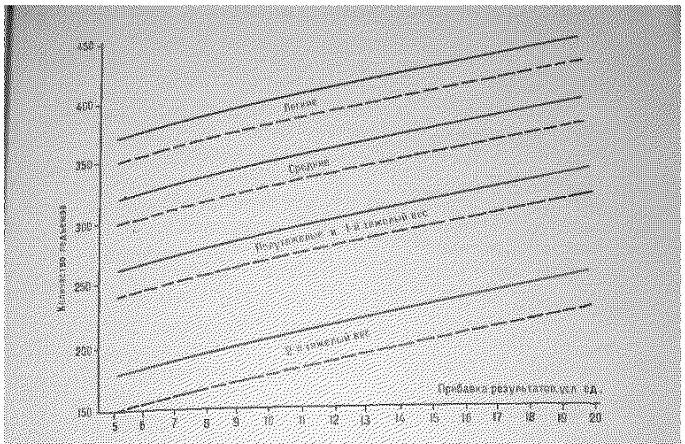

Рис. 19. Зависимость прироста результатов от количества подъемов штанги весом 70% и более в рывковых и толчковых упражнениях (в соревновательном периоде).
Условные обозначения:
——— у мс;
— — — у кмс и спортсменов I р.

- "Everything else being equal, an increase in the training volume facilitates a large increase in muscle mass." (Roman, 1986)

- Muscle mass decreases when the volume is lowered. (Roman, 1986)

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1. **Volume per year, a sum of all barbell exercises**
2. **Volume per month, a sum of all barbell exercises**
3. **Volume per month, press**
4. **Volume per month, squat**
5. **Volume per session, a sum of all barbell exercises**
6. **Volume per session, one exercise class**
7. **Volume per session, press**
8. **Volume per session, squat**

Volume per year, a sum of all barbell exercises

- Between 1964 and 1973 an annual NBL was fairly stable among top Soviet lifters and below 10,000. Since elimination of the press the NBL dramatically increased to 20,000 by 1978. (Medvedev, Verkhoshansky, et al, 1992)

- Norwegian powerlifting team today—NBL 12,000-16,000 per year (counting from 60% 1RM). (Wolf, 2014)

Volume per month, a sum of all barbell exercises

- An 8-week NBL went up by 40% on average from the 1960s to the late 1980s. (Medvedev, Verkhoshansky, et al., 1992)

- Monthly volume for beginners and Level III—no more than 1,000 NBL (700 average). Then grows to 2,000 and more for MSIC[3]. (Medvedev, 1986)

- Powerlifting monthly volume classification (Tsukanov, 2005):

    - NBL 600–low
    - 600-800—medium
    - 800-1200—heavy
    - Over 1,200—maximal

Volume per month, press

- High level lifters had NL 200-250 in the high intensity zones, 70% and up. (Vorobyev & Medvedev, 1972) Double than number when factoring in 50-70% weights.

- 100—monthly presses' volume in a preparatory period. (Roman, 1986) It is "easy strength"—the press is no longer contested and is not a priority.

- In some of Sheyko's plans monthly volume in the prep period exceeds 700 for Level II lifters and exceeds 800 for CMS-MS (includes bench presses, inclines and dips—but not dumbbell military presses or wide grip pushups).

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Volume per month, squat

- Level III-CMS—NL 140-330. (Based on Chernyak, 1978)

- For the elite the volume tripled from the 1960s to the late 1980s from 138 to 413. (Medvedev, Verkhoshansky, et al., 1992)

- Roman (1986):

Rank	PREP	COMP
III, II	337	243
I, CMS	380	276
MS	420	300

- Note that the SQ NBL in modern Russian powerlifting is practically identical to that in the 1980s Soviet weightlifting. Below are average monthly squat volumes by Roman (1986) next to the NBL from two Sheyko's plans for powerlifters of the same level:

Rank	WL	PL
III, II	337	356
MS	420	439

Volume per session, a sum of all barbell exercises

- Experienced weightlifters average 28-35 sets (NBL 84-105[4]). Some as many as 60-70 sets (NBL 180-210). (Rodionov, 1967)

- For experienced athletes (Vorobyev, 1964):

    - An intense session 15-25 sets (NBL 45-75)
    - A typical session 35-40 sets (NBL 105-120)
    - A session with high volume of assistance 40-60 sets (NBL 120-180)

- NBL 120-160. (Roman, 1968)

- 40-50 sets for experienced athletes (NBL 120-150). (Falameev, 1986)

- Typically 50-100 lifts per session; 75 on average. (Chernyak, 1978)

- Up to 50 NBL per session—small volume; 51-100—medium; over 100—large. (Roman, 1986; also used by Sheyko)

- No more than two weekly sessions with >100 NBL for lower ranked lifters. (Roman, 1986)

- Lifters who have low endurance and are unable to perform a high number of sets see slow growth of results. They need to gradually up the volume, initially with lighter weights. (Rodionov, 1967)

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Volume per session, one exercise class

- 10-15 sets per exercise (NBL 30-45) and not more than 20 sets (NBL 60). Up to 25 sets (NBL 75) for a lagging exercise. (Vorobyev, 1964)

- A general trend: no more than NBL 35-40 to avoid injuries[5] and no fewer than 10-15 to assure a training effect. (Chernyak, 1978)

Volume per session, press

- For experienced lifters Roman (1962) recommended around 8 sets of the classic press (together with press variations the sets numbered 12-15 (NBL 36-45) when training with optimal, 75-85%, weights. Reduce the volume in sessions using max and nearmax weights.

- No more than NBL 30-40 in the high intensity zones. (Vorobyev, 1964)

- Roman (1968) proposed up NBL to 50 and more. Analysis of Roman (1962) shows 30 as the bottom limit. Only 25-30% of that volume was with weights lighter than 70% 1RM.

- 8-10 sets and 12-18 sets for high level athletes, including special exercises (typically 1-3 reps per set, high intensity zones only). (Rodionov, 1967)

- High level lifters NBL 20-40 in the high intensity zones. (Vorobyev & Medvedev, 1972) Experienced lifters should start with 70-75% 1RM and train with 80-90% after just a couple of sets. Two series: 4-6 sets in the beginning and 6-8 sets in the end of a session.

- NBL for a given % 1RM for Levels III, II—70%—15, 80%—15-20, 90%—10. (Prilepin, 1975)

- NBL 6-30 (Roman, 1986)—in the era the press is no longer contested and not a priority.

- Typical BP NBL 30-60. Up to 90 and even 110 on high volume days. (Sheyko, 2005)

Volume per session, squat

- Squat session NBL greatly increased from the 1960s to the 1980s. Vorobyev (1964) used to recommend NBL 12-24. Later he upped these numbers considerably. E.g., his 1981 sample training session for a Level I lifter included 42 SQs (followed by 28 GMs).

- 30-40 broken up into two series (Roman, 1968)

- "Barbell squats are included almost into every training session. Up to 60 squats may be done per session but 20-30 are typical. When the number of squats in a session is high, it may be broken up into two series." (Roman, 1986)

- Qualified athletes—30-40 and even up to 50. Beginners tire at 15-20. (Chernyak, 1978)

- For a given % 1RM for Level III and II—70%—20, 80%—15, 90%—10. (Prilepin, 1975)

- Powerlifters, prep period: Level II—30-75, CMS-MS—30-65 (based on Sheyko, 2005)

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INTENSITY

"Our research has shown that improvement of the system of restoration has no influence on the training methods that reflect the magnitude of the resistance and the number of repetitions per set… the optimal number of submaximal and maximal (90-100%) barbell lifts [in the competition lifts]… the distribution of the load by intensity zones." (Medvedev, Verkhoshansky, et al., 1992)

1. **Average relative intensity (ARI)**
2. **Optimal or main training weight**
3. **NL in the 90-100% zone**
4. **Distribution of the volume by intensity zones**
5. **Number of repetitions per set**
6. **Number of repetitions compared to RM**
7. **Rest periods between sets**

Average relative intensity (ARI)

"Determining the optimal average weight that leads to the highest increase of results is one of the main tasks for the coach and the athlete." (Vorobyev, 1981)

ARI is an average weight lifted in a training unit in % 1RM.

Lifts starting with 50% 1RM or 70% are used in calculations. If measurements are made starting with 70%, it is specified as "ARI in the high intensity zones".

For weightlifters from the intermediate level and up the annual ARI has been constant for half a century:

Weightlifting annual ARI = 75±2% (Medvedev, 1986)

This number may be lower or higher depending on the lift. E.g., press—67-77% (Medvedev, 1968), squat—66-71%[6] (Medvedev, 1986).

The powerlifting ARI is slightly lower:

Powerlifting annual ARI = 70±3% (Sheyko, 2008)

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Optimal or main weight

The "main" weight is one that is used more than any other. Some authors offer one number; others a range. "Main" and "optimal" are usually synonymous.

"In science and engineering the solution that meets all the set requirements at a minimal "cost" is called optimal." (Vorobyev, 1981)

The "optimal/main" and "average" may be not far from each other but they are not to be confused. E.g. in Sheyko's system, 70±3% represents the average weight or ARI, while 80% is the main weight, more frequently used than any other. A high number of light lifts pull the average down. (The main intensity range for Sheyko's system is 70-85%).

"Even before the [WWII] and the first years after the war Soviet weightlifting specialists favored training with medium weights." (Dvorkin, 2005)

Over the years relative contribution of different size weights has varied but regardless, medium weights were represented more than any other. Later Luchkin, Vorobyev, Medvedev, Roman and other top Soviet specialists specified the main intensity range for weightlifters as 75-85%. It is common for experienced lifters to perform over half of their volume in this range.

Vorobyev (1964) referred to the 75-85% range as the "main training weight" or "optimal weights". These weights are used because one cannot lift maximal weights often. "Training with limit weights has a greatest effect in increasing muscular strength. However, athletes are only able to lift them 2-3 times, and not every training session. This is why optimal training weights, which enable one to train several sessions in a row without a reduction in work capacity, that is… 75-85% 1RM, are used."

The main intensity range in powerlifting is broader, 70-85%. (Sheyko, 2008)

Falameev (1986) defined training with 75-85% 1RM as the "medium load method" and it is the backbone of effective training. "3-4 rep sets with 75-80% weights in main exercises is the core of year round preparation." Regardless of the phase of training and the athlete's level, they make up over half of the volume. (Medvedev, 1990) The predominant weight for athletes past the beginner stage is 80%. (Medvedev, 1986)

"Weights significantly in excess of [80% 1RM] strictly limit the duration of the exercise's influence, thus reducing the possibility of activating metabolic processes in muscular and other systems of the organism and sharply decrease the load volume [and volume] is needed for providing chronic adaptation processes. Excessively light weights do not allow effective stimulation of strength development since the degree of muscular tension is proportional, within limits, to the weight of the resistance. In this case repetitions to RM would stimulate endurance rather than strength." (Matveev, 1991)

The optimal weight for the back squat is 67-82% 1RM. (Falameev, 1986)

NBL in the 90-100% zone

- Saxonov, Medvedev, Vorobyev (1963) proposed registering the NBL of ≥90% lifts.
- Chernyak (1978) refers to the stress experienced in this "fifth zone" or "top zone" as "psychophysical", as opposed to the "physical" stress of training in lower zones.

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- Later some Soviet specialists tracked the number of heavy lifts starting with 90%—while others from 91% to fit into the new zones.

- "Exceeding the individual optimal number of 90-100% lifts in best case scenario delayed reaching a high result; in the worst case it lead to overstrain and overtraining. An insufficient number of such lifts lowered the effectiveness of the training process." (Medvedev, 1968)

- Elite lifters pressed 90-100% 3-25 times in the month leading up to a successful competition. (Medvedev, 1968)

- For all lifts together the monthly NBL in the 90-100% zone ranges from 5 to 30, depending on the level and weight class. (Medvedev, 1986)

- Lighter lifters' ≥90% NBL is double that of heavier lifters'. (Medvedev, 1986)

- Younger and intermediate lifters' lifters' NBL in the 90-100% zone is higher than that of older and more experienced lifters. (Medvedev, 1986)

- The number varies from lift to lift. (Medvedev, 1986)

- Recommended ratio of 90-100% NBL between the preparatory and the competition months—40:60. (Medvedev, Verkhoshansky et al., 1992)

- "One thing is certain: if one chooses to expand the number of lifts in the 90-100% zone, it should be done mostly with 90-92.5% lifts." (Medvedev, 1986)

- Some athletes, especially heavyweights, are unable to lift 97.5-100% weights in training. They should lift 90-95% weights. (Roman, 1986)

Successful distribution of heavy lifts in the 90-100% zone [7]
(Medvedev, Verkhoshansky et al., 1992)

% 1RM	**90-92.5**	**95**	**97.5**	**100**
Share	65%	20%	15%	0%

- Do not exceed 7 presses at 90% or 3-4 at 95-100% in a session. (Roman, 1968)

- Do not exceed 1-3 lifts in the top zone in an exercise group every 4-5 sessions in the competition period—and even less frequently in the preparatory period. (Chernyak, 1978)

- Younger, less experienced lifters better tolerate a higher ≥90% NBL. Razryadniki do about twice as many as the elite. With 60-70% 1RM vice versa: the elite do about 2/3 of razryadniki's NBL. The share of 75-80% lifts is about the same for both. (Medvedev, 1986)

- Do max SQ two to three times a month. (Vorobyev & Medvedev, 1972)

- Lifts in the top zone on Russian national powerlifting team—90% or opener 2.5-3 weeks out; max on one's weak lift(s)—3-4 weeks out. (Sheyko, 2005)

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Distribution of the volume by intensity zones

Intensity zones in weightlifting [8]
(Chernyak & Medvedev, 1972)

.

0. 40-50%—Minimal
1. 51-60%—Low
2. 61-70%—Small
3. 71-80%—Medium
4. 81-90%—Large
5. 91-100%—Nearmaximal and maximal
6. 101-110%—Supramaximal I
7. Over 110%—Supramaximal II

Distribution of volume by intensity zones in training of top Soviet lifters in successful preparation for important competitions between 1960 and 1971
(Medvedev, Chernyak & Matveev, 1972)

50-60%	61-70%	71-80%	81-90%	91-100%
10	25	**35**	25	5

The above numbers are writ in stone. Later Medvedev (1986) did a statistical analysis of the top Soviet athletes' loads for five Olympic cycles, from 1964 to 1980. The numbers were almost identical:

Distribution of volume by intensity zones in training of top Soviet lifters in five Olympic training cycles
(Medvedev, 1986)

Years	50-60%	61-70%	71-80%	81-90%	91-100%
1964	16	26	**31**	14	13
1968	16	24	**36**	19	5
1972	25	**34**	28	10	3
1976	21	27	**32**	15	5
1980	19	28	**34**	15	4
1988*	8	24	**35**	26	7

* Zatsiorsky (1995)

To hammer the point that these numbers are universal and not limited to the snatch and C&J, elimination of the press from competition after 1972 made no difference—and neither did the proliferation of steroids in the 1970s and 1980s.

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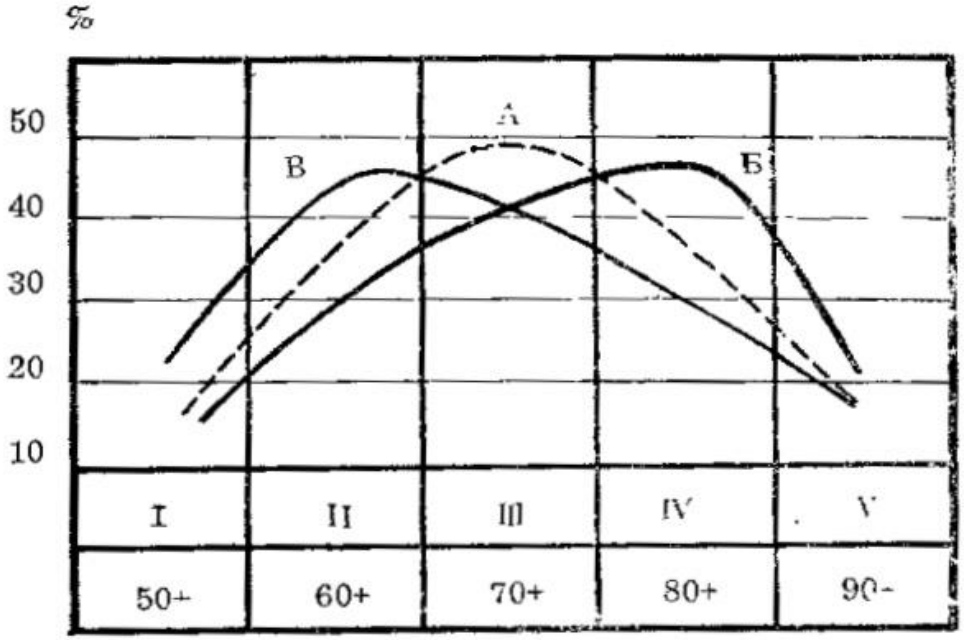
- Normal distribution usually applies.[9] Typically the 71-80% zone is predominant, with 75-80% often making up half the volume.[10]

95-100%
90-95%
85-90%
50-70%
70-75%, 80-85%
75-80%

- Individual distribution can be shifted from medium weights in the 71-80% zone (curve A is statistically normal) to the left or to the right, into the 61-70% zone or 81-90% zone (Matveev, 1977):



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- Most lifters should aim the peak of the curve for the 61-70% zone or the 71-80% zone. (Chernyak, 1978)

- In three experiments Level III to CMS lifters using 65% average intensity (compare to the 75% Russian standard) made *continuous progress for two years*. In addition, the lifters felt less fatigued than the higher intensity group. (Chernyak, 1978)

- There have been precedents of world class lifters using a distribution favoring the 61-70% zone (see the table below). Jan Talts trained his press at 67% average intensity. Alexeev flat out stated: "No one would have had any injuries if they trained the way I suggested: multiple lifts of medium weights that the muscles can handle. I would kill people for heavy gym lifts."

Distribution of volume by intensity zones
of six gold medal winners at a national competition
(Chernyak, 1978)

P, Sn, C&J	**50-60%**	**61-70%**	**71-80%**	**81-90%**	**91-100%**	**ARI**
Prep	25.7	**33.1**	25.7	11.2	4.3	**68.5**
Comp	19	**38.1**	24.4	12.8	5.7	**69.8**

SQ	**50-60%**	**61-70%**	**71-80%**	**81-90%**	**91-100%**	**ARI**
Prep	28	**34.4**	27.6	8.7	1.3	**67.1**
Comp	20.8	**37.3**	23.2	17.7	1.0	**72.1**

- If the goal is to increase muscle mass, up the absolute volume in the 50-60% and 61-70% zones without making significant changes to the absolute volume in the higher intensity zones. The share of the lifts in the two bottom zones may reach 50% of the total volume (up to 75% for the heavyweights) thus significantly lowering the average intensity.

- At times one may forgo the normal distribution and either flatten the peak (a relatively even distribution across the 61-70, 71-80, and 81-90% zones) or opt for two peaks (decrease the volume in the 71-80% zone and increase in the other two). (Chernyak, 1978) For example:

Normal distribution	22	**34**	22
Flat peak	**31**	**31**	29
Double peak	**26**	18	**27**

- Selecting weights only on the top, only on the bottom, both, etc. of a wide 10% intensity zone is another way to individualize. (Chernyak, 1978)

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Some successful distributions of volume by intensity zones
(Rounded up)

Lifts, lifters, sources		50-60%	61-70%	71-80%	81-90%	91-100%	ARI
Top lifters in successful preparation for important competitions in 1960-1971 (Medvedev, Chernyak & Matveev, 1972)		10	25	**35**	25	5	**74%**
Sample press month (Matveev, Roman & Chernyak, 1972)		20	**40**	20	15	5	**69.5%**
Based on Vorobyev (1988)		20	25	**35**	15	5	**71%**
Squat (Chernyak, 1978)		19	30	**40**	10	1	**69.4%**
Razryadniki squat (Pakov, 1978)		15	20	**40**	20	5	**73%**
SQ for an MS in select months (Chernyak, 1978) "A relatively stable ARI does not imply the same distribution across the zones." In one example the lifter focused on 61-70% and 81-90% weights in month one; 71-80% in month two; all of the above in month three.							
		7	**28**	**29**	24	12	**75.7**
		25	**30**	20	19	6	**70.2**
		18	22	**30**	20	11	**73.5**
		15	**29**	**29**	25	2	**72.0**
		7	**31**	**31**	29	2	**73.8**
		8	**32**	**30**	27	3	**73.5**
		20	**35**	26	18	1	**69.5**
Snatch by month for high razryad lifters (Chernyak, 1978b)							
	1-2	10	25	**35**	25	5	**74%**
	3-4	15	30	**40**	10	5	**71%**
	5	18	**35**	30	15	2	**69.8%**
	6	20	**40**	30	8	2	**68.2%**
	7	10	20	**40**	25	5	**74.5%**
	8	8	25	**45**	20	2	**73.3%**
	9	18	**35**	30	15	2	**69.8%**
	10	10	25	**35**	25	5	**74%**
	11	15	30	**40**	10	5	**71%**
Some useful guidelines							
Squat (Chernyak, 1975)		75			25		
Burkov & Nikityuk (1985)		30	60		10		
Medvedev (1986)		40		50	10		
Powerlifting							
One of Sheyko's BP plans		31	30	**34**	5	0.2	**69,0%**
30 elite bench press specialists (based on Belsky, 2000):							
Prep	Competition and accessory lifts,	8	21	**40**	31	---	**74.4%**
	Competition lifts only	14	32	**33**	17	4	**71.5%**
	Accessory exercises only	6	18	**44**	32	---	**79%**
Comp	Competition and accessory lifts,	---	16	**55**	24	5	**76.8%**
	Competition lifts only	7	10	26	**31**	26	**80.9%**
	Accessory exercises only	---	16	**59**	23	2	**76.1%**

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Number of repetitions per set

"Training of bodybuilders is characterized by multiple repetitions with relatively light weights. This allows them to achieve a significant increase in muscle mass. However, in spite of large muscle mass (and in some bodybuilders harmonious development of the majority of muscle groups), they are unable to show high results in lifting weights. Training of high level weightlifters has to do with lifting weights no lighter than 70% of maximum with a strict limitation of repetitions; that is, the training is primarily directed at strength development and maintenance of muscle mass." (Vorobyev, 1981)

Average weight gain from beginner to elite: 14kg for lightweights (in 4 years), 23.5kg for middleweights (in 4.5 years); 60kg for second heavyweights (in 6.5 years). (Kudyukov, 1981)

- "Varying the number of barbell lifts per set throughout the 1-6 range is the optimum that is necessary for weightlifters' training. Increasing the number of repetitions to 7 or lowering them to 1-3 has a negative effect on strength development." (Vorobyev, 1981)

- 5-6 rep sets are best for hypertrophy. In monthly cycles sets of 5-6 reps make up 6-10% of the volume; no less than 15% for those wanting to gain mass and heavyweights. (Vorobyev, 1981)

- "…Berger (1963) researched the question, why six repetitions in each set were more effective than two or ten repetitions. He concluded that training with a heavy weight did not provide an optimal number of repetitions needed to increase strength while training with a light weight and a high number of repetitions was an insufficiently intense stimulus." (Verkhoshansky, 1977)

- "4-6 reps per set definitely increase the muscle mass, while building only slightly less strength than 1-3 repetitions with high intensity." (Roman, 1986)

- "Athletes who want to keep their weight down train primarily with 1-3 reps. But even they should periodically include 5-6 rep sets." (Vorobyev, 1981)

- High level athletes, especially if they try to gain muscle mass, do up to 5-6 reps in special exercises and up to 10-12 in squats. (Falameev, 1974) Reps may also go up to 10-12 in bench presses and incline presses. (Roman, 1986)

- Up to 8 reps per set for heavyweights (compared to 6 for lighter lifters). (Roman, 1986)

- One recovers from strength-endurance training "a lot slower" than from strength training. (Roman, 1962)

- In preparation for the 1988 Olympics the Soviet weightlifting team did half of their squat volume in the 3-4 rep range, followed by 5, 2, and 6. The number of singles and reps above six was negligible. (Zatsiorsky, 1995)

- "Sets of 3-4 reps (75-80% 1RM) in main exercises are the core of the year round training process [half of the volume]."[11] (Medvedev, 1996)

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Number of repetitions compared to RM

- "Weightlifting training must include a significant number of sets with large weights and, therefore, a low number of repetitions. Six sets of three reps are less tiring than three sets of six. Fatigue leads to technique flaws." (Vorobyev, 1981)

- "In training one must avoid sets demanding supralimit efforts… In physiology optimal loads are considered the best for strength development..." (Sheyko, 2005)

- An example why doing high reps to or close to failure is incompatible with serious skill practice can be found in a study by Chen (1998). The subjects did 3x10RM. Their 1RM was tested on the third, sixth, and ninth day. It was reduced by 43%, 22% и 28% respectively!

- Matveev, Roman & Chernyak (1973), Rodionov (1976), and others believed that repping out to RM or close to it leads to fatigue and prevented the athletes from fulfilling the required volume of work. Roman (1968) added, "Multiple repetitions with small and medium weights make less sense… [Even if taken to failure] the tension developed is not identical and smaller than produced by a single lift of a nearmaximal or maximal weight. In addition, last repetitions in a set are performed on the background of reduced due to fatigue excitability of the central nervous system. This makes it difficult to form the subtle conditioned reflexes which drive further strength development."

Experiments with different average rep counts
(Matveev, Roman, Chernyak, 1973)

50-60%	**61-70%**	**71-80%**	**81-90%**	**91-100%**
8	6	4	3	1
"Such a number of reps per set turned out to be excessive and one after another the athletes started complaining about a rapid onset of fatigue and inability to continue the training session. Some athletes received minor injuries of the shoulder, back or wrist.				
7	5	3	2	1
"The athletes quickly fatigued at a high number of reps."				
5	4	3	2	1
Success.				

- Indeed, training to RM was classified as a "non-standard exercise" (Belov, Kuzmin, Risin, 1984). Curiously, Poletaev (2008) believes that this attitude was in part ideologically driver, as training to failure was considered a Western way. Poletaev, himself a former USSR record holder in weightlifting, a sports scientist, and the head coach of women's national weightlifting team, points out that training to RM has been successfully used by Soviet coaches with athletes of all levels, including weightlifters, and has been endorsed by some big names, e.g. Ozolin (1970), Roman (1986), and Verkhoshansky (1988). (At StrongFirst we use occasional training to RM in cycling type plans.)

- It is often thought in the West that the very low reps used by Russian weightlifters with given weights reflect a selection of exceptionally fast twitch athlete. Research by Matveev, Roman, Chernyak (1973) shows that it is not so, at least for lifters below the elite level. They tested RM of 40 lifters, Level III to CMS, in snatches, C&Js, presses, and squats. The results showed that weightlifters' numbers did not differ from the general population. Later research by Rodionov (1976) revealed similar numbers:

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RM for weightlifters

	60%	70%	80%	90%
All athletes for comparison[1] (Matveev, 1991)	**15-20RM**	**10-15RM**	**6-10RM**	**3-6RM**
Level III to CMS weightlifters (Matveev, Roman, Chernyak, 1973)	14-19RM *	12-15RM	6-9RM	3-5RM
Weightlifters (Rodionov, 1976)		12RM	8RM	4RM
* All exercises but C&J				

- "Apparently, the number of repetitions in each set should be sufficient to stimulate a growth of strength and at the same time it must not be too high as it might break down the exercise structure… a half of the repetition maximum… is exactly the number of lifts at which a significant reduction in output is not yet observed." (Rodionov, 1976)

Optimal repetition number per set for weightlifters and powerlifters

	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%
RM (Rodionov, 1976)					**12**	**10**	**8**	**6**	**4**	**2**
Press, experienced lifters (Roman, 1962, 1968)			3-6	4	4	3-6	3-5	2-4	2	
Optimal reps, use 40% of the time (Matveev, Roman, Chernyak, 1973)	5			4		3		2		1
Same as above; range	4-10			2-8		1-6		1-3		1-2
Chernyak (1978)	4-9			2-7		1-5		1-3		1-2
Rodionov (1976)					6	5	4	3	2	1
Prilepin (1975)					3-6		2-4		1-2	
SQ, P, BP (Medvedev, 1986)	4-8		4-6		4-5		3-4		2-3	
	Up to 10 for hypertrophy, especially for squats									
SN, C&J, etc. (Medvedev, 1986)			-6		-6		-4		-2	
BP, SQ (Sheyko, 2008)	5-6	5*	4-6*	4-5*	3-5*	3-4	2-3	2	1	
	* In BP "marathons" up to 10 reps									
DL (Sheyko, 2008 and other top powerlifting coaches from the former USSR)	3-4				2-3			1-2		1

Use of different repetition ranges depending on the training phase for weightlifters of different levels, %
(Medvedev, 1996)

Exercises	Periods								
	Preparatory			Competition			Transition		
	Reps per set								
	1-2	3-4	5-6	1-2	3-4	5-6	1-2	3-4	5-6
Level II, I, and CMS									
Main	29	55	16	47	48	5	17	51	32
Additional	---	42	58	-	54	16	---	39	61
MS-MSIC									
Main	16	56	28	27	60	13	10	55	35
Additional	---	45	55	-	41	59	---	43	57

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- A guideline based on Matveev, Roman & Chernyak (1973): In the 70-90% zones dedicate 40% of the volume to the repetition number deemed optimal.[12]

- You may use different rep ranges for different lifts. In squats athletes can do more reps at a given % 1RM than in presses. (Rodionov, 1976)

- Individualize reps per set keeping the same volume and intensity. (Chernyak, 1978)

- Reps may be lower than the above in warm-ups. When warming up before maxing do triples with 60-70%, then 4-5 doubles and singles. Less confident lifters may do 2-3 extra warm-up sets and get close to the first attempt. (Vorobyev, 1981)

- It is worth noting than Medvedev (1986) recommends doing "additional" exercises for 5-8 reps with 70% 1RM and 4-6 reps with 80 1RM—and lists no other options. Elsewhere he states that one does not vary assistance exercises intensity; only volume.

Rest periods between sets

- "The rest pause is the most important factor in preventing failure in activities demanding complex skills. The rest pause must be viewed a means of reducing the cumulative effect of fatigue." (Vorobyev, 1981)

- "Special studies have not succeeded in finding precise and stable values or rest between sets. It is rather variable—from 2 to 5min… Subjective readiness for lifting the barbell is a rather accurate barometer for selecting the optimal rest period." (Vorobyev, 1981)

- "Multiyear research has demonstrated that rest intervals usually range from 2 to 5 minutes. The next set should be performed when the athlete is subjectively ready for it." (Medvedev, 1969)

- Optimal rest intervals between sets of non-maximal lifts are 2-3min. Intervals less than 1min and greater than 5min lower the effect of the exercise. (Krestovnikov, Koryakina & Ermolova cited by Roman, 1962)

- Recommended rest intervals between sets for powerlifters:
  80% x 5/2—2-3min; 75% x 5/5—4-5min; 90-95% x 1-2—5-7min. (Sheyko, 2008)

- When warming up before rest for 3-4min between sets, the last warm-up set no less than 5min before the max attempt. (Vorobyev, 1981)

- "The most fitting is calm walking between sets while shaking muscles. After barbell press exercises accompanied by high tension of the muscles in the lumbosacral area, free and relaxed forward bends are effective.[13] After barbell back squats—relaxed jumps from leg to leg, shaking one leg at a time, etc." (Rodionov, 1967)

- Optimal rest between sets for a lifter—first active (walking, light jogging, relaxation exercises), then sitting down (active-passive rest). (Matveev, 1991)

- 3-5min rest between sets is optimal, based on observations of national team members and their subjective comments. "Excitability of the vegetative nervous system after a barbell lift remains at a certain level up to 5min and then gradually declines." (Rodionov, 1967)

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VOLUME VARIABILITY

- Certain things repeat over and over, hence the *cycling principle*. (Matveev, 1991)

- "The principle of cyclic organization was already established in the 1960s against the background of analysis of training of sportsmen belonging to the international class. (Matveev, 1964) Their training revealed that training influences and organizational forms repeat themselves at three levels… The duration of waves of the first group is approximately a week. The waves of the second group sum up three to six waves of the first group. Last, all these waves constitute the greatest wave with the duration of a year or half a year. Correspondingly, the training meso- and macrocycles are distinguished." (Viru, 1995)

- Vorobyev (1972) coined the term "variability."

- "Both non-organic and organic nature is characterized by the so-called step like function change. Conversion of chemical energy into mechanical, electromagnetic, and thermal energy also happens stepwise. Discrete changes on sub-cellular and cellular levels are most likely one of the characteristics of live organisms… we propose sudden yet fitting the given athlete's functional abilities changes in load -'jumps'. This principle of organizing the training loads allows one to achieve higher results with a smaller loading volume." (Vorobyev, 1977)

- The "delayed transformation" phenomenon: results lag behind volume dynamics. Best results are reached not when the volume is at its highest but after it has been stabilized or reduced. (Matveev, 1991)

- Vorobyev (1977) questioned delayed transformation.

- "To correctly build the training process one is first and foremost to rationally vary large, small, and medium loads." (Vorobyev, 1981)

1. **Dynamics over the years**
2. **In a year**
3. **In a complete cycle**
4. **From preparatory period to competition period**
5. **From month to month**
6. **Within a month**
7. **Within a week**

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Dynamics over the years

- Increase the NBL by 10-30% compared to the last year for 6 years or so, then stabilize the volume—the 2 stages in a lifter's career. (Medvedev, 1986)

- Annual NBL for beginners—10,000; low ranked lifters—12,000; Level I & CMS—15,000. MS—18,000; MSIC—less. (Roman, 1986) The national team in 1978, 1981-1984—20,000. (Medvedev, Verkhoshansky, et al, 1992) The annual volume doubles from beginner to elite.

- The workload goes up over years through (Vorobyev, 1981):

  1. Increasing the number of sets in a session.
  2. Increasing the number of sessions, while keeping the average number of sets per session constant.
  3. Both of the above.

- Once the target annual volume is reached, vary it minimally. Sample distribution of volume by year in a 4-year cycle: 22, 27, 27, 24% or 24, 26, 24, 26%—relatively level. It is not recommended to plan "stress years" to build a "foundation" for the future—growth of results slows down or even reverses. (Vorobyev & Medvedev, 1972)

In a year

- Once the target annual volume is reached, the volume of the first half of the year should not exceed that of the second by more than 5%. "The intention of some athletes to do peak volumes to build a foundation for further growth of results does not have a serious justification. Progress is then long delayed." (Vorobyev & Medvedev, 1972)

- "Division of the annual training into many cycles is not sensible. In that case, to prepare for every meet and to post a certain result the athlete must taper for it, that is sharply lower the volume of the training load every time. As there are many cycles this decrease takes place often and the athlete does not get a chance to perform the necessary volume of training and thus build the functional base for increasing the results... Usually division of annual training into a high number of short cycles and the desire of the athlete to do his best in every competition lead to a plateau and even a decrease of the results." (Roman, 1968)

- The length of the cycle and the load variability within it depends mostly on the athletes' individual characteristics. Those who can handle sharp load variations tend to stick to 2-3 month cycles with high variability. Those who cannot tend to follow 4-5 month cycles with a gradual load increase and an even decrease. (Chernyak, 1978)

- Only 70-90% of beginners and intermediates post PRs after a 2-month cycle; others take longer. (Chernyak, 1978) "Apparently, a 2-month long phase is insufficient for developing specific qualities in all athletes because of genetic individuality and different levels of giftedness. Probably those athletes who progress quickly can compete after two months and improve their results; the rest just repeat their results." (Chernyak, 1978) Hence the need for at least 3-month long cycles.

- Frequent competition is bad, as athletes do not get to handle large loads needed for significant progress and only stay with medium and low volumes to stay fresh for the meet. (Chernyak, 1978)

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In a complete cycle

- Preparatory period + competition period + transition period = big cycle. (Roman, 1986)

- "The major two tasks of all macrocycles are the creation of movement potential and its realization." (Viru, 1995)

- The preparatory period is one to two months long. (Roman, 1968) In Roman's words, the preparatory period builds a "foundation". It has the highest volume; the intensity in the competition lifts increases slowly (at least in the traditional Matveev's model; see the sidebar for an alternative approach).

- "The goal of the competition period is to reach the highest athletic form and to realize it." (Roman, 1968) A typical length is one month. "The competition period is characterized by a lower volume of loads and their maximal intensity." In Russian lifting slang, the lifter has "freed himself from volumes".

A dissenting opinion

"It is customary to divide preparation within cycles into three periods: preparatory, competition (main), and transition. It must be noted that terms "preparatory, competition, and transition periods" are most appropriate to the beginning stage of a weightlifter's preparation. A competition period as such does not exist for weightlifters. Competitions for each weight class take a day and an interval between them is about a month. This is why weightlifters refer to the pre-competition monthly preparation as the competition period, [but] the term "main period" is more fitting. As experience of the last several years has shown, the athletes who progress are the ones who constantly employ to some degree competition elements (load in the classic exercise and attempts to lift submaximal and maximal weights.) throughout the whole preparation cycle and regularly compete giving it an all out effort. Thus, the training load content in different months throughout preparation cycles do not differ sharply." (Vorobyev & Medvedev, 1972)

"We consider [Matveev's periodization model] a mistake… If every month or two for a year athletes [Rigert, Alexeev, others] show results exceeding national or world records, that means that they do not have preparatory periods in their training. The volume and the intensity are at optimal ratios throughout the entire annual cycle." (Vorobyev, 1977)

"Many authors say that a "foundation" of athletic condition is laid in the preparatory period through an increased volume with relatively low intensity. But… it is impossible to build such a "foundation" and it appears that the very terminology does not fit the dynamic functional state of an athlete. A live system cannot stay in a long static equilibrium." (Vorobyev, 1977)

- "The goal of the transition period is to exclude overtraining, rest before the new training cycle, and maintain the training form on a sufficiently high level." The percentage of special training is reduced, the weight is reduced, there is a greater variety of exercises, active rest. "It is not recommended to sharply reduce the loads or stop training for an extended period of time." (Roman, 1968)

- The transition period volume is very close to that of the competition period. (Medvedev, 1986)

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- The length of the transition period: 5 days after the first training cycle; 7 after the second; 10-14 after the third; 30 days after the last meet of the season. (Roman, 1968)

- Strength and work capacity lasts 7-10 days before dropping. (Vorobyev & Medvedev, 1972) After competition take no more than two days of rest (still do morning recharge). Then low loads for two weeks, hardly any classic lifts. (Vorobyev, 1981)

- One may plan a week of active rest after every two months of training. (Medvedev, 1986)

- A two-week active rest is possible after a half-year training cycle ending in an important competition. (Medvedev, 1986)

- "An athlete who aims to reach high results… needs specialized training all year round with brief passive or active rest. Even after an important competition, when the athlete is psychologically fatigued, it is not worth "check out" from training. Moreover, training with relatively light loads has a greater effect on restoration of the organism than passive or active rest." (Sheyko, 2005)

From preparatory period to competition period

- "The preparatory and the competition periods are not limited by some predetermined timeframe; the transition from one to the other is gradual." (Roman, 1986)

- Low volume loads are used (Falameev, 1986):

  - The beginning of the preparatory period
  - Before competition
  - The transition period

- High-level athletes reduce the volume in the competition month by 20-50%, provided they have been training with high volume. (Medvedev, 1965)

- Volume is reduced in competition month by 20-40% (30% average). The heavier the athlete, the greater the drop. For beginners enough to drop by 17-18%, as their intensity is still low. (Roman, 1986)

- According to the "accepted standards" the prep period volume is be 30-40% higher than comp volume. (Medvedev, 1986) For some elite lifters that number is only 10%. (Prilepin, 1981)

- For the elite the NBL ratio between the prep and the comp periods was close to 60:40 in the late 1980s. (Medvedev, Verkhoshansky et al., 1992)

- "For almost every athlete the volume in the month preceding a competition is rather stable and a strictly individual value. Such stability is not observed if there is no upcoming competition." (Vorobyev, 1981)

- The only time one can succeed increasing the volume in the competition month is if he competed unsuccessfully a month earlier after training with low volume. In all other cases the competition month volume has to go down or stay the same. (Chernyak, 1978)

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- "...athletes are often afraid to lower the workload during the competition month, reasoning that they would get out of shape. And there are other examples when athletes... start sharply lowering the volume... two months before... a non-important meet and taper for every such meet. It is wrong. In the first instance the athlete fails to completely realize his potential. In the second... the athlete hits the same PR every time as he fails to build his base... Either case is a brake for growth of athletic performance." (Roman, 1968)

From month to month

- "Months with high training volumes should be alternated with months with reduced volumes. A gradual volume increase fits only beginners, lifters of low ranks, and qualified athletes after a long transition period." (Roman, 1986)

- When planning monthly volume, factor in the last month's load. If it was high, there are some restrictions. If low or medium, any option is fine. (Roman, 1986)

- High-level athletes' monthly NBL fluctuates between 700 and 2,000. (Chernyak, 1978)

- "To demonstrate high results throughout the year, one must train with a volume that does not demand a long decrease or increase. Most often two to three weeks are more than plenty for that purpose." (Vorobyev & Medvedev, 1972)

Within a month

Weekly volume as a percentage of the monthly volume
(Roman, 1968; Chernyak, 1978)

Week volume, NBL			
Small	**Medium**	**Large**	**Maximal**
< 20%	20-30%	31-40%	> 40%

The above percentages can be used to compare weeks within one month. When you are comparing the last week of one month and the first of the next you should be comparing the NBL (Chernyak, 1978).

Chernyak succeeded in offering sixteen effective distributions using only four numbers: 15, 22, 28, 35%. The following terminology was used to name each one of these loads:

35%—Stress-developmental
28%—Developmental
22%—Maintenance
15%—Deloading

"Load planning with stable structural constants has been tested experimentally on subject athletes of different levels." (Chernyak, 1978)

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Types of monthly volume distribution by week using "stable structural constants"
(Chernyak, 1978)

Variant	Type	Week 1	Week 2	Week 3	Week 4
1	Gradual	35	28	22	15
2a	Sharp-gradual	15	35	28	22
2b		28	35	22	15
2c		22	35	28	15
3a	Gradual-sharp	15	22	35	28
3b		22	28	35	15
3c		15	28	35	22
1-3a	Sharp	35	15	28	22
1-3b		35	22	28	15
3-1a	Sharp	28	15	35	22
3-1b		28	22	35	15
4	Gradual	15	22	28	35
2-4a	Sharp	15	35	22	28
2-4b		22	35	15	28
4-2a	Sharp	22	28	15	35
4-2b		15	28	22	35
		1.5±% variation from the above values is acceptable. Some athletes may tolerate even greater variability than listed.			
Variants best suited for the pre-competition month, deloading on last week, are shaded.					

"It is quite possible that singular athletes have a greater variability range than shown. Our task is to discover the general relationships; not to get hung up on details. Obviously, the listed above variants of load distribution are not the only ones possible. There are others, especially in preparatory months." (Chernyak, 1978)

Gradual and combined variants look like triangles: the volume goes up to hit its peak on some week, then goes down. All gradual and combined variants have a single digit referring to the peak volume week in their names: 1, 3b, 4, etc.

In gradual variants one always changes to an adjacent value from the list of 15, 22, 28, 35%. E.g. from 28% you may go to 22 or 35—but not 15.

In combined variants you may jump as little or as much as you want—but you still have only one monthly peak and look like a triangle.

Sharp variability variants look like two triangles next to each other: the volume goes up and down twice. The first digit still indicates the week with the highest volume. The second digit marks the second highest peak. E.g., take a look at "1-3a": 35, 15, 28, 22. The first week has the highest volume (35). The load drops in week two and goes up again to another, shorter, peak of 28 and drops again to 22.

"Gradual and sharp load variations do no exclude but compliment each other." (Matveev, 1991)

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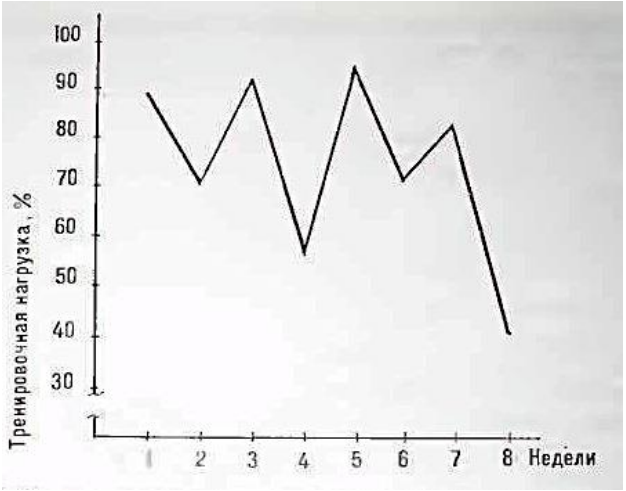
The power of sharp variability

Ermakov (1974) conducted a series of 8-10 week pedagogical experiments comparing different weekly and monthly load distribution. They compared more traditional, smoother, load variation by Matveev (1964) and Ozolin (1970) with much sharper changes. Experimental groups made strength gains 61% greater than the control groups. The researchers also discovered that athletes in the experimental groups were healthier and felt better. Those who used more of a gradual progressive overload sometimes came to training sessions not feeling well and unable to perform the required workload.



Experimental group's volume variation from week to week. (Ermakov, 1974)
100% is one's "maximal volume".

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Additional variants: preparatory period

Variant	Source	Week 1	Week 2	Week 3	Week 4
1	Chernyak, Atanasov & Ermakov (1972)	29	26	25	20 [1]
	Roman (1986)	31	27	23	19
	Sheyko (2013)	46	20	22	12
		34	30	24	12
2	Roman (1986)	20	35	27	18
		23	32	26	19
	Sheyko (2013)	22	38	25	15
		20	34	30	16
3	Roman (1968)	18	25	32	25
		18	31	34	17
	Chernyak, Atanasov & Ermakov (1972)	15	18	39	28
	Sheyko (2013)	22	27	33	18
		17	21	35	27
1-3	Roman (1986)	32	19	27	22
	Sheyko (2013)	36	16	27	21
	Wolf (2014)	37	16	27	20
1-4	Sheyko (2013)	35	22	14	29
3-1	Roman (1968)	25	20	37	18
	Chernyak, Atanasov & Ermakov (1972)	24	19	36	21
	Roman (1986)	27	18	32	23
2-4	Roman (1968)	24	36	15	25
	Roman (1986)	22	33	18	27
	Sheyko (2013)	21	35	13	31
	Wolf (2014)	16	38	20	26
4-2	Roman (1986)	21	28	17	34
	Sheyko (2013)	18	26	12	44

[1] An outlier with extremely low variability.

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Additional variants: competition period

Variant	Source	Week 1	Week 2	Week 3	Week 4
1	Chernyak, Atanasov & Ermakov (1972)	36	29	24	**11**
		46	27	14	13
	Roman (1968)	34	28	21	17
	Roman (1986)	36	28	24	12
	Vorobyev (1981)	36	28	21	15
	Medvedev	62 (about 2/3)		20	17
	Sheyko (2013)	40	27	20	13
2	Roman (1968)	27	35	21	17
	Chernyak, Atanasov & Ermakov (1972)	27	35	24	14
		27	39	25	**9**
	Vorobyev (1977)	29	30	27	14
	Vorobyev (1981)	26	35	23	16
		24	38	25	13
	Roman (1986)	28	33	26	13
	Sheyko (2013)	29	38	22	11
	Wolf (2014)	27	35	25	13
3	Chernyak, Atanasov & Ermakov (1972)	15	27	31	27
1-3	Chernyak, Atanasov & Ermakov (1972)	36	20	30	14
		35	24	30	11
	Roman (1986)	32	26	29	13
	Sheyko (2013)	38	20	28	14
3-1	Roman (1968)	26	21	27	17
	Roman (1986)	29	25	35	11
	Sheyko (2013)	28	24	34	14

- "A magnitude of load jumps increases with the athlete's level." (Falameev, 1986) Beginners and athletes coming back after layoff—minimal load variability. (Roman, 1986)

- Another variable is the athlete's load tolerance or work capacity. Athletes with low work capacity, even experienced ones, should avoid sudden volume swings from week to week. Athletes with high capacity should not be afraid of sudden jumps—yet even they should not do it too often.

- When monthly volume is high, there is not a lot of weekly variability. (Roman, 1968)

- "Same medium workloads sustained longer than two weeks lead to a loss of the traces left by larger loads." (Rodionov, 1967)

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- Supercompensation takes place as the volume is sharply, about 50%, lowered in the last week of the competition month. (Vorobyev, 1977) Last week of the competition month 11-14% of monthly volume is typical. At times be lowered below 10% (there have been precedents below 7%). One of the variables affecting the percentage is your desired NBL in the last week.

- "If there is no competition within two to three months, the principle of variable volume arrangement remains in force. However, there is no need to lower the volume in one of the weeks down to 12%. In this unloading week (and it must be present) one may reduce the volume to 15-18%. After that it is preferable to test one's maxes or have a training session with maximal weights." (Vorobyev, 1981)

- Chernyak (1978) lists the factors affecting the distribution of the monthly NL by weeks:

  - Current month's total NBL
  - Last month's total NBL
  - Psychophysiological and psychological factors
  - Load tolerance

Choice of a competition month taper
based on the previous month's volume
(Based on Chernyak, Atanasov & Ermakov, 1972)

Variant	Last month's volume	Deload from week #
1	Very high	1
2	High	2
1-3	Moderate	3
3	Low (insufficient)	4

- The longer the high volume prep period, the longer the taper. (Roman, 1968)

- Sheyko (2005) plans the highest volume week six weeks before a powerlifting competition, then gradually tapers.

  If your schedule does not cooperate with four-week blocks, 3, 5, and 6-week blocks may be planned. For a 7-week block do 4+3; for 9 do 4+5. (Chernyak, 1978) Possible distributions:

Types of monthly volume distribution by week
within non-standard length blocks
(Chernyak, 1978)

Block length, weeks	Sample load distributions by week, %
3	24, 48, 28
5	15, 27, 13, 30, 15
6	11, 19, 11, 26, 22, 11

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Within a week

"The most important task of the coach and the athlete is to most rationally distribute the load specifically in the weekly cycle, which may be called main. If there are miscalculations in monthly load distribution, they have less of a [negative] effect of the growth of athletic results than mistakes made when building separate weekly cycles." (Vorobyev, 1981)

Frequency

- Intervals between sessions may provide "normal", "excessive", or "partial" recovery. (Matveev, 1991)
- Different types of rest intervals between sessions (Matveev, 1991):
  - *Ordinary*—recovers the same work capacity (except the trace adaptations).
  - *Stiff*—underrecovery, summation of effects of several sessions for a stronger stimulus. "Experience convinces that it is one of the main ways of increasing effectiveness of training sessions enabling one to reach extraordinary results. But it is justified only under strictly defined conditions with the most important being—qualified regulation of loads and recovery." Otherwise overtraining.
  - *Suprercompensatory* interval is so long, that if one used only such intervals, the number of weekly training sessions would be inadequate.
- "It is significant that regulation of the rest intervals [during a session and between sessions] is not only aimed at recovery. It acts as one of the main means of managing the general effect of exercises and loads." Because next session falls on a different phase of recovery. (Matveev, 1991)
- "The degree of the cumulative effect of workouts is somewhat inversely related to the intervals separating them. There are reasons to believe that even a two-three day interval is too great to guarantee optimal accumulation of the [training] effect…" Skills also deteriorate and "other effects are lost". (Matveev, 1991)
- "The logic of shortening the intervals between the sessions is in creating the conditions facilitating summation of sessions' partial training effects… To guarantee extraordinary tempos of training progress one needs training sessions daily and multiple times a day (their number reaches 20 and more in the life of athletes of extra class)." (Matveev, 1991)
- In a microcycle at least one of the intervals must be supercompensatory. (Matveev, 1991)
- The number of training weekly sessions constantly changes. (Chernyak, 1978)
- A common frequency for the elite in the day: 4 days a week. (Rodionov, 1967)
- Until 1972 lifters trained 3-5 times a week; from 1973 they started training 2-3 times a day. (Medvedev, 1986) Note that the press had just been eliminated from competition.
- In the 1970s many leading athletes trained 5-6 times a week and 3-4 in the 15-20 days leading up to competition. (Vorobyev, 1977)

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- "The number of training sessions in weekly and monthly cycles depends on the athletes' qualification. The higher the qualification, the more frequent the training… However, the month before an important competition the number [of weekly training sessions] remains the same—12-18." (Vorobyev, 1977) That is 3-4 times a week.

- "As far as strength gain is concerned, no significant advantage from training more that 5-6 times a week has been observed." (Vorobyev, 1981)

- In the 1980s 12 weekly lifting sessions were the norm among the elite. (Medvedev, 1986) The scientist admitted the role of "restoratives".

- Number of powerlifting training sessions a week by leading Russian PL coaches (reported by Sheyko, 2005):

    - 3—7.1%
    - 4—21.4%
    - 5—64.4%
    - 6—7.1%

- Fragmentation of the strength training volume into more sessions improves results. (Zatsiorsky, 1995)

- In powerlifting training each lift 6 times a week delivers better results compared to 3 times, the volume being equal. (Wolf, 2014)

- Average number of weekly sessions—Level III—3; Level II, I, CMS—4; MS—5 (based on Vorobyev, 1981)

- As you progress, the training session gets longer—from 1-1.5 hours for beginners to 2.5-3 hours for experienced. (Vorobyev & Medvedev, 1972)

- Typical session duration for Level III was 1.5 hours, for Level II—1.5-2 hours, for Level I—1.5 hours to 2:15. (Falameev, 1986)

- Athletes of all ranks through Level I train 2-3 times a week in the transition period. (Falameev, 1986)

On training twice a day

- An early precedent in training 2-3 times a day—1954 world champion D. Ivanov.

- "It has been established that the organism will better convert the daily training load into results if [the load] is performed throughout the day in two-three sessions with rest breaks in between rather than all at once." (Falameev, 1986)

- For qualified athletes: two sessions a day on Monday, Wednesday, Friday. One 60-90min GSP session on Tuesday, Thursday, Saturday (squat jumps, wide grip PBN, seated and straight legged GM, back hypers, hanging exercises for the abs and back). (Medvedev, 1980)

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- Schedule recommended in the late 1980s to the elite by Medvedev, Verkhoshansky et al., (1992): two sessions a day on Monday, Wednesday, Friday; one on Tuesday, Thursday, Saturday. Tuesday, Thursday, Saturday are light days. A total of 9 sessions. This schedule was developed by Chernyak & Melnikova (1974).

- Another schedule recommended in the late 1980s to the elite (successfully used in preparation for the 1988 Olympics): up the load on the two a day days and take the days between off. A total of 6 sessions. (Medvedev, Verkhoshansky et al., 1992) Note that at the same time Hakkinen (1987, 1988a) discovered that while testosterone increased during the second session of the day, it then gradually decreased after several days of hard training. A single day off was enough to reverse this trend.

- "Many people are frightened by training twice a day. It is important to understand that it is easier to tackle a large volume in two sessions (e.g., perform 35 lifts in the morning and 55 in the evening instead of 90 in one session). It is obvious that two sessions allow a higher intensity and the recovery will be quicker and easier than from one session with the same load. Using this method our athletes do up to 120 lifts per day with no problem." (Sheyko, 2005)

- Either the AM or the PM session can be higher in volume. A typical ratio of 1:2 or 1:3 between the sessions but at times can be similar. (Chernyak & Melnikova, 1974).

- Other ratios between the sessions—2:3, 3:7. (Chernyak, 1978)

Adjusting the session number according to the week's volume

- The optimal way to increase the weekly volume is by adding extra sessions. (Chernyak, 1978)

Recommended number of training sessions according to the weekly volume
(Chernyak, 1978)

Weekly volume, NBL	Number of sessions
120-140	2-3
160-320	4
200-400	5
> 350	6

On symmetry between sessions

In the early 1960s high regularity of training (same loads, rest intervals between sessions, etc.) was still considered an asset. Luchkin (1962) believed unequal rest periods between training sessions to be counterproductive. If one trained 6 times a week, alternating days of main exercises and localized assistance exercises was recommended. Ozolin (2006) later recommended the same schedule to power athletes who are not lifters. Pavlov (1949) was quoted: "…dynamic stereotypes are developed when certain stimuli are repeated in a strict order and with strict intervals between them." Vorobyev later not kindly spoke about "Pavlov's apologists".

- Training every other day is most rational before competition. (Rodionov, 1967)

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- "We suppose that at the early training stage a certain uniformity of training loads is justified. It facilitates a quicker adaptation of the athlete to the load and a better acquisition of barbell lifting skills." (Vorobyev, 1977)

Load and rest dynamics in a week

- "There are many options [of load distribution within a week]. The important thing is not to forget waviness of loads." (Sheyko, 2005)

- For the elite a range of volume and intensity variation is very wide. E.g., 20-100 lifts per session (all barbell lifts). (Falameev, 1974)

- "Large loads have the greatest effect on the body and build the base for further growth of athletic performance. Medium loads serve to maintain form. Low loads are for active rest (recovery) and maintenance of form... but also for supercompensation; they create the highest level of functional abilities..." (Roman, 1968)

- "In his studies G. Folbort has proven that exhaustion processes stimulate restoration processes. [Yet] very high fatigue resulting from extensive muscular work is not desirable as recovery processes will stretch out for a long time and the athlete will enter the next training session with reduced work capacity." (Vorobyev, 1977)

- "When workloads are excessive, the organism does not have the time to restore by the next training session: it is unable to quickly remove and neutralize the metabolism products accumulated as a result of muscular activity. As a result, retroinhibition takes place. [A cellular control mechanism in which an enzyme that catalyzes the production of a particular substance in the cell is inhibited when that substance has accumulated to a certain level, thereby balancing the amount provided with the amount needed.] The strength of inhibition by the final metabolic product of a given metabolic pathway in cells of active tissues depends on the degree of exhaustion of the energetic and plastic resources of the organism. This is why it is extremely important to find not the maximal training volume for each athlete but the optimal volume that has the best effect on achieving sports results. In science and engineering an "optimal" is the solution that meets all the set requirements at the minimal 'cost'." (Vorobyev, 1981)

- "Today athletes more and more take training sessions in a phase of some underrecovery. Such training better stimulates recovery functions and has a positive effect of the growth of results. But constant training in the state of underrestoration could have dangerous consequences… chronic fatigue and overtraining." (Vorobyev, 1977)

- "The best conditions for anabolic processes are when small (2-4 ton) loads with low intensity follow large loads." (Vorobyev, 1977)

- Load variation for inexperienced lifters (all barbell exercises in a session): 1-2 tons—small, 4 tons—medium, 6-8—large. (Vorobyev, 1977)

- At a certain level for a greater stimulus an athlete may have two heavy sessions back to back followed by one to two with medium and small loads. (Vorobyev & Medvedev, 1972)

- High level athletes may tolerate up to three sessions with high load back to back before deloading. (Vorobyev & Medvedev, 1972)

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- When training 6 days a week have two sessions with a large load and other days light or medium. (Vorobyev & Medvedev, 1972)

- More that 2 sessions with low volume back to back are not recommended. (Vorobyev, 1981)

- "Based on conducted [by Ermakov] experiments, one may propose the following schemes of load distribution in weekly cycles: large—small—medium—small—large. That is provided the week-to-week load distribution in a month abides by one of the schemes: large—medium—large—small, or large—medium—above medium—small… Such load distribution is 61.3% more effective compared with the traditional gradual increase and decrease of the load, the exercises and the load being equal in the experimental and control groups." (Vorobyev, 1981)

- Classification of volume by session (Vorobyev, 1981):

    - Light—up to 50% of the given athlete's maximal volume
    - Medium—50-70%
    - Large—70-90%
    - Maximal—above 90%

- The above scheme fits razryadniki. A different distribution is proposed for elite athletes who can tolerate much higher loads: large—medium—large—small—medium—large, etc. Or large—large—small—medium—large—large. The above scheme may be used for one week only by less experienced athletes, but then large and small alternation must be brought back. (Vorobyev, 1981)

- A tactic: increase volume towards the end of the week. (Rodionov, 1967)

- A tactic: 2-3 training days in a row followed by a day of rest. (Sundays are off and not factored in). Day 1 and 3 are heavy or medium if three days in a row. If two days in a row use heavy or medium both days; next two training days are light. (Roman, 1968)

- A simpler tactic is to alternate H/M days with L days. (Roman, 1968)

- The most common weekly schedule among world class weightlifters in the late 1960s—main training sessions on Mondays, Wednesdays, and Fridays; assistance on Saturdays. (Rodionov, 1967) Vorobyev preferred Monday, Tuesday, Thursday, Friday, arguing that the second session done without complete recovery develops strength better and develops will power.

- Level II: 35 sets (NBL 105)—large load; 25 sets (NBL 75)—medium, 15-20 sets (NBL 45-60)—small. No difference which day of week heavy load but no less than 7 days between. Every 14-21 days a deload week of 2-3 medium or small loads. (Vorobyev & Medvedev, 1972)

- A deloading tactic: "After a whole series of training sessions with a high workload fatigue accumulates. To get rid of it is best to have an easy hour and a half training session with a 50% 1RM barbell, performing the most favorite exercises without particular strain, one to two reps per set. It will be a good means of restoring previous work capacity. Then the next training session may be conducted on a usual level." (Vorobyev, 1964)

- The session before maxing or near-maxing reduce the volume and do singles and doubles with 60-70% 1RM. (Vorobyev, 1964)

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Session volume as a percentage of the weekly volume
(Chernyak, 1978)

Number of sessions per week	Session volume, NBL			
	Small	**Medium**	**Large**	**Maximal**
2	≤ 40	41-60	61-80	≥ 81
3	≤ 28	29-38	39-48	≥ 49
4	≤ 20	21-30	31-40	≥ 41
5	≤ 15	16-25	26-35	≥ 36
6	≤ 11	12-21	22-31	≥ 21

Chernyak lists three main types of distribution of volume from day to day within a week: sharp, gradual, and combined. The gradual version demands a slow increase or decrease over three or more sessions. Then there are "sharp-gradual" and "gradual-sharp" combinations. Chernyak points out that when there are four or more weekly training sessions, gradual changes are not practiced; only sharp and combined.

There is a good deal of variability in the number of weekly training sessions. E.g., from week to week: 3, 4, 5, 3. A sample distribution for a month by Chernyak (1978):

	Week 1	**Week 2**	**Week 3**	**Week 4**
Week's NBL	220	280	350	150
Sessions per week	3	4	5	3
Load distribution between sessions, %	25, 45, 30%	17, 30, 15, 38%	13, 27, 15, 30, 15%	25, 45, 30%
Load distribution between sessions, NBL	55:99:66	47:84:43:106	46:94:53:105:52	37:68:45

NBL % of the weekly volume
by powerlift by day for 3, 4, and 5 sessions
(Wolf, 2014)

	Days				
	1	**2**	**3**	**4**	**5**
SQ	35	30	35		
BP	30	40	30		
DL	30	35	30		
SQ	35	15	30	20	
BP	20	35	15	30	
DL	30	20	35	15	
SQ	30	10	20	15	25
BP	25	10	30	15	20
DL	20	15	25	10	30

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Some loading options within a week
(Roman, 1986)

Days of the week						
1	**2**	**3**	**4**	**5**	**6**	**7**
3 sessions a week						
L		H		L		
M		L		M		
M		H		M		
H		L		H		
H		M		H		
4 sessions a week						
L		H		M	L	
H		M		H	M	
H		M		H	L	
H		L			H	L
L		M		M	L	
M		H		M	L	
5 sessions a week						
M	L	H		M	L	
M	L	H		M	M	
L	H	L		H	L	
H	L	M	M	H	L	
M	L	M		M	L	
6 sessions a week						
L	M	L	M	L	M	
M	L	H	L	M	L	
M	H	L	M	L	M	
M	M	L	M	L	M	
M	L	H	L	M	M	
L	H	L	H	L	M	

Other

- "When the load is properly selected, after a training session and a shower the athlete should feel a pleasant fatigue and at the same time strength and freshness." (Roman, 1962)

- Fatigue from a training session should be alleviated after 3-4 hours. If trained in the evening, should wake up fresh and energetic. (Rodionov, 1967)

- Strength is down by 20-30% after sleep and it takes 3-5 hours to reach its peak. It decreases again by 1300. (Vasiliev, 1953)

- A first peak is around 0900 (based on 0600 waking up), the second peak is reached around 1800. (cited by Hettinger, 1966)

- Cold shower (especially for trained people) and application of a cold compress to the stomach between sets improves results. Rubbing face with cold water during competition helps. Cold is a stressor. It activates the cortico-adrenal system. (Vorobyev, 1981)

- The maximal work capacity is in the middle of the week: Wednesday (if training MWF) or Thursday (if training TThSa)—optimal for heavy lifts. Slow starters may need to wait till Friday. (Rodionov, 1967)

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- "…the organism gradually develops a reflex ability to perform a given workload on a given day." (Rodionov, 1967)

- "Maximal work capacity is a dynamic stereotype and dynamic stereotypes can change if you make them. That mean that if the most convenient time for your training falls outside [the optimal times of the day], the organism will gradually, say over a month, will move its work capacity peak to that time. The most important thing is not to change that time too often, otherwise the dynamic stereotype will not be reinforced and you will be constantly feeling discomfort." (Sheyko, 2005) Regular early morning training can change the stereotype. (Smirnov, 1955)

- "The load is adjusted according to the way the athlete feels. If technique worsens or work capacity decreases, the number of repetitions and sets is reduced." (Medvedev, 1980)

- SQ training for weak legs: first series heavy, low rep; second series after some exercise not loading the legs much with light weights and higher reps. (Rodionov, 1967)

- Use the same exercise in beginning and end of session but with different volume, intensity, and focus. (Vorobyev, 1964; Roman, 1968; Sheyko, 2005)

Some competition week tactics

- "It is preferable that in the competition month "shock" workouts with large volume and intensity fall on the same day of the weekly cycle as the competition." (Roman, 1986)

- Do not reduce the volume sharply for more than 2 sessions. (Vorobyev, 1977)

- A stress load that takes 5-7 days to recover from—no later than 12-20 days before competition. (Vorobyev, 1977)

- Monday a small load, Wednesday a warm-up: SQ and BP 75% x 3/2. (Sheyko, 2005)

- Two weeks out lower the volume by 1/2 or even 2/3 but keep up the intensity. The duration of the workout and the number of sets is approximately halved. Two days out from the meet do a light workout, the kind that makes you want to keep training. The day before the meet at the exact time of the competition do a thorough 15-20min warm-up up to 50%. (Vorobyev, 1964)

- 3x3 squats are good for "supporting the optimal tonus of the leg muscles". (Vorobyev & Medvedev, 1972)

- Typically 3-4 sessions in the week before competition, alternating days of work and rest. Typical volume distribution: the first workout (six days before the meet) approximately 55% if the weekly volume, the second—30%, and the third—15%. "If the weekly training cycle does not end with a competition, the absolute load values may be significantly higher (provided the given week is not an unloading one). In an unloading week competition week type load distribution is acceptable." (Vorobyev, 1981)

- In the comp period elite powerlifters switch from twice a day to once a day training, four instead of five days. (Sheyko, 2005)

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INTENSITY VARIABILITY

- "Optimal intensity is not stable." (Vorobyev, 1981)

- "Every cell under an influence of monotonous stimuli goes into the state of inhibition." (Pavlov, 1949)

1. **In a year**
2. **From preparatory period to competition period**
3. **From month to month**
4. **Within a month**
5. **Within a competition month**
6. **Within a week**
7. **Within a competition week**
8. **Within a session**

In a year

- Intensity fluctuates 5-7% relative to the average value in an annual cycle. (Vorobyev, 1970)

From preparatory period to competition period

- "The average barbell weight should gradually increase over months of training." (Roman, 1968)

- Analysis of successful lifters' logs by Chernyak (1978) did not confirm the common notion that the average weight goes up in the competition month; even the opposite sometimes happens. Comparative analysis shows that overall the average intensity does not vary much between the preparatory and the competition periods. From month to month here is the fluctuation pattern: if the prep months the average weight was high, then it will go down in the competition months. And if the intensity in the prep period was medium, it will remain so in the comp period. "The average barbell weight reflects the content of training only indirectly. It appears that it is not such a defining characteristic of success in competition."

  Yet shortly after the scientist states that average relative intensity does go up slightly, having analyzed three groups of high level, minimum MS, lifters. The trend: a 0.6-2.6% 1RM ARI increase from preparatory cycle to competition cycle. (Incidentally, the group with the highest increase, from 70.7% to 73.3%, had the best results. This group also had the highest competition period intensity.) Points out that this applies only to athletes with particular circumstances "Practice is multifaceted."

- The ARI in the comp period may be the same as in the prep period or 1% up. (Medvedev, Verkhoshansky et al., 1992)

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- In the preparatory period max and nearmax lifts are spread out across 4 weeks (or 3 if the 4th is deloading). In the competition period they are mostly done in the first three weeks. (Roman, 1986)

- "Training with 80-85% weights leads to an increase of strength that gets ahead of the nervous system readiness. The athlete feels that after two to three months of the preparatory period he is stronger and ready to set personal records. But as soon as comes the time to lift 95-100% weights of the competition period, he starts feeling uncertainty, fear the weights, and fail as a result." To battle that plan two to three weeks on 90-95% singles. If the athlete confidently puts up two or three, he is instructed to do a double with the same weight (rather than go for 100%). (Sheyko, 2005)

From month to month

- High level athlete's monthly ARI fluctuates between 64 and 77.5% 1RM. (Chernyak, 1978)

- "A relatively stable ARI does not imply the same distribution across the zones…. Rational variation of distribution of barbell lifts across intensity zones in monthly cycles… facilitates growth of results." (Chernyak, 1978) States that the following sample training year is representative for the majority of Soviet weightlifters:

- Reducing the intensity for an extended period leads to a loss of strength. On the other hand, one cannot maintain high intensity, especially in competition exercises, for a long time as this could lead to nervous system fatigue and a drop in results. (Roman, 1986)

- If a lifter has to compete every month, he should keep his volume and intensity at average values at all times, with small fluctuations. (Chernyak, 1978)

Annual load dynamics (snatch, C&J, and SQ) for a Master of Sport
(Chernyak, 1978)

Months	**50-60%**	**61-70%**	**71-80%**	**81-90%**	**91-100%**	**ARI**	**NBL**	**Competitions**
	1	**2**	**3**	**4**	**5**			
1.	15[14]	**26**	18	**27**	13	**74.6**	490	+
2.	13	22	**34**	22	10	**76.9**	775	
3.	7	**28**	**29**	24	12	**75.7**	464	
4.	17	**32**	26	18	6	**71.4**	376	+ Successful
5.	**25**	14	20	23	9	**71.6**	510	+ Very successful
6.	25	**30**	20	19	6	**70.2**	335	
7.	18	22	30	20	11	**73.5**	547	+ Added 15kg to total
8.	10	**37**	30	15	5	**72.3**	518	
9.	15	**29**	**29**	25	2	**72.0**	351	
10.	7	**31**	**31**	29	2	**73.8**	370	+
11.	8	**32**	**30**	27	3	**73.5**	295	
12.	20	**35**	26	18	1	**69.5**	420	+ Added 20kg to total

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Sample intensity dynamics in powerlifting
(Wolf, 2014)

Prep				
Weeks	**1**	**2**	**3**	**4**
Variant	**Predominant zone**			
1	91-100	81-90	71-80	61-70
2a	61-70	91-100	81-90	71-80
3-1	81-90	61-70	91-100	71-80
1-3a	91-100	61-70	81-90	71-80
2-4	61-70	91-100	71-80	81-90
4-2	71-80	81-90	61-70	91-100
Comp				
1	91-100	81-90	71-80	61-70
2b	81-90	91-100	71-80	61-70
1-3b	91-100	71-80	81-90	61-70

Within a month

- Intensity fluctuates 11% in a monthly cycle. (Vorobyev, 1970)
- For elite lifters weekly intensity may fluctuate more than ±10% 1RM from the monthly average. (Based on Chernyak, Atanasov & Ermakov, 1972)
- The same patterns as with volume are used—sharp, gradual, and combined—although the variability is a lot lower then in volume. Average intensity stays within the limits of certain medium values. (Chernyak, 1978)
- Aim to vary the training weight from week to week. (Chernyak, 1978)
- There is no obvious trend in intensity variability from week to week. (Vorobyev, 1981)
- "The training load for a session is rather monotonous (most of the lifts are made with the same barbell weight). But the monthly load is varied: the main training weight change and so does the maximal weight for a given lesson. If in a monthly cycle one exercise is repeated in several lessons, each time it is performed with a different weight. The main training weight will not be constant for the duration of a thirty day cycle; it will go down and up." (Falameev, 1986)
- Place the 91-100% weights into the week where you are planning to max. That week is determined by, in the competition month, the number of weeks remaining before the competition. In the prep period it is determined by the athlete's readiness to lift max weights. (Chernyak, 1978)
- "Due to a reduction in strength exercises the training becomes primarily speed-strength. This creates favorable conditions for recovery processes in the body."[15] (Chernyak, 1978)

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Within a competition month

- In the comp month there are four variants: stable in three weeks followed by a sudden drop, 2, 3, 1-3. (Medvedev, 1968)

- Out of 15 top lifters 6 athletes stuck to one variant, 8 used two, three used three, and one used all 4 variants (stable for three weeks, then sudden drop; 1-3, 3, 2). (Medvedev & Chernyak, 1969)

- In 73% of cases top lifters used intensity variants 1, 2, 1-3, 3-1. (Chernyak, Atanasov & Ermakov, 1972)

- A Russian powerlifting team tactic: 90% or opener 2.5-3 weeks out. Max on the weak lift(s) 3-4 weeks out. (Sheyko, 2005)

- Lighter lifters can lift 90%+ weights closer to competition than heavier lifters. (Roman, 1986)

90-100% lifts in three powerlifts in a six-week competition cycle
(Sheyko, 2005)

	Weeks' countdown to competition						**Total**
	6	**5**	**4**	**3**	**2**	**1**	
90%	6	6	6	6	---	Only at the meet	24
95%	---	6	4	---	---		10
100%	---		6	---	---		6
Total	6	12	16	6	---		40

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Within a week

- "The intensity optimum is far from stable. Based on studying this factor on members of the USSR national team, we can presume that intensity fluctuations in training sessions within 20-25% limits are warranted." (Vorobyev, 1970)

- Do not keep the same average intensity in particular exercises from session to session; vary it. (Chernyak, 1978)

Sample intensity dynamics within weeks for a qualified athlete
(Chernyak, 1978)

Week I	Week II	Week III
3 sessions	4 sessions	5 sessions
70%, 72%, 69%	60%, 60%, 66%, 65%	68%, 66%, 65%, 72%, 60%
Sharp variability		Gradual-sharp variability

Sample four snatch sessions of a Level I athlete
(Chernyak, 1978)

Session #	50-60%	61-70%	71-80%	81-90%	91-100%	NBL	ARI
1	2	2	12	---	---	16	71.3%
2	---	3	12	---	---	15	73%
3	3	3	6	6	---	18	73.3%
4	---	3	5	4	---	12	75.8%

- Best effect from two sessions with a medium weight followed by a session with heavy weights. (Roman, 1968)

- Selecting weights only on the top, only on the bottom, both, etc. of a wide 10% intensity zone is one way to individualize. (Chernyak, 1978)

- If lifted heavy on Wednesday, lift medium weights on Monday and light on Friday. (Rodionov, 1967)

- A tactic: squat twice a week, in the beginning of the week heavy in sets of 2-3 reps, in the end of the week lighter in sets of 4-6. (Sheyko, 2005)

Within a competition week

- In the last week before competition the intensity usually decreases by 10-19%—but some athletes keep it at the average monthly level or even increase it by 2-3%. (Vorobyev, 1977)

- Vorobyev (1981) does not prefer a particular type of intensity dynamics in the last 3-4 sessions.

- It appears that strength can be maintained in a short term with a significant, 15%, intensity reduction. (Vorobyev, 1981)

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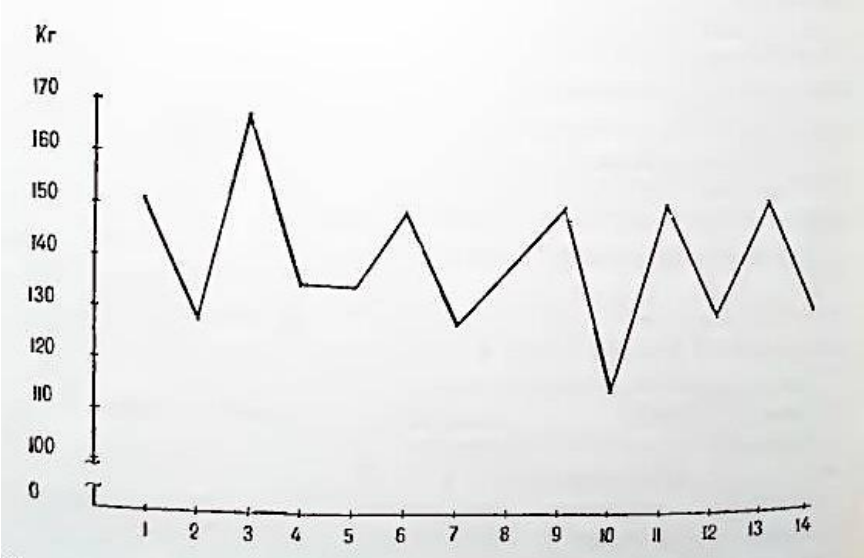

Intensity (in kg) from session to session for Zhabotinsky.

Within a session

- "Experience shows that sport results grow faster if one is training with a particular weight, the so-called main training weight. It is especially pronounced in training the press." This main training weight is a weight you can lift 2-3 times in several sets, 75-85% 1RM. "It is a constant stressor… It is known that the whole situation must repeat itself multiple times for the conditioned reflex to be strengthened. The main training weight will be this definite stressor. It not only promotes precision and coordination of the barbell lift but also the most rapid development of strength of the involved muscles." (Vorobyev, 1964)

- "Training with variable weights is less tiring and one should use it no less than once every two weeks. Training with a given weight (a static main training weight) is more effective for strength development and perfecting the technique but it demands high nervous and will power effort." (Vorobyev, 1964)

- 85-95% weights are best for technique practice. Working in this wide range "excludes the possibility of developing a stable stereotype with any given lifted weight." (Roman, 1986)

- "It has been established that the process of learning and mastering the technique is more effective if a constant barbell weight is used for a period of time and then a variable weight. Strength grows faster as well. For instance, an optimal weight is used and not varied for several sessions when learning technique of an exercise. Then the load in this exercise is slightly varied: one addition is made to the initial weight, then two, then three. The initial warm-up weight is increased as well… As a result, the athlete learns to adjust his efforts to the weight." (Falameev, 1986)

- An assisted single with a supramax weight to build courage. Usually done once. Then help is reduced or faked. (Bozhko, cited by Rodionov, 1967)

- Squat training for weak legs: first series heavy, low rep; a second series with light weights and higher reps after some exercise that does not load the legs much. (Rodionov, 1967)

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VOLUME/INTENSITY

- "Instead of bringing his athlete to a "peak of form", a coach should correctly distribute the load in training sessions. To summarize, the training process is not built on a concept or a "peak of form". [16] (Vorobyev, 2001)

- "The time has come to reconsider aiming the waves of volume and intensity in opposite directions…" (Medvedev, 1968)

1. **Dynamics over the years**
2. **In a year**
3. **In a big cycle**
4. **From preparatory period to competition period**
5. **From month to month**
6. **Within a month**
7. **Special considerations for ladies**
8. **Within a competition month**
9. **Within a week**
10. **Within a session**

Dynamics over the years

- The first year volume growth is the key; it grows faster than intensity. (Vorobyev & Medvedev, 1972)

In a year

- In a year cycle loads follow a normal distribution. (Chernyak, 1978)

- The predominant intensity-volume combination in lifting should be medium-medium (Chernyak, 1979).

Different volume/intensity combinations, use per year
(Chernyak, 1979)

Volume/intensity combination	Used weeks per year
Medium/Medium	12-15
Medium to High/Low	9-10
Low/Medium	6-7
High/Medium	3-4
High/High	2-3
Low/High	3-4
Low/Low	1-2

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In a big cycle

- If you are in the middle of a cycle and feel that you have gotten a lot stronger, reduce the intensity and NBL in two sessions, then max. Next week use the new % 1RM. (Sheyko, 2005)

From preparatory period to competition period

- "According to the established traditions (Matveev, 1969; Ozolin, 1970) during the preparatory period the volume grows quicker than intensity. Then the volume stabilizes and the intensity keeps growing. Later the volume decreases while the intensity keeps growing. Unfortunately, the authors promoting this loading variant do not take into account that the organism adapts only to the load it is performing, only to the conditions it is exposed to. Lifting only a light barbell the athlete will not be able to lift a heavy weight barbell… A transfer of movement quality is limited." (Vorobyev, 1977)

- "Weightlifters do not have a competition period." They compete every month. (Medvedev, 1986)

- "Contemporary preparation practice" shows that athletes who progress the most are those who use optimal heavy loads throughout the cycle and regularly compete all out. Using similar to competition period methods from the very beginning of the prep period. This enables athletes to show by the end of the prep period results exceeding those of the last cycle. (Medvedev, 1986)

- "An athlete starts competing the in the [prep] cycle without decreasing the load and going all out. Competitions serve as a test of whether how correctly the training process has been built… This creates perfect preconditions for higher results towards the end of the [big] cycle." (Medvedev, 1986)

- Medvedev (1980, 1986): in a "tense" prep period highly qualified lifters who did a high volume with optimal intensity approached and in many times exceeded their last comp cycle's results (93% of the lifters got stronger, especially in the back). Those who did the NBL but lowered intensity "were far from their records".

From month to month

Options for changing he load from month to month (Roman, 1986):

- Both volume and intensity increase
- Both volume and intensity decrease
- Volume increases and intensity is maintained
- Volume decreases and intensity is maintained
- Volume increases and intensity decreases
- Volume decreases and and intensity decreases
- Volume is maintained and intensity decreases
- Volume is maintained and intensity increases

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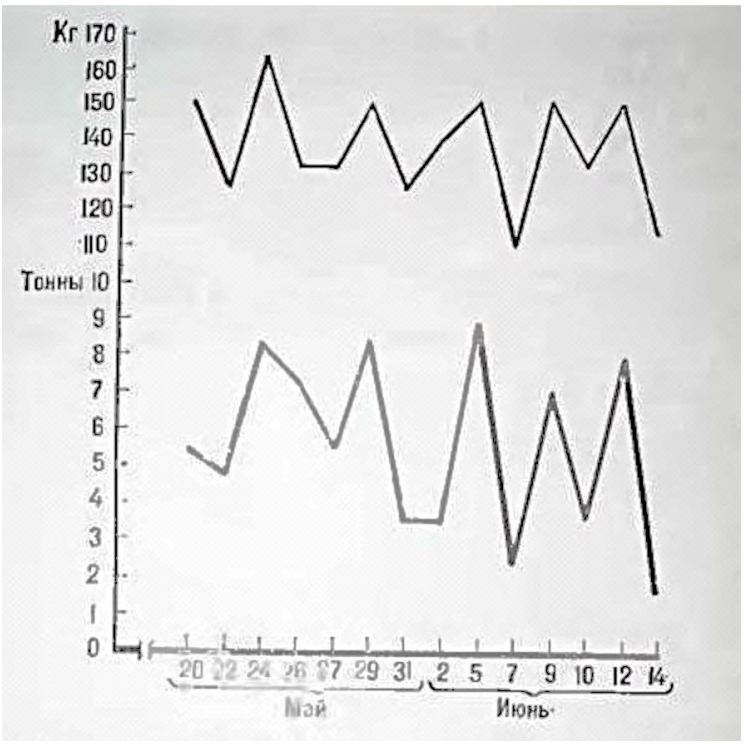
Within a month

- Two weeks of loading and two weeks of unloading to restore the endocrine and immune systems. (Selouyanov)

- Volume and intensity, as a rule, should go up and down together, although in short (e.g. weekly) cycles they can go in opposite directions. (Vorobyev & Medvedev, 1972)

- To appreciate the effectiveness of sharp variability, consider the pedagogical experiment Ermakov (1972) conducted with two groups of beginner and low ranked weightlifters. The eight-week volume and intensity were the same. The competition four-week blocks were the same as well (variant 1 for both NBL and ARI.) The only difference was in the four-week prep period. The control group used a monotonously increasing variant 4 for both volume and intensity—typical for Matveev's traditional model. The experimental group used sharp variability for both—3-1 for volume and 2-4 for intensity. In the end of two months the control group gained 5.9kg on its total of the press, snatch, and C&J. The experimental group gained more than twice as much—12.9kg.



Intensity (top) and volume dynamics in Zhabotinsky's 1967 monthly cycle before breaking four world records.

- "Intensity variation in weeks and within a week may correspond to the volume variation or not; however, as a rule, in the preparatory period it does." (Roman, 1986)

- "In our opinion, different NBL and ARI monthly combinations may be repeated multiple times throughout a year." The length of particular combinations depends on the athlete. E.g., lifters with good strength but lacking in technique are advised to spend three months with high NBL and relatively low ARI. If athletes are lacking in the strength department, raise the ARI, but not for longer than one to two months. (Chernyak, 1978)

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- One way to plan razryadniks' prep period (Sheyko, 2005):
    - Week I—max volume.
    - Week II, III—80-85% 1RM leading up to 90% next week.
    - Week IV—90% 1RM. Max intensity, min volume.

Special considerations for ladies

- "…no significant gender peculiarities [in hormonal responses to exercise] can be revealed by cross-sectional comparison of data obtained in male and female persons. Testosterone level is lower in women but relative changes are rather similar. (Viru, 1995)

- Guidelines for women weightlifters (Rumyantseva & Gorulev, 2005):
    - In the, ovulating, premenstrual, and menstrual phases reduce the volume in exercises loading the pelvic floor, especially squats.
    - Highest strength and speed-strength performance, maximal efficiency and speed of recovery is in the post-menstrual and post-ovulating phases. These are the phases to plan the highest volume and intensity loads.
    - Performance is not at its highest in the ovulatory phase.

Within a competition month

- For elite athletes successful in competition in 76% cases an increase or decrease in volume was matched by same in intensity. In 13% cases the intensity went up when the volume went down. In 11% intensity went up when the volume stabilized. (Medvedev, 1968)

- In most cases intensity increases are accompanied by volume increases. (Chernyak, Atanasov & Ermakov, 1972)

Load dynamics for two high class athletes before a very successful competition
(Chernyak, Atanasov & Ermakov, 1972)

Lifter A	**Preparatory period**				
	Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**
Volume	**3-1a**	24%	19%	36%	21%
Intensity	**4-1a**				
	Competition period				
	Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**
Volume	**1-3b**	35%	24%	30%	11%
Intensity	**1-3b**				

Lifter B	**Preparatory period**				
	Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**
Volume	**3**	15%	18%	39%	28%
Intensity	**4**				
	Competition period				
	Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**
Volume	**1**	46%	27%	14%	13%
Intensity	**1**				

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Sample powerlifting competition cycle
(Sheyko, 2005)

Weeks	Mon	Tue	Wed	Thur	Fri	Sat
6	SQ 80% x 3/2 85% x 2/2 90% x 1/2 BP aux 60% x 5 70% x 5/4	DL aux 60% x 3/2, 70% x 2/3, 75% x 1/3	BP, DL 70% x 3/2 80% x 3/3 85% x 2/2 90% x 1/2	Rest, sauna	SQ, BP 70% x 3/2 80% x 3/5	DL aux, BP aux 60% x 4/2, 70% x 3/2, 75% x 2/4
5	SQ, BP 80% x 3/2 90% x 1/2 95% x 1/2	BP aux, DL aux 75% x 2/5	BP 75% x 3/3 85% x 2/4 DL 80% x 2/3 90% x 1/2 95% x 1/2	Rest, sauna	SQ, BP aux 80% x 2/5	BP aux 75% x 2/4 DL 80% x 2/4
4	SQ, BP 80% x 2/2 90% x 1/2 95% x 1 100% x 1/2	Rest	DL 80% x 2/2 90% x 1/2 95% x 1 100% x 1/2	Rest, sauna	SQ, BP 80% x 3/3 85% x 2/3	BP aux, DL aux 80% x 3/5
3	SQ, BP 80% x 2/3 90% x 1/2	Rest	BP aux 80% x 2/5 85% x 2/3 90% x 1/2	Rest, sauna	SQ, BP 70% x 3/2 80% x 2/3 85% x 2/3	BP aux, DL aux 70% x 3/2 80% x 4/3
2	SQ, BP 70% x 3/2 80% x 2/3 85% x 1/3	Rest	BP, DL 70% x 3/2 82% x 2/4	Rest, sauna	SQ, BP 70% x 3/2 80% x 2/4	Rest
1	BP 70% x 2/3 80% x 1/3 DL 70% x 2/3 75% x 1/3	Rest	SQ, BP 75% x 2/3	Rest	Rest	Meet: 92% 97% 100-102%

- The weeks are listed in a countdown to competition order.
- The reps are listed before the sets. E.g., "80% x 2/3" means three sets of two reps.
- Peak on fourth and third week, then supercompensate through a reduction of both volume and intensity.

Within a week

- The endocrine system can be overtrained, therefore no more than 1-2 hard workouts a week. (Selouyanov)

- Intensity is cycled autonomously from volume (Chernyak, 1978), as can be seen from the examples of two qualified athletes' training on the graph below.

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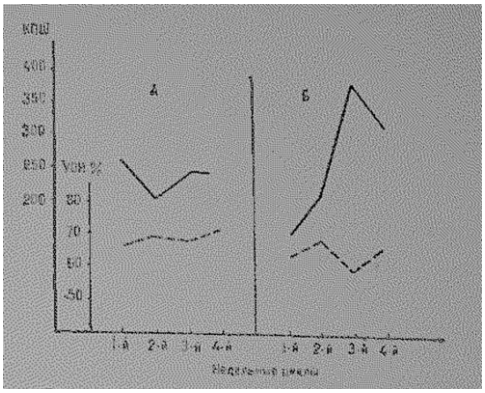
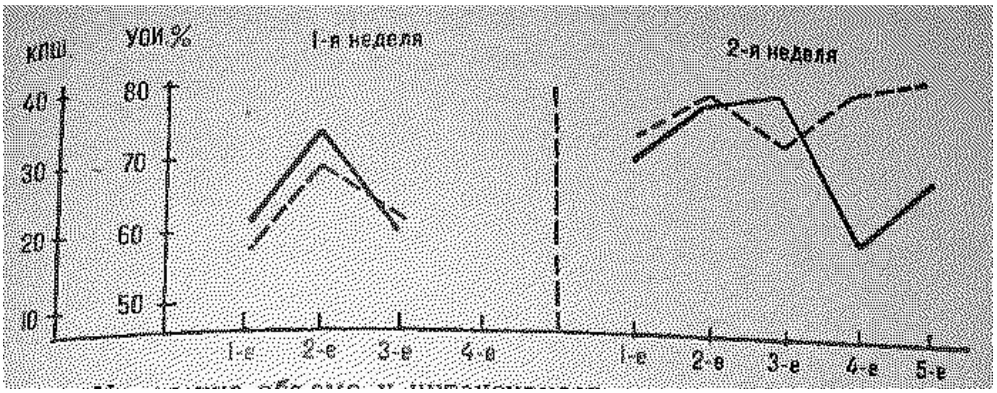


The solid line represents the NBL; the dotted line the ARI.

- High volume and high intensity rarely occur in the same exercise in the same session. Medium-mediums are very common though. (Chernyak 1978)

- A rule of thumb: for a given exercise one may (but does not have to) wave the volume and the intensity together if both of their values are medium. If one of the variables has a high value, the other must be decreased to low or medium. (Chernyak 1978)

Sample snatch loading over two weeks
(Chernyak, 1978)



The solid line represents the NBL; the dotted line the ARI. In week one the volume and intensity closely follow each other and stay around medium values. In week two both the NBL and the ARI reach peak values—but not at the same time.

Within a session

- "At times, when things are not going well, one must overcome himself no matter what. But this should not be made into a system." (Rodionov, 1967)

- If you cannot handle the planned load—decrease the NBL but not the ARI. Applies to assistance too. (Medvedev, 1986)

- If after your top warm-up set you realize that you are not going to make the day's plan, adjust the load according to these guidelines (Sheyko, 2005):

    - In the preparatory period lower the intensity and maintain the volume;
    - In the competition period lower the volume and maintain the intensity.

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EXERCISES

1. **Number**
2. **Partial loads**
3. **Order**
4. **Variety**
5. **Measuring assistance exercises' intensity**

Number

- World record holders have used 2-7 exercises per session. There is no correlation between the number of exercises and the results. (Vorobyev, 1977)

- 5-6 exercises in a session, 10-15 in a month. (Vorobyev & Medvedev, 1972)

- Empirically 4-6 exercises per session are optimal. "Varying the number of exercises in the four to six range leads to a subjectively better tolerance of training loads. A low number of exercises [in a session] creates monotony which in turns and leads to a greater fatigability." (Vorobyev, 1981)

- An occasional unloading session may include only 2-3 exercises. When the athlete is fatigued from previous training, the training is made "emotional" by increasing the number of exercises to 8 while reducing the number of sets and reps. (Vorobyev, 1981)

- 2-3 exercises in low volume sessions; 5-8 in high volume sessions. (Roman, 1986)

- On average lifters use 18±2 barbell exercises. (Vorobyev, 1981)

- 3-8 exercises per session, up to 20 per week, up to 30 and more per month. A higher number is planned in the preparatory period; a lower in the competition period. (Medvedev, 1986)

- Medvedev (1980) successfully used 48 "exercises perfecting physical qualities and technical mastery" (specialized variety) in a 3-week period with elite lifters. 93% of the subjects got stronger; 85% asked for even more exercises. A high number of exercises (and particular load planning) enabled a very high volume—more than double the average for 2 weeks.

- The number of exercises was increased in the preparatory period and reduced in the competition period, especially in the last two weeks. (Chernyak, 1978)

- In the last 5-6 sessions before the competition the number of exercises is reduced to 3-4. (Vorobyev, 1977)

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Partial loads

"Partial loads" refer to the percentage the volume of each exercise group, e.g. press, makes up of the total load. Partial loads were adjusted from the average figures to emphasize the individual lifter's weakness, e.g. a higher relative share of squats to bring up the leg strength.

- Lighter weightlifters use classic lifts more than heavier lifters. (Vorobyev & Medvedev, 1972)

- As a rule of thumb, as the lifter's qualification grew, the share of the competition lifts in the total workload went down as the share of special exercises went up. (Chernyak, 1978)

- As a percentage of the total load, Level II and I lifters do 2 times as many classic lifts as the CMS and 3 times as many as the MSIC. (Medvedev, 1986)

Share of SQ, GM, Pulls, Presses as a Part of the Total Load
(Falameev, 1986)

Athlete's Rank	%
Level III, II	42-50%
MS, MSIC	58-65%

- Falameev (1974) states that the skills are not solidified until one has reached the Master of Sports level. Only then he recommends pulls from a deficit or from blocks.

- In powerlifting Sheyko (2005) does not recommend using the deadlift style opposite to one's competition style (sumo vs. conventional), arguing that it takes years to hone one style.

- In the beginning most time is spent on the classic lifts; only 20-30% on assistance. Later, when the skills are decent, the share of the classic lifts is reduced to 50% and even less. In high level athletes with great technique it can be 25%—and even less than 20% for some outliers. (Vorobyev, 1981)

- 50% of assistance exercises are squats and pulls. (Vorobyev, 1981)

- For high level lifters squats and pulls are 30-40% of <u>all</u> load and up to 50-60% at times 3-4 months before competition (no longer than 1 month). (Vorobyev, 1981)

- Average partial loads from MS and up: snatch exercises—22%, C&J exercises—25%, snatch pulls—10%, clean pulls—10%, squats—22%, presses and other exercises—11%. (Medvedev, 1986)

- When press was contested, press, snatch, and jerk exercises made up 70% of the total volume, with the following breakdown: press 30%, snatch 22%, C&J 18%. (Roman, 1971)

- Partial loads were varied. E.g., month 1—high relative volume of C&J exercises; months 2 and 3—squats; month 4—C&J again; month 5—snatch and C&J, etc. "Apparently, such load organization excludes accommodation to the same partial load volumes and facilitates priority development of the lagging physical qualities and skills." (Chernyak, 1978)

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- "The dynamics of the volume and intensity of partial loads from session to session are independent." (Chernyak, 1978)

Partial loads in five sessions of a Level I weightlifter
(Chernyak, 1978)

Session #	NBL per exercise group							Session NBL
	Sn	C&J	SQ	Sn Pull	Cl Pull	GM	P	
1	31	21	---	21	---	---	30	103
2	23	---	18	---	18	---	---	59
3	---	18	15	18	---	---	---	51
4	13	15	24	---	---	25	---	77
5	15	15	---	---	15	---	---	45
"None of the listed exercises was performed at every session, therefore certain muscle groups got a rest at least during one session."								

- After a meet do only 20% of classic lifts; before a meet the most, up to 50%. (Vorobyev & Medvedev, 1972)

- When it comes to volume of particular exercise groups, there are two types of variability. The first is a sharp change of volume from small to large. The second is exclusion of an exercise from some training sessions altogether. Both are used by most athletes by one may be dominant, given one's physiological or psychological individuality. (Chernyak, 1978)

- Arrange a training session to alternate exercises with heavy and light weights. (Rodionov, 1967)

- "To sustain high work capacity during a training session the weights should not be kept constant in different exercises. If in some exercises the emphasis is made on lifting heavy, over 80% 1RM, weights, then in others it should be 60-70% and 70-80% weights." (Chernyak, 1978)

Sample partial weekly loads in powerlifting
(Wolf, 2014)

SQ	BP	DL
32	38	30
34	34	32
30	36	34

Order

- The optimal exercise order: speed, strength, endurance. (Vorobyev, 1981)

- Although the order from speed to strength to endurance is optimal (research shows it has the greatest effect on the functions of the vegetative organs and systems), some time should start with powerlifts, isometrics, etc. to avoid habituation. (Vorobyev, 1981)

- Alternating between harder and easier exercises is advisable. (Roman, 1962)

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- Separate lifts that can have negative skill carryover to each other such as presses and jerks with a third lift. (Roman, 1968)

- Sequence exercises in a manner that emphasizes different muscle groups. (Roman, 1968)

- Until 1966 the number of exercises in a session remained static for 4-6 sessions in a row. In 1977 no more than 2 sessions back to back, max 3. (Vorobyev, 1977)

- Some lifters do snatch assistance after C&J and vice versa; others do assistance for the same lift they just did. Some do assistance for both in all sessions. (Rodionov, 1967)

- Doing assistance for another lift is effective and less tiring. Most USSR champs did it this way. (Rodionov, 1967)

Variety

- Press every other day; on alternate days use light weights with local exercises for the pressing muscles. (Roman, 1962)

- A tactic: a series of a competition lift working up to a heavy weight; later in the session a series with lighter weights and higher reps using an assistance exercise for that lift. (Roman, 1962)

- Bench 3 times a week changing the grip width every session. (Roman, 1962)

- Chernyak (1978) acknowledges the effectiveness of "combined exercises"—what Americans know as "complexes"—but warns about possible overtraining if one performs such exercises frequently. He also points out that such exercises demand lighter weights that usual.

- Use combined exercises more. (Medvedev, 1996)

Measuring assistance exercises' intensity

There are two methods of measuring the relative intensity in assistance exercises. (Chernyak, 1978) The first method used the competition lift's max as the reference point. E.g., snatch pulls are prescribed as % of the snatch 1RM: 110%, etc. (Hence supramaximal intensity zones.) This method is used when the athlete does not know his max in the given exercise(s), for reasons of safety or some other reasons. E.g., good mornings were typically prescribed as a percentage of the max snatch.

The second method measured the intensity as a percentage of the max in the given exercise. At times both methods were used for one exercise, e.g. the squat. The practice of measuring and prescribing the squat as a percentage of the max C&J by Roman & Falameev (1969-1970) has been criticized by Chernyak (1978) as a practice that held back development of better methodologies.

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SET AND REPS

Roman (1962)

Roman advocated benching three times a week working up to max and changing the grip width every workout. Start with a light weight for 4-5 reps and keep adding 5-10kg per set, lowering the reps as the weight goes up, eventually working up to 1RM. Typically 5-6 sets were done with the average NBL of 20. Roman also liked more formalized 654321 countdowns: 75% x 6, 80% x 5, 85% x 4, 90% x 3, 95% x 2, 100% x 1. The percentages are based on the max in the particular style of the BP.

Vorobyev (1964)

A qualified athlete's press session:

1st series:

- (1RM-40-45kg) x 3-4
- Add 10kg x 3
- Main training weight (1RM-10-15kg) x 2-3/5-6 (reps/sets)

2nd series:

- BP 3-4 x 4-5

Vorobyev & Medvedev (1972)

Sample razryadnik training, 60-90% 1RM:

- SQ: X x 5, (X + 20) x 5, (X + 30) x 4/3, (X + 20) x 4/2 (NBL 30)
- BP: X x 4, (X + 10) x 4, (X + 20) x 4/4, (X + 25) x 2/2 (28)

Sample MS training, 70-95% 1RM:

- SQ: X x 6, (X + 40) x 6, (X + 60) x 4/3 (24)
- Incline BP: X x 3, (X + 20) x 3/2, (X + 30) x 2/3 (15)
- GM: X x 3, (X + 10) x 4, (X + 20) x 6/2, (X + 30) x 6/2, (X + 40) x 4/2 (29)

Light and medium sessions are similar for razryadniki and MS.

Medium session, 65-90%:

- P: X x 3, (X + 10) x 3, (X + 20) x 3, (X + 35) x 2/3 (15)
- Lunge: X x 3/2, (X + 40) x 2/2, (X + 50) x 2/2 (14)

Light session, 60-80% :

- SQ: X x 3, (X + 20) x 3/3 (12)
- Incline BP: X x 3, (X + 15) x 2, (X + 30) x 2/5 (15)

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Rodionov (1966, 1967)

Note that although some of the following schemes feature up to 3 reps per set with 90%, which is higher than recommended by most Soviet authorities, in his later research Rodionov (1976) concluded that 2 was the optimal rep number for 90%. Make changes accordingly.

In all of the following reduce the number of sets with most weights if the exercise is done for more than one series in a session.

A classic scheme for optimal weights:

- 60x2-3, 70x2-3, 80-85x2-3/4-6
-

A typical heavy squat scheme:

- 70x3, 80x3, 90x2/3-4

Following are schemes using maximal and circamaximal weights.

Three classic heavy pyramids. The first employs easy back-off sets to provide active rest before the next exercise. The second and the third feature multiple back-off sets with heavier weights.

- (60x2-3), 70x2-3, 80x2-3, 87.5-90x1-2/5-6, 80x2-3, (70x2-3)
- 60x2-3, 70x2-3, 80x2-3, 87.5-92.5x1-2/3-4, 85x2-3/3-4
- 60x2-3, 70x2-3, 80x1-2, 90x1, 95x1, 85x2-3/3-5

Following are two options for working up to PR attempts. The choice of the 90 or 95% weight preceding the max set is up to the athlete. The way this weight feels will help in selecting the PR attempt. The back-off sets in brackets are optional. They may be replaced with 70x2-3/2-3.

- 70x2-3, 80x2-3, 90x1, 100-102.5x1, (85-90x2-3/3-4), 70x2-3
- 75x2-3, 85x2-3, 95x1, 100-105x1, (85-90x2-3/3-4), 70x2-3

Next is an effective scheme for improving your skill. You get to critically compare your technique in both sets with each weight except 90%.

- 65x2-3/2, 75x2-3/2, 85x2-3/2, 90x1, 75x2-3/2, 65x2-3/2

Multiple lifts with the competition opener in the next scheme help an athlete whose technique tends to deteriorate when the poundage approaches maximal.

- 60x2-3, 70x2-3, 80x1-3, 90x1-2, 95x1/3-4, 85x2-3, (75x2-3)

In the following two schemes a 95% single is done on the background of fatigue from multiple sets with 85 or 90%.

- 65x2-3, 75x2-3, 85x2-3/4-6, 95x1
- 60x2-3, 70x2-3, 80x2-3, 90x1-3/4-6, 95x1

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Vorobyev (1981)

3 Level I sessions:

- BP 60x4/2, 80x3/4
- SQ 60x6, 70x6/6
- GM 50x6, 70x6, 80x4/4

- SQ 70x3, 90x2/3

- Incline BP 60x3/3
- FSQ 70x3/2, 80x3/5
- Clean pull with a shrug 100x2/5

Roman (1986)

Snatch sessions (snatch first). In brackets are the main training weights.

Beginners:

- 55x4, 65x4, (75x3/5)

- 50x3, 60x3, (70x 4/5)

Level III, II:

- 55x3/2, 65x4, (70x4/5)

- 50x4, 60x5, (70x4, 75x3/2, 80x2/2, 85x1/2)

Level I, CMS, MS (and occasionally lower ranked athletes):

- 60x2/2, 65x2, (70x2, 80x5/2, 85x4/2)

- 55x3/2, (70x2/2, 80x2/2, 85x1/5)

- 50x3, 60x3 (70x 4, 5, 2, 4, 1, 5)

Going for a max:

- 55x4, 65x2, (75x2, 80x1, 90x1, 95x1, 100x1, 102.5-105x1)

- 50x 3/2, 60x2/2, (70x1/2, 80x1/2, 85x1, 92.5x1, 97.5x1, 100x1/1-2)

Snatch late in the session after other pulls:

- (72.5x4/4)

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Sheyko (2005)

- 50x5, 60x4, 70x3/2, 80x3/2, 90x2/3
  NBL= 27, ARI=71.5%

- 55x5, 65x4, 75x3/2, 85x2/4
  NBL=23, ARI=72.4%

- 50x5, 60x4, 70x3/2, 80x3/6
  NBL=33, ARI=71.2%

- 50x5, 60x5, 70x5/2, 75x4/4
  NBL=36, ARI=68.1%

- 50x6, 60x6, 70x6/5
  NBL=42, ARI=65.7%

Sheyko (2013)

Squat:

- 50x5, 60x5, 70x3+5+7+9+8+6+4
  NBL=42, ARI=67,2%

- 50x5, 60x5, 70x3+7+4+8+2+6
  NBL=40, ARI=66,3%

Bench press:

- 50x5, 60x4, 70x3, 80x3, 85x2/2, 90x1/2, 85x2/2
  NBL=25, ARI=72%

- 50x5, 60x4, 70x3, 80x3/2, 85x2/3, 80x3/2
  NBL=32, ARI=73,4%

- 50x6, 60x5, 70x4, 80x3/2, 85x2/2, 75x3, 65x5, 55x7
  NBL=40, ARI=65,9%

  50x6, 60x5, 70x4, 80x3/2, 85x2/2, 80x3/2, 70x5, 60x7, 50x9 NBL=52, ARI=65.4%

- 50x7, 60x6, 70x5, 75x4, 80x3, 85x2/2, 80x3, 75x4, 70x5, 65x6, 60x7, 55x8, 50x9
  NBL=71, ARI=63,8%

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Deadlift:

- Series #1: Box DL—50x3, 60x3, 70x2/4
  Series #2…
  Series #3: DL lockout, bar below the knee—65x3, 75x3, 85x3/2, 90x2/3 (measure the intensity as a percentage % of the competition style DL from the platform)
  NL=32, ARI=73.8%

- Series #1: 50x3, 60x3, 70x3, 80x2/4
  Series #2…
  Series #3: DL with chains—50x3, 60x3, 70x3, 75x2/4
  NL=34, ARI=68.2%

- Series #1…
  Series #2: DL with pause(s) 5cm/2" above and/or below knee—
  50x3, 60x3, 70x3, 80x2/5
  NL=19, ARI=70.5%

Wolf (2014)

Sets and reps at 80% 1RM according to Prilepin's table.

After a warm-up of 60x6, 72.5x4:

- 80x2/7
- 80x3/5
- 80x4/4
- 80x5/3 (prep period only)
- 75x3, 77.5x3, 80x3, 82.5x3, 85x3 (average 80)
- 75x3, 80x3, 85x3, 82.5x3, 77.5x3 (average 80)

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TRAINING BEGINNERS

- "In unqualified athletes strength (and weightlifting results) quickly increase from the simplest training methods." (Falameev, 1974)

- "It is known that the tendon-ligamentous apparatus is perfected much slower than strength. This is one of the important reasons not to force strength development." (Medvedev, 1986)

- "The process of forming a movement skill is always tightly fused with development of physical qualities." (Falameev, 1986)

Volume

- Train 3 times a week on the same days. This regularity eases the adaptation in the beginning. Later, however, such regularity leads to accommodation. Eventually training is done on different training days. (Falameev, 1974)

- A typical beginner schedule—3 times a week plus GPP once a week. (Roman, 1986)

- Optimal for beginners and low ranked lifters NBL in classic lifts is 10-20, in assistance exercises 8-18 and up to 24 in the prep period. (Falameev, 1974)

- Sets in a session for beginners 20-22 (compare to 40-50 for experienced athletes). (Falameev, 1974)

- Select 8-10 exercises for a month; do about 4 each session. 5-6 sets of 3-5 reps (20-25 barbell sets total). Train 3 times a week in 1.5-2 hour long sessions. (Vorobyev & Medvedev, 1972)

- In each exercise 6 sets of 3-4 reps with a constant weight. In squats 4-6 sets of 4-8 reps. In presses 4-6 sets of 4-10 reps. (Medvedev, 1986)

- Beginners' volume of the classic lifts is 80% of the total; a little later 60%. (Vorobyev & Medvedev, 1972)

- The first four months do high volume of practice lifts—NBL 1,150 in a month; 110 per session. Beginners and lower ranked lifters train with low intensity, so putting up the required volumes is not a problem. (Roman, 1986)

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Intensity

- Track the weekly load in NBL but do not start tracking ARI until Level III. (Sheyko, 2005)

- "Technique should be learned with a medium weight… Neither weak nor strong stimuli are optimal for forming a conditioned reflex. Only medium weights are able to deliver [technique learning] and facilitate better development of the special physical qualities needed for lifting a barbell with a maximal weight." (Lukyanov & Falameev, 1969)

- 60-80% 1RM is the perfect range for beginners. (Rodionov, 1966)

- An optimal weight is one a beginner can "confidently lift up to 5-6 times." (Vorobyev, 1981; Medvedev, 1986; Sheyko, 2005) Call this weight "X".

- For the press a weight that is easily pressed 3-4 times and even (!) 5 times. No more than 5-6 sets. (Roman, 1962)

- Lift light weights for 3-4 reps; medium and heavy for 1-2 reps. For awhile one makes gains with medium, 60-70%, weights, then gains slow down and the weights need to be upped. (Vorobyev, 1964)

- Technique is mastered faster with a static weight in sets of 3-4 reps. (Vorobyev & Medvedev, 1972)

- Usually beginners start a session with 50-60% weights (compared to 70-75% for experienced athletes). (Vorobyev & Medvedev, 1972)

- "The most important in training of novices is not intensity but learning and solidifying the technique of the competition lifts. Usually technique is learned with moderate weights that the athlete is able to lift 4-5 times." (Sheyko, 2005)

- "For beginners and intermediates the correlation between the training intensity and the results is insignificant, so there is no need to increase the number of lifts with 70% and heavier weights. Results grow even with lighter weights." (Roman, 1986)

- Beginners should mostly use 3-6 reps with minimal, small, and medium weights. Up to 10 reps for some exercises for certain muscle groups. (Roman, 1986)

- "It must be pointed out that beginner and intermediate athletes should not do a maximal number of reps possible [with a given weight], but 2-3 times fewer. In other words, a weight that could be lifted, say, 12 times, is lifted 4-6 times. As research and experience show, such effort is quite sufficient for non-stop growth of their results.

  "If beginner or intermediate athletes used a maximal number of repetitions per set (RM), they would get tired after several such sets and would be unable to continue training with a sufficient volume. Moreover, even a low number of sets of a maximal number of repetitions would lead to injuries and interfere with regular training." (Roman, 1986)

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Recommended repetitions for beginners and Level III and II lifters
(Roman, 1986)

	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Classic lifts		4-6	3-4	2-3	1-2	1
Power clean, power snatch			4-6	3-4	1-2	1
Push press		4-5	3-4	2-3	1-2	1
BSQ, FSQ		6-10	4-6	3-4	1-2	1
BP	6-10	5-8	4-6	3-4		

- "Some coaches employ the multiple repetition method (5-8 reps per set) with beginners and low ranked lifters. It appears to us that there is no need to introduce such a training element into the lessons of lower ranked weightlifters." (Falameev, 1974)

- Rest for 3min between sets. (Vorobyev & Medvedev, 1972)

- Compete in technique once a month. (Sheyko, 2005)

Sets & reps

- Work up in doubles or triples starting with "X", 60-70% 1RM. (Rodionov, 1966)
  For the press:

  - X x 2, (X+5kg) x 2, (X+10kg) x 2, (X+15kg) x 2
  - X x 3, (X+10kg) x 3, (X+20kg) x 3/3
  - X x 3, (X+10kg) x 3/2, (X+20kg) x 3

- A method of improving jumping ability for non-weightlifters: back SQ with low to medium weights and a maximally explosive concentric. 7-8 sets of 3-5 reps (up to 20-30 NL) 3-4 times a week. (Medvedev, Moroz, Falameev, 1986)

- "Easy strength" bench presses: "The training effect from using an 80% weight is the same, independent of the number of lifts in the 9-21 range and this is why there is absolutely no need to do more than nine [total] repetitions when training three times a week." (Saxonov & Prilepin, 1970)

- Effective loading parameters for the squat, snatch, and other lifts for beginners to MS level (Medvedev & Dvorkin, 1984):

  1. 70-80% 1RM x 3-4 reps per set, NBL 18. (Most effective)
  2. 50-65% 1RM x up to 6 reps per set, NBL 36. (Second best)
  3. 80-90% x 1-2 reps, NBL 12. (Least effective)

  Younger and weaker lifters had better results with lighter weights.

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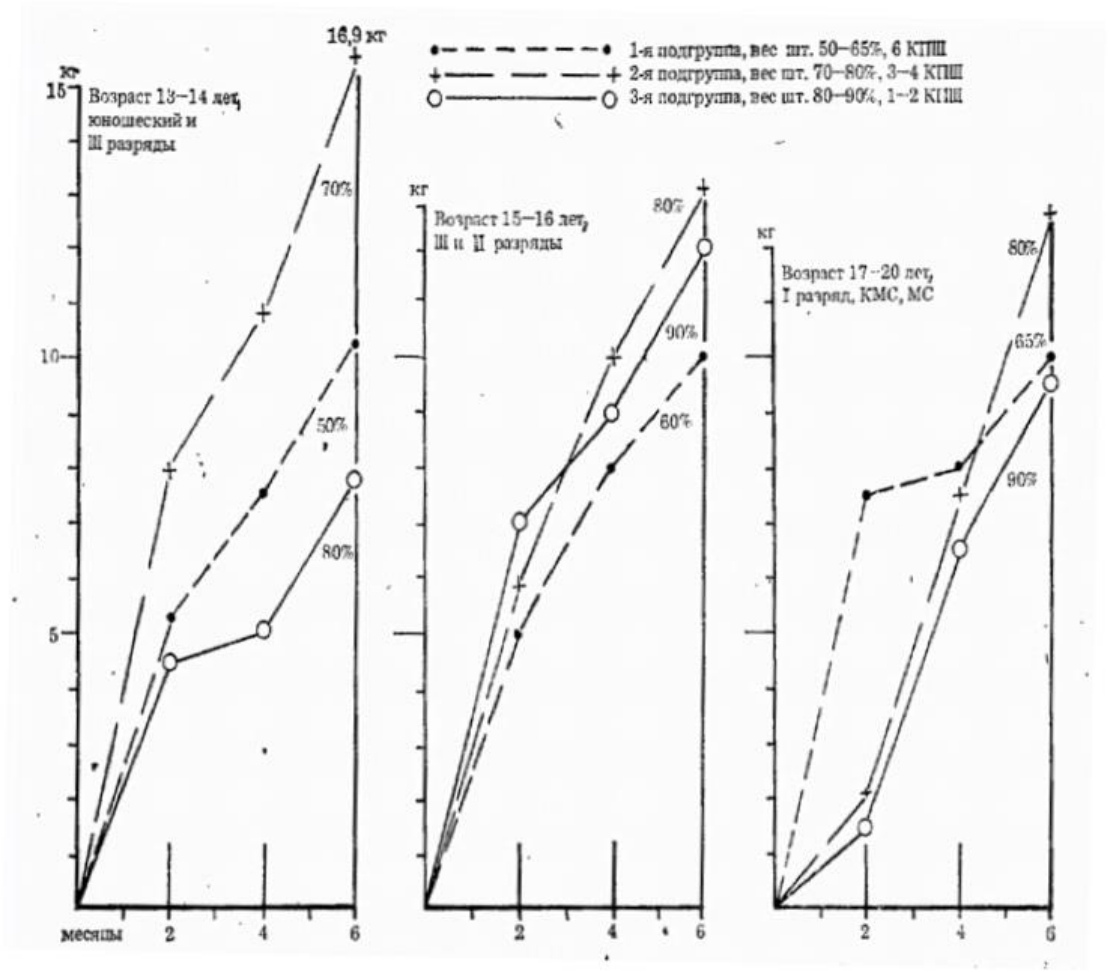

Variability

- Variability is needed but it is limited. (Vorobyev, 1977)
- Load variability in weeks and months is "insignificant". (Medvedev, 1986)
- Low level athletes may use counterphases of volume and intensity in a week. (Vorobyev & Medvedev, 1972)
- In beginners and razryadniki the intensity is not varied much from session to session; the NBL is. (Falameev, 1974)
- Monthly NBL distribution for the first four months (Roman, 1986):

    1. 1,150
    2. 1,250
    3. 1,200
    4. 1,000 (sort of a comp month)

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Then "rest" in month five: twice a week GPP and weightlifting once a week. NBL 500 a month, 125 a week with minimal weights.

The next four months the volume is a bit lower (weights are heavier now): 1,100 a month in prep, 900 in comp. Per session—90 in prep, 75 in comp.

Note that the scientist divides training into prep and comp period from the start. First competition is about six months after starting. (Roman, 1986)

- Volume is reduced in competition month by only 17-18%, as the intensity is still low. (Roman, 1986)

- Beginners' training sample month volume distribution by weeks (Roman, 1986):

  1. 250
  2. 300
  3. 250
  4. 200

  A later prep month (1,100):

  1. 275
  2. 310
  3. 250
  4. 265

  A comp month (900):

  1. 240
  2. 300
  3. 210
  4. 150

- A beginner's week: the first session is the highest volume, the second lowest (high, low, medium). E.g., the first week of prep period (275 NL): Monday—110, Wednesday—75, Friday—90. The second week (310): 120, 90, 100. (Roman, 1986)

Sample weekly volume distribution for beginner powerlifters
(Sheyko, 2005)

Day	Week 1	Week 2	Week 3	Week 4
1	M	L	M	L
2	L	H	L	H
3	M	L	H	M

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PLAN STRONG™ 70

HIGH INTENSITY ZONES AND A LIMITED NUMBER OF WEIGHTS

Plan Strong™ 70 uses the high intensity zones only, from 70% 1RM, and a limited number of weights. It is easier to implement and can be used when light weights are not an option, e.g. in bodyweight strength training where an athlete's bodyweight pullup may represent 70% 1RM or more. It is up to the coach to assign lighter warm-up weights. Some athletes choose not to touch light weights at all. E.g., John Davies, a 1940s' world champion, took 80% as his starting weight. Back when the press was a competition lift many plans by Vorobyev and Roman started with 70% after a general warm-up; lighter weights were not used at all. In special exercises like the power snatch Rodionov (1966) recommended starting with 80%. If ARI measured in all exercises and zones, it tends to increase as the athlete's weight decreases.

Interestingly, when measured in the classic lifts in the 70% and higher zones only (HARI), the intensity does not depend on the weight class and is an individual value. And individual differences are a lot lower than when all exercises and zones are measured—5% vs. 10%. (Medvedev, Verkhoshansky et al., 1992) E.g., Alexeev (1976) added many lighter lifts to the previously done high intensity work thus lowering his ARI—but not the HARI.

BIG CYCLE

PREPARATORY PERIOD

Step I: Monthly volume
Step II: Volume dynamics within a preparatory month
Step III: Monthly intensity (91-100% NL, distribution of the volume by intensity zones, HARI)
Step IV: Intensity dynamics within a month
Step V: Volume dynamics within a week
Step VI: Intensity dynamics within a week
Step VII: Load in a session
Step VIII: Interaction of different lifts within a week and within a session
Step IX: Programming assistance exercises. Training more than main one lift prioritizing the same muscle groups.

COMPETITION PERIOD

Step I: Monthly volume
Step II: Volume dynamics within a competition month
Step III: Monthly intensity (90-100% NL, distribution of the volume by intensity zones, HARI)
Step IV: Intensity dynamics within a competition month
Step V: Volume, intensity, and their dynamics within a week

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BIG CYCLE

Preparatory period (1-2 months)
+
Competition period (1 month)
+
Transition period (1 week to 1 month)
=
Big cycle

As a rule of thumb, the preparatory period has higher volume and lower intensity than the competition period.

Competitive powerlifters, plan your competition period before the prep period and then work back, as you are likely to take some maxes or near-maxes in the last two weeks of the prep period.

The transition period is typically a week of "random acts of variety".

PREPARATORY PERIOD

A two-month prep period is preferable and some athletes are unable to consistently improve on a one-month prep period. Nevertheless, when you are starting to use this system, stay with one-month prep periods for several cycles to learn your lessons faster.

If there are two months in the preparatory cycle, either month can be higher in volume. Well-trained lifters can handle a 100% increase in volume from one month to the next. A rock bottom minimum NL change from month to month, up or down, is 20%.

Consider some of the great many possibilities in arranging a big squat or deadlift cycle consisting of 2 preparatory months and 1 competition month:

150-200-125 (+33%, -38%)
150-250-125 (+67%, -50%)
150-200-150 (+33%, -25%)
200-250-125 (+25%, -50%)
250-200-125 (-20%, -38%)
250-175-125 (-30%, -29%)

A minimum NL change of 20%, increase or decrease, applies to any two adjacent training units: months, weeks, days.

If there is more than one prep month, increase or decrease the HARI by at least 1% 1RM between them and change the shares of individual zones.

As a rule of thumb, the lifter should be able to comfortably handle his old 1RM while training with high volume towards the middle or the end of the second preparatory month.

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Step I: Monthly volume

The main planning unit is a month (4 weeks).

Monthly NL per class of lift in the preparatory period, ≥70% 1RM
(StrongFirst)

- Squats and/or[2] deadlifts—**150-250**
- Presses, pullups—**200-300** and higher

Select the monthly volume for each class of lift—squat/deadlift, press, (pullup)—from the above ranges. Let your recent training history dictate whether to choose a number on the low end, the high end, or in the middle.

Calculate 35% of the number you have chosen—this is the stress week. Ask yourself: can the athlete safely handle this volume for one week out of a month? If it seems unlikely, lower the total and/or reduce the volume variability (lower the share of the top week below 35%).

Step II: Volume dynamics within a month

Variability (both for volume and intensity) may be compared to investment. High variability corresponds to a riskier investment with a possibility of a greater return—and a higher risk of loss (overtraining). Lower variability may be compared to a safer investment with lower return.

Volume variability within a month is described by:

- The difference in NL between the highest and the lowest week (and, to a smaller degree, between the medium weeks). The higher is the difference, the greater variability.

  E.g., 40% and 15% are more variable than 35% and 15% or 40% and 20%.

  40-15=25
  35-15=20
  40-20=20

- The change between each two adjacent weeks. Gradual variants are least variable. Sharp variants are most variable. Gradual-sharp and sharp-gradual variants are in between.

  Variant 1 35-28-22-15%, is the least variable. The changes from week to week are 20, 21, 32%. For contrast, in one of the most variable variants, 2-4b, 22-35-15-28%, the changes from week to week are 59, 57, 87% respectively.

In your first cycle of this type of training or after a lay-off pick 3b, 3c or 4. Later rotate all the variants (not necessarily in equal proportions or in the listed order).

Generally, use the 15, 22, 28, 35% constants in one of the sixteen combinations:

[2] For competitive powerlifters these numbers apply to each lift: the squat and the deadlift.

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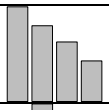
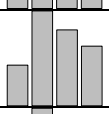
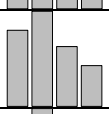
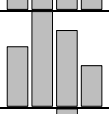
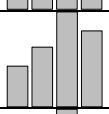
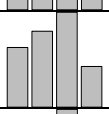
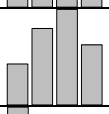
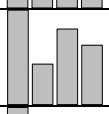
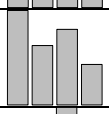
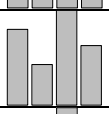
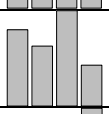
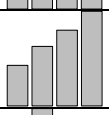
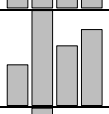
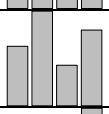
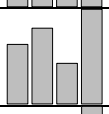
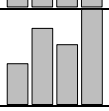
Types of monthly NL distribution by week
using "stable structural constants"
(Chernyak, 1978)

Variant	Dynamics	Percentage of the monthly total			
		Week 1	Week 2	Week 3	Week 4
1		35	28	22	15
2a		15	35	28	22
2b		28	35	22	15
2c		22	35	28	15
3a		15	22	35	28
3b		22	28	35	15
3c		15	28	35	22
1-3a		35	15	28	22
1-3b		35	22	28	15
3-1a		28	15	35	22
3-1b		28	22	35	15
4		15	22	28	35
2-4a		15	35	22	28
2-4b		22	35	15	28
4-2a		22	28	15	35
4-2b		15	28	22	35

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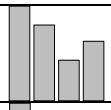
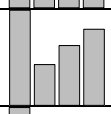
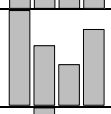
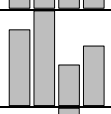
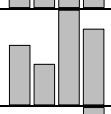
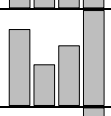
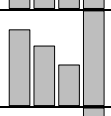
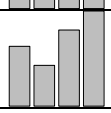
Eight more combinations of four percentages are possible but are rarely used given two deloading weeks back to back. They are useful for intensity variability though (assigning 81-90% and 91-100% weights to weeks):

1-2		35	28	15	22
1-4a		35	15	22	28
1-4b		35	22	15	28
2-1		28	35	15	22
3-4		22	15	35	28
4-1a		28	15	22	35
4-1b		28	22	15	35
4-3		22	15	28	35

The method allows great flexibility to fit your athletic, professional and personal circumstances.

E.g., you have a T&F competition on week 3. Make week 3 lowest in volume, 15%, and taper towards it using the variant most suitable for your circumstances. E.g., 4-2a (22-28-15-35%) or even 4-1 (28-22-15-35%).

When 4-week blocks do not fit one's circumstances

A three-week block can be put together using one of these sets of percentages, from the least to the most variable:

25, 33, 42
20, 35, 45
22, 28, 50
20, 30, 50
15, 35, 50
15, 30, 55

The volume of the three-week block must be ≤60% or ≥90% of the previous month (a 20% difference recalculated for a shorter block).

By the same token the next month after the three-week block must be ≤107% or ≥160% of the three-week block.

If the circumstances demand a five-week block, plan a four-week block, then add a fifth week with an NL change of at least 20% from the fourth week.

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E.g., the plan calls for 200 lifts for a month. Choose variant 2-4b, 22-35-15-28%. Last week's 28% converts to NL 56.

56x0.8=45.
56x1.2=67

Fifth week's NL must be ≤45 or ≥67. Your decision to go up or down should be based on the content of the following month.

Ladies, consult the guidelines for women weightlifters (Rumyantseva & Gorulev, 2005):

- In the, ovulating, premenstrual, and menstrual phases reduce the volume in exercises loading the pelvic floor, especially squats.
- Highest strength and speed-strength performance, maximal efficiency and speed of recovery is in the post-menstrual and post-ovulating phases. These are the phases to plan the highest volume and intensity loads.
- Performance is not at its highest in the ovulatory phase.

If an athlete needs higher or lower variability than provided by the 15, 22, 28, 35% constants, customize the percentages using the following guidelines.

A classic preparatory period variant has to have one small week, one large week, and two medium weeks. When the highest volume week is in the maximal range, one of the medium weeks may become small. Or both medium weeks may stay in the bottom half of the medium range, 20-25%.

Weekly volume as a percentage of the monthly volume
(Roman, 1968; Chernyak, 1978)

Small	Medium	Large	Maximal
< 20%	20-30%	31-40%	> 40%

If you take a close look at Prof. Chernyak's constants, you will see that 15% is squarely in the middle of the small range (typically cut off at 10%) and 35% is in the middle of the large range. Round up or down 22% and 28% to 22.5% and 27.5% and they are right in the middle of the 5% medium sub-ranges of 20-25% and 25-30%. Hence the following guidelines:

Guidelines for customizing volume distribution among weeks
(StrongFirst)

	Week's volume			
	1 (low)	**2 (med)**	**3 (med)**	**4 (high)**
Standard	14-16%	22-23%	27-28%	34-36%
More variable	11-13%	20-21%	29-30%	37-46%
Less variable	17-19%	24-25%	25-26%	31-33%
When the highest volume week is in the maximal range, one of the medium weeks may become small. Or both medium weeks may stay in the bottom half of the medium range, 20-25%.				
May combine standard, more variable, and less variable percentages.				
A minimum of 20% difference between two adjacent weeks.				

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There must be a minimum of 20% increase or decrease between two adjacent weeks.

E.g., you have selected a low variability set of numbers: 19-24-26-31. Weeks with 24% and 26% may not be adjacent.

26x0.8=21. Therefore, you must go down to 21% or less after the 26% week.

24x1.2=29. You must go up to at least 29% after 24%.

Hence this combination of numbers excludes almost half the variants possible: 1, 3b, 1-3a, etc.

Note that all weeks do not have to fall into the same category ("standard", "more variable", "less variable"); your variant may be a combination.

E.g., Chernyak, Atanasov & Ermakov (1972): 24-19-36-21—less-less-standard-more.
E.g., Sheyko (2013): 34-30-24-12—standard-more-less-more

For months with very high volume lower variability is recommended for the month or at least for the highest volume week.

E.g., monthly NL 400 is high for the given athlete. 35% of that is 140. 31% is 124, a noticeable reduction of NL 16 in the high zones. The coach chooses 17, 25, 27, 31%.

Step III: Monthly intensity (NL in the 91-100% zone, distribution of the volume by intensity zones, HARI)

Some specialists included 90% into the top zone; others started with 91%. This distinction becomes trivial when using a limited number of set weights, which allows one to blur the edges. For our purposes, "90-100%" and "91-100%" are synonymous.

The number of 90-100% 1RM lifts is highly individual and varies from lift to lift.

There are many successful lifters, especially in powerlifting, who do no lifts in this zone in the preparatory period and very few in the competition period. Experiment to find your optimum using the following guideline may as the starting point. Per lift per week:

- NL 0-3 in the prep period and the last two weeks of the comp period;
- NL 2-5 in the first two weeks of the comp period; 0-3 in the third.

Once you have established the NL in the 90-100% zone, convert the reps into percentage of the monthly total.

E.g., you have decided to do 7 squats in the top zone in the upcoming month. Your monthly SQ NL is 200. 7: 200=3.5. Round up this number: 4% of your volume belongs in the top zone.

Assign the percentage to a table and place the remaining percentage in the 71-80% and 81-90% zones.

E.g.:

70-80%	***81-90%***	***91-100%***
97		*3*

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An option: flexible planning of top zone lifts

As an option, the coach may write the training plan and make the calculations with no lifts in the top zone and then identify several training sessions when the athlete may choose to perform some top zone lifts if he is feeling very strong.

In a session do not exceed:

- Presses: 7 presses at 90% or 3-4 at 95-100%
- Squats and deadlifts: 3 with 90% and 2 with 95-100%

Standard set/rep schemes:

- 90x
  - 1/3 (reps/sets)
  - 1, 2
  - 1, 2, 1
  - 1, 2, 1, 2
  - 1, 2, 1, 2, 1

- 90x1, 92-5x1/1-3

- (90x2, 92-5x1) x 1-2

- 90x2, 92-5x1/2, 90x2

- 90x1, 92-5x1, 96-102x1 (never more than once every two weeks and only if the athlete is 100% certain of making the lifts)

Following these heavy lifts the athlete proceeds to finish the session. Maintain the planned NL—get rid of the same number of lifts you have added—but lower the intensity if fatigued.

E.g., the plan called for 70x3, 75x2, 85x2, 3, 2, 3, 80x3, 4 (NL 22) with a note that the athlete may perform several heavy lifts if she is feeling very strong.

The first double with 85% felt strong, so she did a 90% single. It felt strong and she did a double. She had room to spare and did 95x1. It felt strong but she doubted she would be able to exceed or even match her 100%, at least not without great stress. She did another 90% single. The athlete did the math: she had completed NL 12 out of 22. She chose to drop the planned 85% lifts and complete the remaining NL 10 with 80:

70x3, 75x2, 85x2, 90x1, 2, 95x1, 90x1, 80x2, 3, 2, 3 (NL 22)

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Next assign the remaining lifts among the 71-80% and 81-90% zones using the following guidelines:

Guidelines for monthly volume distribution in the high intensity zones
(StrongFirst)

% 1RM	70-80%	81-90%	91-100%	HARI
Average	55	35	10	**80%**
Primary	**50-60**	**30-40**	**0-10**	**79-81%**
Expanded	20-90	10-60	0-20[17]	**75-85%**

The "primary" numbers indicate the ranges where most of the successful Soviet weightlifting plans fit in. The "expanded" range where a smaller number of successful plans existed is listed to give you more options, especially when you have a limited choice of weights to work with. Feel free to push into the HARI on low end of the expanded range, all the way down to 75%. But be very cautious about pushing the top end above 81%.

E.g., in the notorious Smolov's squat cycle the HARI of the prep period is "only" 79.6%.

Yet many young intermediate athletes thrive on a high ratio of heavy lifts (when coached correctly). E.g., Prilepin (1975) recommended the following loads to razryadniki: 70% 1RM—23.8%, 80%—36.8%, 90%—25.4%, ≥91%—14%. That added up to a HARI of 82.3%

Select 1-2 weights per zone in the two bottom zones. In the 91-100% zone select 0-3 weights.

When training with a barbell, use only large plates, except in the 91-100% zone.

E.g., a man with a 570-pound deadlift should use an unlimited number of 45s and one set of 25s: 405 (71%), 455 (80%), 495 (87%), 545 (96%). However, he might decide to be more surgical in his selection of nearmax weights. E.g., based on his experience, going above 93% burns him out. He chooses 93% or 530 pounds, even though it requires an assortment of small plates.

E.g., a lady's max deadlift is 240 pounds. She tries limiting herself to 45s, 25s, and 10s: 185 (77%), 205 (85%), 225 (94%). She may decide to stay with this limited selection—or conclude that exclusively training with weights no lighter than 77% is too much and add another selection that requires more plate choices, 165 (69%).

For kettlebell presses your choices are already preselected:

High intensity zones' kettlebell training weights as % of 1RM
(Starting from 70% 1RM; from 67% or 2/3 1RM when necessary)

1RM ↓	12 kg	14 kg	16 kg	18 kg	20 kg	22 kg	24 kg	28 kg	32 kg	36 kg	40 kg	44 kg
48kg									67%	75%	83%	92%
44kg									73%	82%	91%	
40kg								70%	80%	90%		
36kg							67%	78%	89%			
32kg						69%	75%	88%				
28kg					71%	79%	86%					
24kg			67%	75%	83%	92%						
22kg			73%	82%	91%							
20kg		70%	80%	90%								
18kg	67%	78%	89%									

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Note that in several cases exceptions have been made to accommodate the limited choices and 67-69% weights, lighter than the 70% bottom, have been used. The same tactic is acceptable for barbell training if this enables one to stick with larger plates. Balance out this very light selection with a sufficient number of heavier lifts near the top of the 70-80% zone.

E.g. a lady with an 18kg max press has the following choices that do not neatly fit the three zones:

1RM	*12kg*	*14kg*	*16kg*
18kg	*67%*	*78%*	*89%*

It is apparent that to stay in the ballpark of the desired 80% average intensity she has to go easy on the 67% lifts, emphasize 78%, and add a healthy dose of 89% to pull up the average. The 89% weight ends up doing double duty for both the 81-90% and 91-100% zones.

The above example emphasizes that a weight like 89% that borders two zones has to do double duty for both zones it straddles. This tactic eliminates uncomfortable debates of this nature: "Should we start counting the top zone from 90% or 91%?" Just aim for the target HARI—while being careful not to exceed a chosen number of circa-90% lifts.

To continue with the above example, if our athlete chooses to stay in the primary HARI range of 79-81% she will need to experiment with the numbers:

67%	***78%***	***89%***	***HARI***	***Commentary***
30	*60*	*10*	***75.8%***	*Way below the primary range 79% bottom.*
10	*80*	*10*	***78%***	*Close to the target of the primary range; acceptable for the expanded range.*
10	*70*	*20*	***79.1%***	*Jackpot—but the athlete should use her judgment whether such a high number of 89% lifts are doable for her.*
5	*80*	*15*	***79.1%***	*She decides that the above number of heavy lifts is excessive for her, reduces it to 15%, and beefs up the middle weight at the expense of the bottom one. The resulting HARI is the same but its composition is more fitting for the given athlete.*

If you are able to press a given kettlebell more than once yet are unable to press the next one up, estimate your 1RM. Use your experience or a conversion table such as the one by Matveev (1991):

95% x 2-3RM
90% x 3-6RM
85% x 5-7RM
80% x 6-10RM
75% x 8-12RM
70% x 10-15RM

Then calculate the % 1RM for lighter kettlebells.

E.g., you can press 40kg x 2RM. According to the table, it should correspond to 95% 1RM. (40 : 95) x 100 = 42. Your estimated 1RM is 42kg.

32kg : 42kg = 76. 36kg : 42kg = 86. Therefore, 32kg = 76% 1RM and 36kg = 86% 1RM.

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If your 1RM places you in a position of a very limited weights selection, diversify with doubles. Note that there is no correlation between your single kettlebell press 1RM and double kettlebell press 1RM.

*E.g., your single kettlebell MP max is 20kg and you do not have 14kg or 18kg kettlebells, only a pair of 12s, one 16, and one 20. The only choice you have in the 70% and up range is 16kg/80% 1RM. 12kg is too light, 60%. But if you test yourself with 2*12kg, it turns out that it is your 12RM. Based on Matveev's table, this is your 70-80% 1RM. Using your experience, you decide it is close to 70%. Now you have an additional weight to work with, 70% 1RM.*

Even if you do not have a sufficient selection of weights you may diversify your choices with doubles using the above method.

Next assign each weight a percentage of the total using the earlier established guidelines.

Let us stick with the above example of 20kg x 1RM and the following distribution:

70-80%	**81-90%**	**91-100%**
98		2

Considering that you have no weights in the 81-90% zone, just 80% on the edge of that zone, you will need to play with the 70% and 80% choices to achieve the desired HARI. Let us try assigning half of the volume to 70%:

% 1RM	70%	80%	100%	**HARI**
	2*12kg	**16kg**	**20kg**	
	50	48	2	
	Calculations:			
	50 x 70 = 3,500	48 x 80 = 3,840	2 x 100 = 200	
	3,500 + 3,840 + 200 = 7,540			
	7,540 : 100 = 75.4			**75.4%**

The resulting average intensity is near the bottom of the acceptable range. You decide that it is too low. Move more NL from 70% to 80%:

% 1RM	70%	80%	100%	**HARI**
	2*12kg	**16kg**	**20kg**	
	38	60	2	
	Calculations:			
	38 x 70 = 2,660	60x 80 = 4,800	2 x 100 = 200	
	2,660 + 4,800 + 200 = 7,660			
	7,660 : 100 = 76.6			**76.6%**

Success.

If you have more than one weight per zone, list all the weights.

E.g., for a 44kg x 1RM press:

73%	82%	91%	100%	**HARI**
32kg	**36kg**	**40kg**	**44kg**	
40	40	16	4	**80.6%**

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Step IV: Intensity dynamics within a month

Intensity dynamics refers to HARI's increases and decreases from week to week. Generally change the intensity from week to week and month to month by 1-5% 1RM.

Like volume dynamics, it is described by a "variant" like 1-4 or 3a.

> *E.g. the HARI changes in this manner from week to week: 81.2%, 79.4%, 77.6%, 80.4%. The highest value is on week one and the second highest on week 4. Hence the intensity variant is 1-4. To be more exact, 1-4b, as week two is higher than week three.*

A variant does not presume that the percentages are 15, 22, 28, 35%; it only describes the dynamics.

> *E.g. in 4-2a and 4-2b, the highest number falls on week four and the second highest on week two. The lowest falls on week three in 4-2a and week one in 4-2b.*

Unless the ratio of 90% and heavier lifts is unusually high, the distribution of the 81-90% lifts is likely to determine the intensity variant for the month. Although these lifts are lighter, they strongly influence the average because they are many.

Intensity is varied by assigning lifts in the 91-100% zone and in the 81-90% zone to weeks according to the same procedure as for assigning the total volume but using a different variant and, as an option, different percentages.

An option is to assign all lifts to two or three weeks only for more variability. (And when the number of lifts in the top zones is low, it is a necessity.)

> *E.g., you have NL 10 in the 91-100% zone. You choose to assign 3 to week two and 7 to week four (30:70 ratio)—variant 4-2.*
>
> *Now you want to assign the 50 lifts in the 81-90% zone according to the same variant between three weeks. You choose the 25, 33, 42% distribution and assign the NL from week to week respectively: 13, 16, 0, 21—variant 4-2a.*

There is no obvious trend in intensity variability from week to week.

Intensity dynamics are independent from that of volume and all combinations build strength. (Although few are optimal for displaying that strength, which is why the competition period plays by its own rules.)

When assigning 91-100% lifts consider your ability to make these lifts on the background of the recent loads. But do not automatically place most 91-100% lifts into the lowest volume weeks as this would set up an inverse relationship between volume and intensity where there should be none. (Also, if you plan your weeks with an increase of volume towards the end, going heavy on Wednesday of a high volume week takes place before the lion's share of the week's volume.)

A tactic: repeat your latest 1RM set at the end of the competition month "under load" in the middle or the end of the prep period—a PR of sorts.

You may use the 15, 22, 28, 35% constants—or make assignments without adhering to percentages, just paying attention to maintaining variability. The usual rule of thumb applies: a rock bottom minimal increase or decrease from week to week is 20%.

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E.g., the NL in a given zone is 30 in week three. 20% of it is 6. Hence in weeks two and four (or four and two) the NL must be ≤36 and ≥24.

You may also use the same ratios you have used for designing the variant for the total volume. See the table "Guidelines for customizing volume distribution among weeks".

First assign the 91-100% zone—select a variant and the percentages. If the number of these lifts is low, calculating the percentages is unnecessary.

E.g. you have chosen NL 7 for top zone lifts in a month and variant 3a. It is easy to simply assign 0-1-3-2 lifts to weeks without getting into percentages.

Then assign the ≥95% lifts out of the 91-100% NL total to given weeks without any consideration for the chosen variant. Any timing is possible—but you must remember that you must make the planned lifts without excessive excitation.

In the following example there are 25 lifts in the top zone. Note that they are assigned according to the 15…35% constants—while the NL 5 of 100% lifts are arbitrarily assigned to given weeks.

% 1RM	72%	82%	91%	100%	**HARI**
	16kg	**18kg**	**20kg**	**22kg**	
	50	40	10		**78%**

	NL 250			
	Week 1	**Week 2**	**Week 3**	**Week 4**
3b	22%	28%	35%	15%
NL	55	70	87	38
	91-100% (NL 25)			
2-4b	22%	35%	15%	28%
	5	9	4	7
	Out of that number 100% (NL 5)			
NL	—	2	—	3

Next apply the above procedure to the weights in the 81-90% range. The variants and percentages for 81-90% and 91-100% zones may or may not match.

% 1RM	75%	83%	92%	100%	**HARI**
	18kg	**20kg**	**22kg**	**24kg**	
	60	30	7	3	**79.3%**

	NL 200			
	Week 1	**Week 2**	**Week 3**	**Week 4**
3b	22%	28%	35%	15%
NL	44	56	70	30
	92-100% (NL 20)			
2-1	28%	35%	15%	22%
NL	6	7	3	4
	Out of that number 100% (NL 6)			
NL	1	2	—	3
	83% (NL 60)			
1-3b	35%	22%	28%	15%
NL	21	13	17	9

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When you are assigning 81-90% lifts, generally do not place 35% of these lifts into the 15% and 22% weeks and do not place 28% of these lifts into 15% weeks, as you are likely to run out of space or at least end up with an insufficient number of lighter lifts needed for progressive rehearsal.

The same recommendation (not rule) applies when percentages other than the classic 15, 22, 28, and 35 are used for the total volume, the 81-90% lifts, or both: do not assign the highest number of 81-90% lifts to either of the two lowest total volume weeks.

If you have two weights in the 81-90% zone, generally assign them according to the same variant (although it is advisable to change the percentages).

E.g., variant 2-4b for both 83% and 87% weights. For the former 22-35-15-28%; for the latter 19-40-15-26%.

An alternative tactic for planning two weights in the 81-90% zone is to first lump them together and assign the total according to the same variant and percentages. Then asymmetrically distribute the two weights between the weeks.

E.g., the plan calls for NL 50 in the 81-90% zone: 35 with 83% and 15 with 87%.

Assign all 50 lifts according to the classic variant 2-4b: 22-35-15-28%; respectively NL 11-17-8-14.

Since week four calls for several lifts in the 91-100% zone, take it easy with 87% weights; a single in the warm-up will suffice. Skip 87% in week three altogether and distribute the remaining NL 14 between weeks one and two: 6, 8.

The NL in the bottom zone is calculated by subtracting the middle and top zone lifts from the total in each week.

To stay with the earlier example:

	NL 250			
	Week 1	**Week 2**	**Week 3**	**Week 4**
3b	22%	28%	35%	15%

	NL 200				**Total**
	Week 1	**Week 2**	**Week 3**	**Week 4**	
3b	22%	28%	35%	15%	
NL	44	56	70	30	**200**
	92-100% (NL 20)				
2-1	28%	35%	15%	22%	
NL	6	7	3	4	**20**
	Out of that number 100% (NL 6)				
NL	1	2	—	3	
	83% (NL 60)				
1-3b	35%	22%	28%	15%	
NL	21	13	17	9	**60**
	Calculations to arrive at 75% assignments:				
	44-6-21=17	56-7-13=36	70-3-17=50	30-4-9=17	
	75% (NL 120)				
	17	36	50	17	**120**

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Make sure to add up the numbers in each column and row to verify your calculations. E.g. adding up all the numbers in the 75% row we get NL 120. Given the total NL of 200 and a 60% share of 75% lifts, the math is correct.

If you have two weights in the 70-80% zone, assign the weight that represents the majority of that zone according to the overall volume variant, then determine the NL of the remaining weight by the subtraction procedure described above.

Finish monthly planning by calculating the HARI in every week, identifying the overall intensity dynamics (and changing it if necessary), and laying out a detailed plan.

Your numbers are:

	NL 200			
% 1RM	**Week 1**	**Week 2**	**Week 3**	**Week 4**
100%	1	2	—	3
92%	5	5	3	1
83%	21	13	17	9
75%	17	36	50	17
Total	**44**	**56**	**70**	**30**

Calculations for week one:

% 1RM	75%	83%	92%	100%	**HARI**
	18kg	**20kg**	**22kg**	**24kg**	
NL 44	17	21	5	1	
	(17:44) x 100 =39	(21:44) x 100 =48	(5:44) x 100 =11	(1:44) x 100 =2	
	39%	**48%**	**11%**	**2%**	
	39x75=2,925	48x83=3,984	11x92=1,012	2x100=200	
	(2,925+3,984+1,012+200=81.2) : 100				**81.2%**

Calculations for all four weeks net the following. You can see that the given method has enabled you to sharply vary the intensity from week to week. In addition, the dynamics of the volume (3) and the intensity (1-3) are independent, as desired. The overall intensity variant (1-4b) does not exactly matches that of the 81-90% zone (1-3) but they are similar.

	Week 1	**Week 2**	**Week 3**	**Week 4**	**Month**
HARI	81.2%	79.4%	77.6%	80.4%	**79.3%**
	1-4b				

Here is your monthly plan, with all the data:

% 1RM	75%	83%	92%	100%	**HARI**
	18kg	**20kg**	**22kg**	**24kg**	
	60%	30%	7%	3%	**79.3%**
NL 200	120	60	14	6	

	Week 1	**Week 2**	**Week 3**	**Week 4**	**Month**
3b	22%	28%	35%	15%	
NL	44	56	70	30	**200**

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HARI	*81.2%*	*79.4%*	*77.6%*	*80.4%*	***79.3%***
1-4b					
	92-100%				
2-1	*28%*	*35%*	*15%*	*22%*	
	100%				
NL	*1*	*2*	*—*	*3*	***6***
	92%				
NL	*5*	*5*	*3*	*1*	***14***
	83%				
1-3b	*35%*	*22%*	*28%*	*15%*	
NL	*21*	*13*	*17*	*9*	***60***
	75%				
NL	*17*	*36*	*50*	*17*	***120***

Barbell plans are made in the same manner, in kilos or pounds, using a limited selection of weights made up with large plates (and often some smaller plates in the top zone).

In this example a 570-pound deadlifter uses an unlimited number of 45s, one set of quarters, and whatever plates it takes to make up his max pull.

% 1RM	*71%*	*80%*	*87%*	*96%*	*100%*	***HARI***
Lbs.	***405***	***455***	***495***	***545***	***570***	
	33	*40*	*20*	*7*		***79.6%***

This methodology is not recommended for kettlebell front squats: the upper body frequently becomes the limiting factor.

For handstand pushups find a ROM or HSP variation for each intensity zone:

Approximate % 1RM to RM conversion
for athletes with average endurance with a given RM

70-80%	81-90%	91-100%	ARM
8-12RM	4-7RM	1-3RM	6-10RM

Approximate % 1RM to RM conversion
for athletes with low endurance with a given RM

70-80%	81-90%	91-100%	ARM
6-10RM	3-5RM	1-2RM	5-8RM

E.g., you are a "fast twitch athlete" and you can do 2 reps with your hands on boxes, a full ROM. That is your 95% and the variation for the top zone. With your hands elevated on blocks that place your upper arms parallel to the deck you can do 5 reps—your choice for the middle zone. You can do more than 10 reps with your hands on the floor but when you elevate your hands on 2" blocks, you are down to 9. That is your bottom zone resistance.

You could roughly estimate your HARI, which is bound to be quite inaccurate.

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To stay with the above example:

	On 2" blocks	**Upper arms parallel to deck**	**Full ROM**	**HARI**
	9RM	*5RM*	*2RM*	
Est. % 1RM	*71% 1RM*	*85% 1RM*	*95% 1RM*	
Distribution	*60*	*30*	*10*	**77.6%**

Or you might choose to calculate your HARI in RM rather than % 1RM: average RM or "ARM". Athletes with average endurance with a given 1RM should aim for 6-10RM average; those with low endurance 5-8RM—approximately 75-85% 1RM of the target HARI.

The procedure is illustrated in the following example.

Regardless of the actual monthly NL, use 100 in the calculations.

	On 2" blocks	*Upper arms parallel to deck*	*Full ROM*	**ARM**
	9RM	**5RM**	**2RM**	
% and NL	*60*	*30*	*10*	
	60:9=6.7	*30:5=6*	*10:2=5*	
Total sets if taken to RM	*7x*	*6x*	*5x*	
	7+6+5=18			
	100 total reps : 18 sets = 5.6			**5.6RM**

For weighted pullups, dips, pistols, handstand pushups, etc. add the bodyweight and the kettlebell's weight to arrive at the 1RM.

E.g., a 220-pound athlete's tactical pullup 3RM is with a 24kg kettlebell. You may decide to use kilos or pounds. If the former, convert his bodyweight into kilos; if the latter, convert the kettlebells' weights into pounds.

220lbs. + 53lbs. = 273 x 3RM.
Estimate that 3RM=90% 1RM. (273 : 90) x 100 =303. Estimated 303 x 1RM.

Add the bodyweight to the weight of each available kettlebell and divide the result by 303 to arrive at % 1RM that kettlebell represents. E.g., for a 16kg kettlebell: 220 + 35 = 255. 255 : 303 = 84. When the athlete is doing pullups with 16kg, his is training at 84% 1RM.

Bodyweight (220) = 73%
12kg/26lbs. (246) = 81%—Optional. One weight per zone is usually enough.
16kg/35lbs. (255) = 84%
24kg/53 lbs. (273) = 90%
32kg/70 lbs. (290) = 96%
36kg/80 lbs. (300) = 99%

% 1RM	*73%*	*84%*	*90%*	*96%*	*99%*	**HARI**
	Bodyweight	**16kg**	**24kg**	**32kg**	**36kg**	
	60	*30*	*8*	*2*		**78.2%**

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Step V: Volume dynamics within a week

The more training days and sessions you fragment the load into, the greater the results. But since it is rarely practical, training a lift 3-4 days a week is standard.

For training 4 days a week the recommended schedule "R" (after Rodionov) is Monday, Wednesday, Friday, Saturday.

It is a standard practice to change the number of training sessions depending on the week's volume. If you are cutting back to three, get rid of the Saturday session. If you are adding a fifth, do it on Tuesday.

Another option for training 4 days a week is schedule "V" (after Vorobyev): Monday, Tuesday, Thursday, Friday. When cutting back to 3 times a week, eliminate either the Tuesday or the Thursday session. Add a fifth on Saturday on very high volume weeks.

If your SQ/DL volume is low, 2-3 sessions a week will suffice. Lift on Mondays and Fridays. On weeks with high volume add a third session on any other day but preferably Wednesday.

Recommended single lift weekly schedule "R", 4 days a week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
X	Add on high volume weeks.	X		X	X Remove on low volume weeks.	

Recommended single lift weekly schedule "V", 4 days a week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
X	X		X	X	Add on high volume weeks.	
	Remove either Tuesday or Thursday session on low volume weeks.					

Recommended single lift weekly schedule, 3 days a week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
X		X Remove on low volume weeks.		X	Add on high volume weeks.	

Recommended single lift weekly schedule, 2 days a week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
X				X		
On higher volume weeks add a session on any other day but, preferably, Wednesday.						

The above recommendations apply to a single lift. Stagger schedules for multiple lifts to train them together on some days and separately on others (a type of variability). Plan at least one day off lifting a week.

If we are to settle on the Monday, Wednesday, Friday, Saturday schedule for the press, consider some of the many possibilities with 3 SQ/DL sessions a week:

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	Mon	Tue	Wed	Thur	Fri	Sat	Sun
P	**X**		**X**		**X**	**X**	
SQ/DL		X	X			X	
	X		X		X		
		X			X	X	
		X		X	X		
		X		X		X	
	X	X			X		

When a given lift is trained two days in a row, keep the following in mind:

- If day one is low in volume and low to moderate in intensity, it improves performance on day two.

- If day one is low in volume and high in intensity, it improves performance on day two in a session of any volume and low to medium intensity.

- If day one is high in volume, a low volume/low intensity day two amplifies the anabolic its effect.

Although these three patterns are useful in planning, do not always abide by them for the sake of variability and because they reduce the internal load.

The next option increases the internal load:

- If on the first day one of the load variables, volume or intensity, is high and the other is medium, the second day's training will be conducted in a state of fatigue. The training effects of both sessions will be summated. It is an effective advanced tactic but it should not be used too frequently.

NL per class of exercise per day, ≥70% 1RM
(StrongFirst)

- **10-20**—minimal
- **20-30**—optimal
- **30-50**—maximal

If a lift's daily volume exceeds 30, separate it into 2-3 series separated by unrelated exercises.

E.g. 50 MPs can be split into 30+20, 20+15+15, etc. with SQ, DL, PU etc. in between.

You may also separate the daily load into 2-3 sessions with at least 30min of rest between them. The ratio of volume between two daily sessions is typically 1:3 or 1:2 and at times 1:1. Either the earlier or the later session may be higher in volume. The earlier offered guidelines on the effect of a training session on another one conducted the next day generally apply to the scenario with two sessions in one day.

Do not plan two days with twice-a-day training back to back. Ideally, take a day off after a day with two training sessions or at the most follow it up with a day with only one easy session.

Assign the week's volume to separate days using one of the following ratios. Use less variability in your first cycle. Stay with one distribution for a month, then change it.

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Distribution of a weekly volume by days, from least to most variable, %
(StrongFirst, based on a number of sources)

Training days per week			
2	**3**	**4**	**5**
40, 60 35, 65 30, 70 25, 75 20, 80	25, 33, 42 20, 35, 45 22, 28, 50 20, 30, 50 15, 35, 50 15, 30, 55	15, 22, 28, 35 10, 20, 30, 40 Or any other combination used in planning weeks	10, 15, 20, 25, 30

Note that the above numbers by no means imply that you must use them in that order. (Even though increasing the volume towards the end of the week is a common prep period tactic.)

Other combinations are possible if adjacent numbers with a change less than 20% are separated.

> *E.g., in Chernyak's 25, 30, 45% combination the first two numbers may not be adjacent: 30-25=5. (5:30)x100=17%. They need to be separated by 45%: 25-45-30 or 30-45-25.*
>
> *In the 15, 20, 30, 35% combination, if one to go from 35 to 30, the difference is only 14% (5:30=14). Separating 30% and 35% with either 15% or 20% eliminates the problem.*

When the number of training units exceeds four, variability is decreased as the change between some adjacent numbers drops below 20%. The answer is to separate these percentages.

> *E.g., insert any other number between 25% and 30% in the 10, 15, 20, 25, 30 five day line-up. If one to go from 30 to 25, the difference is only (5:35)x100=17%. Separating 30 and 25 with either 15 or 20 fixes the problem.*

As on a larger scale (weeks within a month), arranging days with varying volumes within a week in different order will have different training effects. Even though some arrangements appear more effective than others, all should be used for the sake of variability. And because some variants enable a higher external load (performance) while others maximize the internal load (stress).

> *E.g., you train three times a week on Mondays, Wednesdays, and Fridays. If you arrange the loads in the low-high-medium order, you will set yourself up for peak work capacity on the high volume day (a low volume session two days earlier has such effect).*
>
> *If you go low-medium-high, the primary tactic among Soviet lifters in the 1960s, you will do the opposite and maximize the stress produced by the high volume day, as you will enter it in a state of fatigue from the medium day. This arrangement also allows you to hit some heavy weights on Wednesday if the plan calls for it.*
>
> *High-low-medium, a tactic popular in the US, on the surface seems to be superior as it allows you to perform your peak volume after an extra day of rest on the weekend. Yet Soviet research showed that the peak work capacity is not in the beginning of the week but in the middle. Still, it is a valid arrangement.*
>
> *High-medium-light is a standard order for the competition week taper, rarely used in the prep period.*

You may use a different number of training days in different weeks of a given month, a standard practice in Soviet weightlifting.

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E.g., 4 days for the 35% week, 3 days for 22 and 28% weeks, and 2 days for the 15% week.

As an option, if your life is busy and complicated, just perform the week's volume on any days that are available. You may compliment or replace regular sessions with GTG.

E.g., you are travelling on Friday through Sunday. Fulfill all your volume on Monday, Tuesday, Wednesday, and Thursday.

E.g., this week you can only train on Wednesday and on Saturday and Sunday you are attending a seminar. Have a high volume session or two on Wednesday and "grease the groove" on breaks during the weekend seminar.

Maintain the same volume dynamics for one month. If you have a second prep month, you may change the dynamics and/or the training schedule or not.

Step VI: Intensity dynamics within a week

Intensity is varied independently from volume.

A standard preparatory period tactic, used together with an increase of volume towards the end of the week, is placing the heaviest lifts to the middle of the week after a low to medium volume and intensity session one or two days earlier. Stagger different lifts' highest intensity days.

Possible schedules for facilitating a high intensity mid-week session for a lift

<table>
<tr><th>Monday</th><th>Tuesday</th><th>Wednesday</th><th>Thursday</th><th>Friday</th><th>Saturday</th><th>Sunday</th></tr>
<tr><td>L</td><td></td><td>H</td><td colspan="4" rowspan="3">Increase the volume towards the end of the week.</td></tr>
<tr><td></td><td>L</td><td>H</td></tr>
<tr><td>L</td><td>H</td><td></td></tr>
<tr><td></td><td>L</td><td></td><td>H</td><td colspan="3" rowspan="2"></td></tr>
<tr><td></td><td></td><td>L</td><td>H</td></tr>
</table>

Do not keep the same average intensity in a given exercise from session to session; vary it by selecting different main weights. An effective tactic is selecting different main weights for each session in a week, except one. In that one session use variable weights, typically with weight ladders.

You may maintain the same intensity dynamics for a month (recommended, at least until you are experienced with the method) or change it from week to week.

Consider the following table that compares the training effects of different volume and intensity combinations and order in training 3 times a week on Mondays, Wednesdays, and Fridays. In the far right column is the overall load expressed in relative units. "Low" volume or intensity is assigned 1 point, "medium" 2 points, and "high" 3 points. For each training day the volume and intensity points are multiplied by each other. In the end of the week the sums are totaled to express the overall load of the given week.

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Volume	Intensity	Training effect	Overall load
L-M-H	L-H-M	Ideal for lifting heavy on Wednesday. Friday's load, a combination of high volume and medium intensity, is significant. A preferred choice.	1+6+6=13
	M-L-H	A hard choice, with both volume and intensity peaks falling on the same day. Very effective but to be used sparingly.	2+2+9=13
	L-M-H	Both volume and intensity go up towards the end of the week. Even harder than the last option; the most stressful combination.	1+4+9=14
	H-M-L	Volume and intensity are at counter phases. A common tactic in the West, it trades volume for intensity and vice versa. Although seems "logical", it reduces the internal load. Use sparingly.	3+4+3=10
	M-H-L	Some facilitation of high intensity Wednesday. Friday's load is at counter phases, hence least stressful. Similar to H-M-L in its effect.	2+6+3=11
	H-L-M	No obvious relationship between volume and intensity and hence a little more variability than the last choice.	3+2+6=11

"Low", "medium", and "high" intensity, in the context of the given methodology, refer to the 70-80, 81-90, and 91-100% zones respectively. That means you will be emphasizing one zone in each session.

When distributing top zone lifts between days, generally do not exceed:

- Presses and pullups: NL 7 @ 90% or NL 3-4 @ 95-100%
- Squats and deadlifts: NL 3 @ 90% or NL 2 @ 95-100%

If you are working with two or more weights in a given zone, distribute them between days unequally.

Consider a weekly pullup plan:

% 1RM	77%	89%	95%	100%	Total
	Bodyweight	16kg	24kg	32kg	
NL	73	37	2		112

You decided to train 3 times a week, used low volume variability from day to day (20, 35, 45%) and arrived at NL 23, 39, 50. (You would have been better off training 4 times a week but it is not in the cards, given your busy life.) You decided to organize your volume in a L-M-H order: 23, 39, 50 on Monday, Wednesday, and Friday respectively. You have chosen the L-H-M order for intensity:

	Monday	Wednesday	Friday
NL	23	39	50
Main zone	70-80%	91-100%	81-90%
Weight, %1RM	Bodyweight (77%)	16kg (89%)	24, 32kg (95, 100%)

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Start with assigning the top intensity—only 2 singles. Obviously they go to Wednesday.

Friday takes 50 total lifts and emphasizes the 81-90% zone. You have 37 such lifts in your budget. You choose not to assign all of them to Friday for three reasons.

First, you need a couple of 81-90% lifts for ramping up to max and near-max lifts on Wednesday. Second, your kettlebell selection, at 89% 1RM, places you on the very top of the zone. Even though the pullup is a lot less demanding than the squat, 37 reps at close to 90% is an excessive load. Third, Wednesday is supposed to be the highest intensity day. But with only 2 reps in the top zone you need a lot of help from the middle zone to bump up the average. If you divide the 81-90% about equally between Wednesday and Friday, you will get a lot higher average intensity on Wednesday because that day's volume is lower. Since you know that dividing equally is frowned upon, you go with a ratio ranging between 2:3 and 1:4.

Since you want to keep the NL on both days similar, go with a minimal 40:60 ratio—NL 15+22.

Now all you have to do is write these numbers down and fill in the rest with 70-80% lifts:

	Monday	**Wednesday**	**Friday**
Total	**23**	**39**	**50**
91-100%	---	2	---
81-90%	---	15	22
Subtract the NL in the top 2 zones from the total and arrive at the 70-80% NL:			
70-80%	23	22	28

What if experience tells you that you do not handle many heavy lifts in one day?—Go with less variability and spread the 89% 1RM lifts across the 3 days. Use one of the ratios earlier recommended for dividing weekly volume between days. If you pick 20, 30, 50%, NL 37 will break down into 7, 11, 19, which you will find much more manageable (the most aggressive choice is on the bottom of the table; the most gradual on the top).

	Monday	**Wednesday**	**Friday**
Total	**23**	**39**	**50**
91-100%	---	2	---
81-90%	7	11	19
Subtract the NL in the top 2 zones from the total and arrive at the 70-80% NL:			
70-80%	16	26	31

Your week has been planned.

Step VII: Load in a session

The load is typically organized as a pyramid or a half-pyramid. Flexible rep and weight ladders are built in to wave the effort throughout a session.

Use the repetition ranges recommended below with a bias towards the optimal number in the middle. The same lifter may use one table for some lifts and the other for others.

Vary the reps from set to set.

It is better to do 4, 6, 4, 6 that 5, 5, 5, 5.

When your rep choices do not allow you to exactly fill the day's volume with a given weight, use lower or higher reps then specified in the tables.

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Reps per set for lifters with average to high endurance with a given RM
(StrongFirst, based on a number of sources)

% 1RM	70%	75%	80%	85%	90%	95%
RM (Rodionov, 1976), ≥	12RM	10RM	8RM	6RM	4RM	≥2RM
Optimal	6	5	4	3	2	1
Range	**4-8**	**3-6**	**3-5**	**2-4**	**1-2**	**1**
Warm-ups; comp week	2-3	1-2		1		

Reps per set for lifters with low endurance with a given RM
(StrongFirst, based on a number of sources)

% 1RM	70%	75%	80%	85%	90%	95%
RM (Matveev, 1991, lowest values)	10RM	8RM	6RM	5RM	3RM	2RM
Optimal	5	4	3	2	1-2	1
Range	**3-6**	**3-5**	**2-4**	**2-3**	**1-2**	**1**
Warm-ups; comp week	1-3	1-2		1		

To summarize: **in the 70-90% 1RM zones one should do 1/3-2/3 of RM**. (StrongFirst)

We can speculate that there is a "law of medium reps" similar to the law of medium loads. A low number of reps, relative to the 1RM, prolongs the training session and probably does not provide enough of a training stimulus. When reps get too close to the RM, fatigue quickly sets in and prevents one from fulfilling a high volume of training while compromising technique and safety.

Individualize the reps per set keeping the same volume and intensity. (Chernyak, 1978). You may choose to use the first table for some lifts and the second for others.

E.g., in squats athletes can do more reps at a given RM than in presses. (Rodionov, 1976)

Before your planned 70% and up lifts perform joint mobility exercises followed by a ramp-up with lighter weights, e.g., 50% x 4, 60% x 3 (if the given exercise makes it possible). These requirements are individual.

Keep ramping up with 70% and higher weights until you quickly reach the heaviest weight for the day. You may choose to do more than one set with a warm-up weight.

If a second series or session is performed on the same day, shorten the warm-up according to your individual needs.

Let us stay with the above pullup example. Our athlete is "fast twitch" and uses the bottom table.

	Monday	***Wednesday***	***Friday***
Total	***23***	***39***	***50***
24, 32kg (95, 100%)	---	*2*	---
16kg (89%)	---	*15*	*22*
Bodyweight (77%)	*23*	*22*	*28*
Ramping up to the session's heaviest weight	*B (bodyweight) x 3…*	*B x 2, 3, 2* *16kg x 1/2 (reps/sets)* *24kg x …*	*B x 4* *16kg x …*

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Once you have worked up to the top weight of the day you can:

a) Stay with this weight until you have completed all the reps with it and then perform the remaining lighter sets if they are listed (the most common tactic);

	Monday	***Wednesday***	***Friday***
Total	***23***	***34***	***50***
24, 32kg (95, 100%)	*—*	*2*	*—*
16kg (89%)	*—*	*15*	*22*
Bodyweight (77%)	*23*	*22*	*28*
	B x 3, 4, 5, 3, 5, 3	*Session or series # 1* *B x 2, 3, 2* *16kg x 1/2* *24kg x 1* *32kg x 1* *16kg x 2* *#2* *B x 3* *(16kg x 1, 2) x 3* *B x 3, 5, 4*	*B x 3, 4* *16kg x 1, 2, 1, 2, 1, 2, 1, 2, 1, 2, 1, 2, 1, 2, 1* *B x 5, 3, 4, 3*
		When a day's volume exceeds 30, divide it either into 2 series separated by a non-related exercise or into 2 sessions separated by at least 30min.	

b) Sandwich some lighter sets between heavier sets. You can alternate sets of different weights, alternate series of several sets, or use a combination of both.

Examples of how Friday's load can organized according to option B:

Session or series #1 *(B x 3, 16kg x 1) x 2* *(B x 4, 16kg x 2) x 2* *(B x 5, 16kg x 2) x 2* *#2* *B x 4* *(16kg x 1, 2) x 4*	*Session or series # 1* *(B x 3, 16kg x 1, 2) x 3* *#2* *B x 3, 4, 3* *(16kg x 1, 2) x 2* *B x 4, 5* *16kg x 1, 2, 1, 2, 1*

Use option A at least twice as often as option B.

c) An option to exercise even less frequently is to do the heaviest lifts after a fatiguing series of lighter lifts. Back-off sets afterwards are optional.

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E.g.,

B (77%) x 2, 3, 4, 2, 3, 4
16kg (89%) x 1, 2, 1
24kg (95%) x 1
B (77%) x 3, 4, 3

Rest as long as you need between sets. First walk and perform "fast and loose" drills, then sit down.

If after your ramp-up set you realize that you are not going to make the day's plan, adjust the load according to these guidelines:

- In the preparatory period lower the intensity and maintain the volume;
- In the competition period lower the volume and maintain the intensity.

You may also choose to stop for the day and make another go at it tomorrow.

Note that this problem is rare with the given methodology, given that the reps are very conservative compared to the RM. If the cycle has been laid out correctly and you are having difficulty on a given day, most likely you are not taking enough rest between sets. Oftentimes taking extra rest between sets, 5min and longer, will enable you to fulfill even a challenging workload on a day you are not feeling your best.

If, on the other hand, you are in the middle of a preparatory cycle and you feel that you have gotten a lot stronger, reduce the intensity and volume for two sessions, then max—emulate a competition week. Estimating your new max is another option for an experienced lifter. Next week recalibrate the load to match your planned average intensity using the new 1RM in your calculations.

E.g., your kettlebell military press 1RM in the beginning of a preparatory month was 44kg and you planned your intensity in this manner:

73%	*82%*	*91%*	*100%*	***HARI***
32kg	***36kg***	***40kg***	***44kg***	
40	*40*	*16*	*4*	***80.6%***

Suddenly on the second week you put up an easy double with 44kg. Estimating this to be 95% of your 1RM, you arrive at 46kg as your estimated 1RM. Suddenly your distribution averages out only 76.9% (compared to the 80.6% target). Still within acceptable parameters—but far off the plan:

70%	*78%*	*87%*	*95%*	***HARI***
32kg	***36kg***	***40kg***	***44kg***	
40	*40*	*16*	*4*	***76.9%***

You play with the numbers and arrive at the planned HARI:

70%	*78%*	*87%*	*95%*	***HARI***
32kg	***36kg***	***40kg***	***44kg***	
20	*40*	*33*	*7*	***80.6%***

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Note how significant the changes had to be: more than doubling the number of 40kg lifts and almost doubling the number of 44kg lifts—while halving the 32kg NL.

Step VIII: Interaction of different lifts within a week and within a session

You are training 2 or 3 lift categories:

4. Squat/deadlift
5. Press
6. Pullup (optional)

The volume and intensity in all classes of lifts are independent of each other. Use different volume and intensity dynamics within a week and within a session for different lifts.

All combinations of the 2-3 main lifts are possible in a session—1, 2, or 3 of them. Vary these combinations.

The above lifts can be performed in any order in a session, the only exception being no bench presses after a high volume of pullups. Even though some sequences are more effective than others, all should be used for the sake of variability.

Alternating between harder and easier exercises is advisable.

Presses and pullups may be alternated every set or every several sets (series)—unless your press is the bench press. If it is, do all your BPs before starting pullups, otherwise fatigue in the upper back and shoulder stabilizing muscles will make it difficult to keep your shoulders depressed and retracted in the bench press.

An effective tactic is alternating one or two sets of an exercise done for a low volume on the given day with a series of several sets of a high volume exercise.

E.g., (MP 32kg x 2, 3, 5; PU 32kg x 2, 3) x 3

Plan to emphasize different intensity zones for different lifts in the same session; this not only adds variability but also increases the work capacity.

Regarding interaction of lower and upper body exercises on two consecutive days:

- A low to medium volume with medium to high intensity on day one for the upper body improves performance for the lower body on day two and vice versa.

- A high volume lower body session on day one will have a negative effect on upper body performance on day two. However, if the upper body session is high in volume and hypertrophy oriented, it will benefit from the anabolic hormones released by the lower body session.

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Step IX: Programming assistance exercises. Training more than one main lift prioritizing the same muscle groups.

In the Plan Strong™ methodology the only assistance exercises used are multi-joint exercises that respond to the "squat/hinge", "press", and "pullup" descriptions and preferably those that fit the specialized variety (SV) definition. (Local single joint exercises may be added at the discretion of an experienced coach.)

The less experienced the lifter, the fewer assistance exercises he or she needs and the lower should be their share of the total volume. In your first couple of cycles you are advised to stick to one variation per lift category and no assistance.

Then move on to 70-80% share of the main lift in the NL for the given exercise category (squat/hinge, press, pullup). Eventually you may advance to 50% if your main lift's technique is stellar.

But before you add SV exercises consider that Falameev (1974) warned not to add pulls from blocks until one reached the MS level and Sheyko (2005) does not recommend using a deadlift style opposite to one's competition style (conventional vs. sumo) at all.

Reduce the number and share of assistance exercises in the comp month, especially the last two weeks.

> *E.g., the DL is your only main squat/hinge lift. Treat all squats as assistance for it. Assign 50-80% of your squat/hinge volume to your competition style deadlift—and the remaining 20-50% to various SQs, GMs, and other types of DLs.*
>
> *E.g., the handstand pushup is your only main press. Assign 50-80% of your press volume to the HSP—and the remaining 20-50% to kettlebell and barbell MPs, close grip barbell BPs, etc.*

Change the ratio of the main lift to specialized variety every month, e.g., 70:30, 80:20, 75:25 if we stay in the range recommended to intermediate lifters.

In the beginning use only 1-2 assistance exercises per lift. In the future you may increase (possibly greatly) and individualize that number. Greater exercise variety enables an advanced lifter to further increase the volume due to the phenomenon of fatigue specificity.

Change the assistance exercise(s) every month.

In your first several cycles of Plan Strong™ training mostly—or even totally—restrict your assistance exercises to the 70-80% intensity zone.

> *As an example, consider the numbers of 30 elite bench press specialists (based on Belsky, 2000):*

Period	***Lifts***	***70-80%***	***81-90%***	***91-100%***	***HARI***
Preparatory	*Competition and accessory*	*55*	*44*	*1*	***79.6%***
	Competition only	*60*	*32*	*8*	***79.8%***
	Accessory only	*58*	*42*	*—*	***79.2%***
Competition	*Competition and accessory*	*65*	*29*	*6*	***79.1%***
	Competition only	*31*	*37*	*32*	***85.1%***
	Accessory only	*71*	*27*	*2*	***78.4%***

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There are two methods of measuring the relative intensity in assistance exercises. The first method uses the main lift's max as the reference point. The second method measures the intensity as a percentage of the max in the given exercise. If the max—or estimated max—in a given specialized variety lift is within 5% of that of the main lift, use the first method. Otherwise use the second method.

E.g., your main lift is the DL (500 x 1RM) and the Zercher squat (350x1RM) is an assistance lift. You have assigned 60% of the NL to the DL and 40% to the ZSQ and are a HARI below 80%.

You identify a limited number of weights in the three high intensity zones for the deadlift and in the first (and possibly second) zone for the squat. If you stick with 45-pound plates and one set of quarters, you have two choices for the ZSQ: 275 (78%) and 315 (90%). You remember that you have dedicated 40% of the NL to the squat. You can either assign all this volume to 275 or give a small share to 315. You decide to assign the former 35% and the latter 5% of the hinge/squat total. You play around with numbers and arrive at:

	1RM	***73%***	***78%***	***81%***	***90%***	***91%***	***HARI***
DL	***500***	*365*		*405*		*455*	
ZSQ	***350***		***275***		***315***		
		25	***35***	*30*	***5***	*5*	***78.9%***

If you feel that you have gotten stronger in an assistance exercise, go for a 1-3RM in it in order to recalibrate the loads. Select a week and a day when you are relatively fresh. To stay with the planned distribution of the NL by intensity zones, max on a day when top zone NL are planned for the main lift and do it at the expense of these lifts.

To revisit the above DL and ZSQ example, you are feeling a lot stronger in the Zercher. You are waiting for the next session that has several 455 pound 91% DL singles:

DL 365x2, 3; 405x2; 455x1/2; 405x3, 4; ZSQ 275x3, 4 (NL 23)

…and instead do:

ZSQ 275x1, 2, 1; 315x1; 335x1, 355x1, 360x1RM… (NL 8 so far)

…You have more that used up your top zone allowance—4 lifts from 90% and up instead of the planned 2. You could go to the trouble of calculating exactly what you need to do now to match the HARI of the given session—or you just play it by ear and do the rest or most of the remaining lifts with the lightest, 73%, weight available:

…DL 365x3, 5, 3, 4 (total NL 23) or …DL 365x3, 405x2, 365x3, 4, 3 (total NL 23)

Recalculate the % 1RM for 275 and 315 pounds in the ZSQ. Both dropped 2% 1RM: from 78% to 76% and 90% to 88% respectively. Reorganize the furniture to stay with the target 78.9% HARI. (If you do not, you are down 0.8% 1RM to 78.1%; a very significant difference given a narrow range of intensity fluctuation).

Move some lifts from the lighter ZSQ weight to the heavier one:

	1RM	***73%***	***76%***	***81%***	***88%***	***91%***	***HARI***
DL	***500***	*365*		*405*		*455*	
ZSQ	***360***		*275*		*315*		
		25	***28***	*30*	***12***	*5*	***78.9%***

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As it turns out, you had to more than double the number of 315 ZSQs to keep the planned HARI. This is an illustration of why one should not be lazy to update the calculations after a PR to stay on target.

Assign your assistance lifts to back-off sets and/or to separate session(s).

Let us return to the tactical pullup example. The coach has selected a 60:40 ratio of tactical pullups and other pullup variations. Given the weekly NL of 112, the athlete needs to do 45 reps with pullups on rings, parallel grip pullups, wider than usual tactical pullups, etc.

	Monday	***Wednesday***	***Friday***
Total	***23***	***34***	***50***
24, 32kg (95, 100%)	*—*	*2*	*—*
16kg (89%)	*—*	*15*	*22*
Bodyweight (77%)	*23*	*22*	*28*
	B x 3, 4, 5, 3, 5, 3	*Session or series # 1* *B x 2, 3, 2* *16kg x 1/2* *24kg x 1* *32kg x 1* *16kg x 2* *#2* *B x 3* *(16kg x 1, 2) x 3* *B x 3, 5, 4*	*Session or series # 1* *B x 4* *(16kg x 1, 2) x 4* *#2* *B x 3* *16kg x 1, 2, 1, 2, 1, 2, 1* *B x 5, 3, 4, 5, 4*

Monday's session is all in the bottom intensity zone, so he can dedicate totally it to SV. E.g., pullups on rings with a pause on the top. 23 reps down; 22 to go.

One option is to assign the 22 SV reps to the last series on Friday (bold): B x 5, 3, 4, 5, 4. That makes it one rep short. You could move one rep from the very first set on Friday's first series to the last series—or just not worry about it and be content with 44 SV reps instead of 45. A very small difference.

Another option is to dedicate some of the variety to heavier, 16kg, pullups. E.g., you could assign the entire second series on Wednesday to SV. The total is NL 24, 2 more than needed. One option is to change one of the doubles with 16kg to the classic tactical pullup; another is to be content with 24, slightly more than the plan.

Exercises that promote skill development of the main lift may also be inserted into the ramp-up to heavy weights.

E.g., the behind the back deadlift not only improves your leg drive but teaches you how to better use it in the conventional deadlift. Your DL max is 500 and behind the back DL max is 450. You do:

BBDL 315x3 (70%), 365x2, 3 (81%); classic DL 425x1 (85%), 445x1, 2, 1 (89%)

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The above tactics apply a simple scenario when an athlete has only one main lift in a category. Most athletes who are not competitive lifters are advised to exercise this simple choice.

If you are a powerlifter who has two competition lifts using the same muscle groups, the squat and the deadlift, use the same tactics as in planning SV exercises, with the exception of limiting the second exercise to the 70-80% zone. Remember to change the SQ/DL ratio (partial loads) from month to month, at times heavily biasing one or the other. Serial specialization is a very effective advanced strategy: push one lift hard while maintaining the other, then flip.

In the following example using all five intensity zones the athlete treats both the bench press and the barbell military press as main lifts and does not limit either to 70-80% intensity.

Prep	**450**							
	Week 1		**Week 2**		**Week 3**		**Week 4**	
2c	22%		35%		28%		15%	
NBL	99		157		126		68	
BP:MP ratio 65:35 (partial loads)								
	BP	MP	BP	MP	BP	MP	BP	MP
	64	35	102	55	82	44	44	24
BP+MP 81-90% (NBL 68)								
	17		11		20		20	
	Out of that BP NBL 44							
4-2	22%		15%		28%		35%	
	10		7		12		15	
	Out of that MP NBL 24							
3-1	28%		15%		35%		22%	
	7		4		8		5	

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COMPETITION PERIOD

Unlike the traditional Soviet methodology, Plan Strong™ includes the competition day load into the calculations and keeps the preparatory period nomenclature for naming the weeks in an ascending order rather than using a countdown to the competition week.

Step I: Monthly volume

As a rule of thumb, reduce the volume in the competition month by 20-50% (typically around 30%) compared to the last prep month.

Let us map out a competition month for an athlete with a 40kg x 1RM kettlebell military press. Last month's volume was 320. You decide on a 25% reduction, 320 to 240.

Step II: Volume dynamics within a competition month

In the competition month the volume is tapered. Depending on the last preparatory month's volume, one may taper from the very first week or later:

Variant	Last month's volume	Deload from week #
1	Very high	1
2	High	2
1-3	Moderate	3
3, 3-1	Low (insufficient)	4

The NL corresponding to "very high", "moderate", etc. are relative and individual.

As a rule of thumb, use the classic 15-22-28-35% constants:

Variant	Week 1	Week 2	Week 3	Week 4
1	35	28	22	15
2b	28	35	22	15
2c	22	35	28	15
3b	22	28	35	15
1-3b	35	22	28	15
3-1b	28	22	35	15

For the following reasons you may choose different percentages:

- When the percentages are converted into NL, it turns out that they are too close to the NL in the prep month's last week. There must be at least a 20% difference.
- You are looking for more or less variability than offered by the 15, 22, 28, 35% constants.

An example of greater variability is 40-27-20-13. An example of lesser variability is 17-21-26-27%.

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- You are set on a very particular competition week's NL and it does not convert to the classic 15%. This is typical for experienced competitive powerlifters. The solution is to first plan the competition week, then convert its NL into % of the monthly total.

To develop your own variant or tweak an existing one use the procedure described in the preparatory period section and refer to the table entitled "Guidelines for designing a custom variant of volume distribution among weeks".

Note that all weeks do not have to fall into the same category ("standard", "more variable", "less variable"); your variant may be a combination.

E.g., Roman (1968): 34-28-21-17—standard-standard-more-less.
E.g., Roman (1986): 29-25-35-11—more-less-standard-more.

First, select the "bookends", the weeks with the lowest and the highest volume, 1 and 4, then fill in the medium weeks, 2 and 3.

Back to our athlete with a 40kg press.

In his last prep month the athlete did the following:

Prep	***NL 320***			
	Week 1	***Week 2***	***Week 3***	***Week 4***
3c	*15%*	*28%*	*35%*	*22%*
NL	*48*	*90*	*112*	*70*

You consider the last month's NL of 320 "high", so you choose variant 2 for volume dynamics.

You calculate the NL range for the first week's volume (at least 20% difference from that of the last week of the prep month). The last week's NL was 70. 70x0.8=56. 70x1.2=84. Convert these NL into % of the month's total:

(56:240) x 100 = 23%.
(84:240) x100 = 35%.

Hence the first week's volume must be ≤23% or ≥35%.

≥35% demands abandoning variant 2 (there is no going up after 35%). ≤23 translates into 2c: 22-35-28-15:

Comp	***NL 240***			
	Week 5	***Week 6***	***Week 7***	***Week 8***
2c	*22%*	*35%*	*28%*	*15%*
NL	*53*	*84*	*67*	*36*

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Step III: Monthly intensity (91-100% NL, distribution of the volume by intensity zones, HARI)

Compared to the last prep month, increase the HARI by at least 1%. Change the share each weight contributes to the total.

Different combinations of percentages may lead to the same HARI, e.g.:

71-80%	***81-90%***	***91-100%***	***HARI***
60	*40*	*0*	***79%***
65	*30*	*5*	***79%***

Increase the NL in the 91-100% zone. A 2:3 ratio of 90-100% NL between the preparatory and the competition months is standard. (Note that this refers to the number of lifts; not their share of the total.)

The number of 91-100% lifts varies greatly between lifts and between lifters. E.g., a wide range of NL 3-25 is recommended for the press. A long period of trial and error is needed to optimize your numbers. Use the following as the starting point:

- NL 2-5 in the first two weeks of the comp period; 0-3 in the third.

A strong bias toward the bottom of the 91-100% zone is advised. As a rule of thumb, assign at least 2/3 of the zone's NL to 90-92.5% weights.

You might need to adjust the number of lifts in the top zone after mapping out the comp week.

First identify the number of lifts in the top zone. Use the recommended 2:3 ratio between prep and comp months' volume in the top zone as the starting point. In his last prep month the same athlete with 40kg x 1RM kettlebell military press did the following:

% 1RM	*70%*	*80%*	*90%*	*100%*	***HARI***
NL 320	***28kg***	***32kg***	***36kg***	***40kg***	
Prep	*38%*	*50%*	*12%*	*–*	***77.4%***
			NL 40		

Multiple 40 by 1.5 and arrive at NL 60 for the comp month.

Next convert this NL into %. Your planned comp month's NL is 240. 60:240=25. 25% is beyond even the expanded range that tops out at 20% (although there are some outliers in the thirties, we do not recommend emulating exceptions to the rule). Try 20%: 240 x 0.2 = 48. It will do—different enough from the last month.

Here is what we have so far:

90%	*100%*	*110%*
36kg	***40kg***	***44kg***
20%		
NL 48		

Based on your experience, you assign the top zone NL to particular weights, then convert the NL into % of the 240 monthly volume:

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	90%	100%	110%
	36kg	**40kg**	**44kg**
NL	48		
	37	10	1
%	15	5	–

You remember the guideline to raise the HARI by at least 1% RM compared to the last prep month. It should be easy, given that you have upped the share of the 90% and heavier lifts from 12% to 20%:

% 1RM	70%	80%	90%	100%	110%	**HARI**
	28kg	**32kg**	**36kg**	**40kg**	**44kg**	
Prep	38	50	12	—	—	**77.4%**
Comp	40	40	15	5	—	**78.5%**

Adding a small number of overloads, e.g. 2-3 singles, of 105-120% 1RM—not heavier!—is recommended for all lifts in the comp period. Examples of overloads are:

- Loaded clean for the military press
- Loaded hang for the pullup
- Deadlift lockout
- Squat walk-out
- Bench press support

E.g., how to use loaded cleans when ramping up for a 1RM test: 70% x 3/3, 80% x 2/2, 90% x 1, old 100% x 1, loaded clean with 110% your old 1RM x 2/2, new 1RM.

Do not count the overloads as a part of the total volume.

Professionally assisted singles with current and target maxes are recommended for lifts in which it is practical, e.g., the kettlebell military press, the bench press, and the weighted pullup. (See SFGII instructor manual for more information about ACE or "artificial controlling environment".) The assistant must provide enough help to make it a 90-95% effort, no more and no less. Help must be given only through the sticking point and in a manner that does not alter the groove and does not get in the lifter's way. E.g., for the single kettlebell military press the assistant stands slightly behind the athlete and the kettlebell. For the weighted pullup he presses up at a forty-five degree against the athlete's mid-back. The athlete should not psych up for the effort. The assistant keeping his hands on the bell after the clean and on the athlete's back from the very beginning of a pullup communicates "I have got your back" and reduces the athlete's excitation.

1-2 ACE singles are typical. Log them as 95% lifts.

Assisted lifts may also be sparingly used in the prep period, usually with one's current max. This enables the athlete to "own" the weight.

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Step IV: Intensity dynamics within a competition month

Unlike the prep period, in which there is no correlation between the volume dynamics and the intensity dynamics, in the comp period in 3/4 of cases match an increase or decrease in volume by same in intensity.

In 3/4 of cases use intensity variants 1, 2, 1-3, 3-1. In other words, assign the heavier, >80% 1RM, lifts to weeks according these variants. (There may be a slight intensity spike in the competition week because, unlike the traditional method, we include the competition day's load into the calculations.)

For introducing variability while maintaining the direction of the load, assign the 91-100% weights in the following manner.

Assign the desired number of 91-100% lifts NL to the comp week (usually NL 3-4), then distribute the remaining NL between the first three weeks using one of these familiar sets of constants:

Distributing the volume between 3 training units,
from the least to the most variable, %

25, 33, 42
20, 35, 45
22, 28, 50
20, 30, 50
15, 35, 50
15, 30, 55

Sometimes, especially in the deadlift, there will be zero lifts in the top zone during one and even two weeks of the comp month. If such lifts are present only in two weeks, simply subtract the NL in the comp week from the total to arrive at the other week's NL. Use one of the following ratios:

Distributing the volume between 2 training units,
from the least to the most variable, %

40, 60
35, 65
30, 70
25, 75
20, 80

When assigning the remaining lifts to the first three weeks, the easiest thing to do is to match the dynamics of the main variant's first three weeks.

> *E.g., you have planned NL 10 in the top zone for the comp month. You set aside NL 3 for the comp week and distribute the remaining NL 7 between the first three weeks according to the 20, 30, 50%: NL 1, 2, 4.*
>
> *Your main variant is 2b. Disregard the last week, 28-35-22-~~15~~, and note the dynamics between the first three weeks: the highest number is on week two, followed by week one, and finally week three. Use the same pattern to assign the top zone lifts: 2, 4, 1.*

An experienced coach may be more surgical about the heavy lifts' dynamics.

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To stay with the above example, you know that you tend to burn out hitting top zone weights more than three weeks in a row, so you decide to keep the first week free and to spread the 7 heavy lifts across weeks two and three, 5 and 2, a 70:30 ratio. (This turns out to be variant 2c.)

One tool that allows introducing more variability while still maintaining the direction of the loads demanded by the competition period is assign the top zone lifts according to a variant similar to the main variant. Disregard the lifts in week four; just match the dynamics of the first three weeks with the chosen variant:

Main variant	Similar variants
1	2b, 2c, 1-3b
2b	1, 1-3b, 2c
2c	1, 1-3b, 2b
3b	2b, 2c, 3-1b
1-3b	1, 2b, 2c
3-1b	2b, 2c, 3b

Let us stay with the same scenario of NL 10 in the top zone for the comp month, out of that number NL 3 in the comp week and the remaining NL 7 distributed between the first three weeks according to the 20, 30, 50% ratios: NL 1, 2, 4.

The main variant is 2b. You know that the given athlete benefits from a spike of intensity in the week before the meet and chooses variant 1-3b for the top zone lifts.

The classic 1-3b, which means that the highest number is in week three, the second highest in week two, and, the third in week two. With the given NL in translates into 4-1b.

Within the 91-100% lifts focus on the max and near-max lifts. First, write down the 2-3 attempts on the test day. Then distribute the remaining ≥95% lifts across the first three weeks.

In spite of their high intensity, these lifts will not be decisive in determining the month's intensity dynamics because of their low number. You should be guided by your experience in how each of your lifts responds to heavy singles a given number of weeks away from the test day.

These dynamics are lift specific and lifter specific and take trial and error to optimize. Some tactics to consider:

- MP, BP, PU: week 1—90%, week 2—95-100%, week 3—90%
- SQ, DL: week 0 (last week of the prep period)—95-100%, week 2—90%
- Any lift: week 2—90%, week 3—95%

The questions you need to answer over a period of multiple cycles are, for each lift:

- Do 95-100% lifts help or hurt the athlete? If they help, how many weeks from the meet should they be done?

- Does the athlete benefit from 90% lifts every week? If yes, how many?

- What is the optimal number of 90% lifts in week three? (It can be zero, especially for squats and deadlifts.)

- In the military press and pullup, does the athlete benefit from one or two 90% singles on Monday of week four?

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Next distribute the 81-90% weights.

They are to be underrepresented in the comp week when the emphasis is on easy rehearsals and warm-ups with 70-80%.

For 81-90% assign 10% (or, once you are experienced in writing these plans, surgically select the NL in this zone for the comp week and convert it into a percentage) to the comp week and distribute the remaining lifts between the first three weeks using one of the above sets of three percentages. Match the variant used for the total volume. (Using a similar variant, as for 91-100%, should be considered only by experienced coaches, as it is easy to run out of space.)

Back to the 40kg presser. Time to work out the intensity dynamics: assign the 48 lifts with 90% and heavier weights by week.

First assign the lifts for the comp week. Based on your experience, you decide to do a 90% single early in the week. Then you map out your test day. You need to press 90% only once when working up to your max, then take your old max for a single, and go for a 44kg PR. That nets NL 4 for the entire week.

Distribute the remaining 44 lifts between the other three weeks. You choose an aggressive set of constants, 15-35-50% and break up the NL 44 into 7, 15, 22.

In week one you want some contrast with the last prep week that had 11 reps with 90%. You conclude that any of your three numbers fit the bill.

For your 90% and heavier lifts you choose variant 1. Selecting 1 next to volume's 2c maintains the general load direction while introducing some variability. In addition, the algorithm used in distribution assures that the percentages are different. (If you were to reverse-engineer the percentages from the NL 48 of 90% and heavier lifts, you would get 46-31-15-8%.)

	NL 240			
	Week 5	**Week 6**	**Week 7**	**Week 8**
2c	*22%*	*35%*	*28%*	*15%*
NL	*53*	*84*	*67*	*36*
	≥ 90% (NL 48)			
1				
NL	*22*	*15*	*7*	*4*

Next assign the 100% and heavier lifts (11 and a part of the above NL 48). The comp week is already settled: 40kg x 1, 44kg x 1. 9 lifts with 40kg remain to be spread across the first three weeks.

Use your experience to decide whether lifting your old 1RM a week before the test is going to help you or hurt you. You think would be better off without it. That means you have to assign the 9 lifts to the first two weeks.

Although NL 3 and 6 would make for better variability, 6 lifts with your old max in one week are a lot, so 4 and 5 it is. You can assign 4 to week one and 5 to week two or vice versa; either will work.

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	NL 240			
	Week 5	**Week 6**	**Week 7**	**Week 8**
2c	22%	35%	28%	15%
NL	53	84	67	36
	≥ 90% (NL 48)			
I				
NL	22	15	7	4
	Out of that number 100-110% (NL 11)			
NL	4	5	—	2

Note that the heaviest lifts' NL are staggered to the right; it is done to prevent accidentally counting them twice when totaling all the numbers—these lifts are counted as a part of the ≥ 90% lifts.

Time to assign the 80% lifts using the same variant as the main one, 2c, but different percentages.

The comp week gets 10% of the total (NL 10); the remaining NL 86 are assigned among the first three weeks using one of these sets of percentages:

25, 33, 42
20, 35, 45
22, 28, 50
20, 30, 50
15, 35, 50
15, 30, 55

You choose a non-aggressive version, given a high number of 80% lifts (a highly variable version would overwhelm one week with 80% lifts). 25, 30, 45% translate to NL 21, 26, 39. Hence the plan:

	NL 240			
	Week 5	**Week 6**	**Week 7**	**Week 8**
2c	22%	35%	28%	15%
NL	53	84	67	36
	≥ 90% (NL 48)			
I				
NL	22	15	7	4
	Out of that number 100-110% (NL 11)			
NL	4	5	—	2
	80% (NL 96)			
I				
NL	39	26	21	10

(If you were to reverse engineer the percentages in all four weeks for 80% weights, you would get 41-27-22-10%. Similar yet different enough from the 90%+ dynamics of 46-31-15-8%. And substantially different from the total volume dynamics. You could have chosen different variants for 80% and 90%+ weights as well, e.g. 1 for one and 2b for the other.)

The above is a very challenging load layout for an experienced athlete who can handle major load swings. Week two is a killer, with the highest volume, a lot of lifts in the top zone, and more than its fair share of 80% lifts. An athlete less capable to handle load variability might use the

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opposite tactic and "smooth out" the load by giving the second week less than its fair share. He could do it by shuffling the percentages or by opting for a different variant, say 1.

Now subtract all ≥80% 1RM lifts from the total and you will arrive at the NL of 70% 1RM lifts. Check to make sure the totals in all rows and columns add up.

	NL 240				**Totals for control**
	Week 5	**Week 6**	**Week 7**	**Week 8**	
Total	**53**	**84**	**67**	**36**	**240**
≥90%	22	15	7	4	**48**
80%	19	36	27	14	**96**
Calculate 70%	53-22-19=12	84-15-36=33	67-7-27=33	36-4-14=18	
70%	12	33	33	18	**96**

If you have more than one weight selection in the 81-90% zone, use the same variant but preferably different percentages.

E.g., you have 50 lifts in the 81-90% zone, 40 with 82% and 10 with 88%. The variant for the total volume—the one you will match—is 2b.

First you assign the optimal number lifts in this zone to week four. You consider the standard 10% and conclude that NL 5 is not enough. You choose two singles with 82% and one single with 88% for your competition day ramp-up, then two doubles with 82% for Monday: NL 6 @ 82% and NL 1 @ 88%.

Within the confines of variant 2b and the recommended sets of constants you want to place as many 88% lifts into week three as possible. So you choose 42-33-25% (although with such a low NL several combinations of percentages would produce the same results). With the remaining NL 9 that converts to, by week, NL 4, 3, 2.

For the sake of variety choose a different set of percentages for the 82% lifts, say 22, 28, 50%. With NL 34 (what is left after week four) and variant 2b it translates into, by week, NL 10, 17, 7.

If you have more than one weight in the 70-80% zone, assign the heavier one(s) to all four weeks according to the main variant, then calculate the number of the lifts with the lightest weight by subtraction.

E.g., your plan has two weights in the bottom zone, 78% and 67%:

	Week 5	**Week 6**	**Week 7**	**Week 8**	**Total**
3-1a	28%	15%	35%	22%	
NL	**70**	**38**	**87**	**55**	**250**
	≥89%				
2-4a	15%	35%	22%	28%	
NL	2	5	2	4	**13**
	78%				
3-1a	28%	15%	35%	22%	
NL	52	28	66	41	**187**
	Subtract for 67%				
	16	5	19	10	**50**

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Finish by calculating the HARI for each week. Evaluate the resulting ARI, variability, and dynamics. Adjust the plan if necessary.

Let us return to an earlier example of a 40kg x 1RM MP and calculate the HARI for each week.

	NL 240			
% 1RM	**Week 5**	**Week 6**	**Week 7**	**Week 8**
110%	---	---	---	1
100%	4	5	---	1
90%	18	10	7	2
80%	19	36	27	14
70%	12	33	33	18
Total	**53**	**84**	**67**	**36**

Calculations for week one:

% 1RM	**70%**	**80%**	**90%**	**100%**	**110%**	**HARI**
NL 53	12	19	18	4	---	
	(12:53)x100 =23	(19:53)x100 =36	(18:53)x100 =34	(4:53)x100 =7*		
* Although (4:53)x100=7.54, we have rounded the number down. In this type of calculations one almost always has to compromise in this manner to arrive at a sum of 100.						
	23%	36%	34%	7%	---	
	23x70= 1,610	36x80= 2,880	34x90= 3,060	7x100= 700	---	
	(1,610+2,880+3,060+700):100=82.5					**82.5%**

Calculations for all four weeks net the following numbers:

	Week 5	**Week 6**	**Week 7**	**Week 8**
HARI	82.5%	76.7%	76.2%	77%
	Variant 1-4b			

You have achieved sufficient intensity variability from week to week with an overall tapering dynamics. How does one explain the spike of intensity in week 4?—by the competitive attempts and heavy warm-ups for them. If you eliminate the test day from the calculations, as Soviet weightlifting coaches did, the HARI would drop considerably, down to around 75%, and the resulting variant would be 1 and not 1-4b.

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Step V: Volume, intensity, and their dynamics within a week and a session

Due to a drop in monthly volume, the training frequency in the comp period is often reduced, e.g., from 3-4 times a week to 2-3.

A rhythm of loads is important in the comp period. Ideally train every other day: Tuesday, Thursday, Saturday if you are a powerlifter competing on Saturday; Monday, Wednesday, Friday if you are testing your military press on Friday.

Each week sharply taper volume and intensity towards the last session of the week that falls on the same day as the competition or the test.

Weekly loads in competition month

Day 1	**Day 3 (if applies)**	**Day 5**
Tuesday	Thursday	Saturday
	Or	
Monday	Wednesday	Friday
High volume, moderate to high intensity	**Low volume, low to moderate intensity**	**Competition simulation**

If you have more than one session before the competition day, aggressively taper the volume throughout the week, approximately halving it from the first to the second session.

Taper the intensity at the same time. Every session before the competition is a progressively easier "warm-up": "a couple of easy doubles or triples".

In the press the intensity tends to be higher than in the squats. E.g., in the MP you might go up to 85-90% on day one and 70-75% on day two. In the SQ it could be 75-80% and 70% respectively.

Typical repetitions per set are lower than usual, especially in the last session before the test.

If the month's volume is very low (typical for the DL), you will have only one session, on Tuesday, before the test day on the competition week. Make it similar to the Thursday session of a three-day a week schedule.

Use the same schedule and dynamics for every week of the comp month: the highest volume and moderate intensity on day one, lower volume and intensity on day two, the lowest volume and the heaviest lifts of the week on day three.

The number of the warm-up sets on the competition day, the reps at given weights, and jumps between them are individual.

A sample competition week for the kettlebell military press:

Tuesday: 70% x 2, 3, 2, 3; 80% x 2/2, 90% x 1 (NL 15)
Thursday: 70% x 2/3, 80% x 1/2 (NL 8)
Saturday (max): 70% x 2/2, 80% x 1/2, 90% x 1; 2 attempts (NL 9)

Week's NL 32

A sample competition week for the back squat:

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Tuesday: 70% x 3/2, 80% x 1/3 (NL 9)
Thursday: 70% x 2/3 (NL 6)
Saturday (max): 70% x 2/2, 80% x 1/2, 90% x 1; 3 attempts (NL 10)

Week's NL 25

A sample competition week for the deadlift:

Tuesday: 70% x 1, 2, 1, 2, 75% x 1/3 (NL 9)
Saturday (max): 70% x 2/2, 80% x 1, 90% x 1; 3 attempts (NL 8)

Week's NL 17

Sample powerlifting competition week
(Sheyko, 2005)

Mon	***Tue***	***Wed***	***Thur***	***Fri***	***Sat***	***Sun***
BP 70% x 2/3, 80% x 1/3 *DL 70% x 2/3, 75% x 1/3*	*Rest*	*SQ, BP 75% x 2/3*	*Rest*	*Rest*	*Meet:* *92%* *97%* *100-102%*	

In addition to the above, a specific to the contested/tested lift warm-up to 50% 1RM exactly 24 hours before the meet or test (on Friday) is recommended.

Competition week

<table>
<tr><th colspan="2">Day 1</th><th colspan="2">Day 3 (if applies)</th><th colspan="2">Day 5</th></tr>
<tr><td colspan="2"></td><td colspan="2">Approximately 50% of the first day's volume</td><td colspan="2"></td></tr>
<tr><td colspan="2">MP/BP up to 85-90%
SQ up to 75-80%
DL up to 70-75%</td><td colspan="2">MP/BP up to 70-75%
SQ, DL up to 70%</td><td colspan="2">Competition or test</td></tr>
<tr><td colspan="6">All repetitions are in the warm-up range:</td></tr>
<tr><td>70%</td><td>75%</td><td>80%</td><td>85%</td><td>90%</td><td>95%</td></tr>
<tr><td>1-3</td><td colspan="2">1-2</td><td colspan="3">1</td></tr>
<tr><td colspan="3" rowspan="2"></td><td colspan="2">Day 4 (Optional)</td><td rowspan="2"></td></tr>
<tr><td colspan="2">A specific to the contested lift warm-up to 50% 1RM exactly 24 hours before the meet.</td></tr>
</table>

It is time to write up the 40kg presser's comp week and then the rest of the weeks.

You have your comp week's NL:

70% x 18
80% x 14
90% x 4
100% x 1
110% x 1

First write out the test day. Use your experience to tell you how many ramp-up sets you need at each weight. E.g.:

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70% x 2/2, 80% x 2, 90% x 1; 100% x 1, 110% x 1

Then put together the session before the test, the one on Thursday. It is a warm-up to 70-80%:

70% x 2/3, 80% x 1/2

Review what is left for Tuesday: 70% x 8, 80% x 10, 90% x 1. You realize that 80% x 10 is too many for you if supercompensation on Saturday is the goal. 6 seems about right. Spread the 4 extras across the first three weeks and get the same number of "GTG" 70% lifts in return.

Your Tuesday now has 70% x 12, 80% x 6, 90% x 1. It is easy to write the daily plan using the lower rep guidelines:

70% x 3/2, 80% x 2/2, 90% x 1, 80% x 2, 70% x 3/2

You can simply trade the 80% lifts for 70% lifts equally in each of the first week or unequally to strategically load or unload given weeks. In our example week two is already very hard, so you give 2 reps to the first week and to 2 to the third week:

	NL 240			
	Week 5	**Week 6**	**Week 7**	**Week 8**
Total	53	84	67	36
≥90%	22	15	7	4
80%	19	36	27	14
	+2		**+2**	**-4**
	=21		**=29**	**=10**
70%	12	33	33	18
	-2		**-2**	**+4**
	=10		**=31**	**=22**

Take a close look at the modified plan and try it on for size:

	NL 240			
	Week 5	**Week 6**	**Week 7**	**Week 8**
≥90%	22	15	7	4
≥100%	4	5	—	2
80%	21	36	29	10
70%	10	33	31	22
Total	**53**	**84**	**67**	**36**

The first week is intense, with a high number of 90% and heavier weights but very manageable due to low volume.

The second week is tough: a significant number of 90% lifts, a lot of 80% lifts, and a high overall volume. With time you will learn whether such a hard week this close to the test is optimal for you or is it better to soften it up by shifting more heavy lifts to first week. Weeks three and four seem spot on. Here is your month:

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	Week 5	Week 6	Week 7	Week 8
2c	22%	35%	28%	15%
NL	53	84	67	36
	≥90% (NL 48)			
1	46%	31%	15%	8%
NL	22	15	7	4
	Out of that number 100-110% (NL 11)			
1-2				
NL	4	5	—	2
	80% (NL 96)			
2c	22%	38%	30%	10%
NL	21	36	29	10
	70% (NL 96)			
NL	10	33	31	22

Putting it all together

Let us go through a complete process of planning a competition month for a 500-pound deadlifter who is successful with very low volume and has just completed the following prep month:

% 1RM	73%	81%	91%	HARI
500x1RM NL 100 Prep	365	405	455	
	60%	35%	5%	**76.7%**
			NL=5	

Remembering the 2:3 ratio between prep and comp months' volume in the top zone you arrive at 7 or 8 lifts in the comp month.

You decide to take a standard 25% volume cut in the comp period, from NL 100 to 75.

8 lifts is 11% of 75.

At this point you could write down the exact poundages for the top zones, but it is fairly difficult without seeing the rest of the plan. It will suffice to group all lifts starting with 91% together and assign them 92.5% or 93% intensity (most lifts will be with 91%; several maxes and near-maxes will pull this number slightly up).

Change the ratio between 365 and 405 in favor of the latter to have variability and to up the HARI:

% 1RM	73%	81%	91%	>91%	HARI
Comp			93%		
500lbs.	365	405	455		
NL 75	50%	39%	11%		**78.3%**

The distribution is noticeably different from the one in the prep month and so is the HARI.

You review your last prep month:

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Prep	NL 100			
	Week 1	Week 2	Week 3	Week 4
4-2a	22%	28%	15%	35%
NL	22	28	15	35

First, establish the borderline NL for the first week (a minimal 20% difference from the last prep month's last week's NL). 35x0.8=28. 35x1.2=42. NL 28 corresponds to 37% of 75 and 42 to 56%. The latter would make for a very extreme distribution (over half of the month's volume is concentrated in one week), so it is out. Hence your first week's volume must be NL 28/37% or less.

For the given ultra low volume lifter monthly NL 100 is "moderate", which suggests variant 1-3 for the comp month's volume dynamics:

1-3a	35	15	28	22
1-3b	35	22	28	15

As long as you do not exceed 37% in week one, you are fine.

To complicate the task, let us say that, based on your experience, you choose NL 17 for your comp week. 17 is 23% of the 75 monthly total. You have some decisions to make. As a rule of thumb, the comp month is in the 11-15% range, in some cases as low as 9% and as high as 17%. 23% is very high—but not unprecedented (36-20-30-27 by Chernyak, Atanasov & Ermakov, 1972). Considering the lifter's very low overall volume, it is not surprising.

Variant 1-3 demands that the first week's volume is higher than the comp week's and the highest volume on week 3. So you have consulted your table for the range of percentages for each week, combined it with the restriction that week one may not exceed 37%, and have come up with:

1-3a	31-37	12-19	25-30	23

The next step is to identify the percentage in the 25-30% in week three. Remember that there must be no less than 20% change from week to week. However, we cannot calculate it by multiplying 23x1.2, as the change takes place from week three. 20% of that higher volume is higher than 23. Still, it is a good start. 23x1.2=28. So we know it will be higher than 28. Try 29: 29x0.8=23. Jackpot. Since you are trying not to go overboard on volume in week three, you are not even considering 30%. 29 it is.

Your options narrow:

1-3a	31-37	12-19	29	23

29+23=52. You have 48 to divide between the first two weeks. You can put together many combinations to make up that total:

31+17
32+16
33+15
34+14
35+13
36+12

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Any of the above options meet the requirements of variability—and high variability. Any of the numbers in the second week's 12-17 range do the same for the change from week two and week three. When in doubt, just go with the classic 35%. Hence your variant is:

1-3a	*35*	*13*	*29*	*23*
	Standard	*More variable*	*More variable*	*Standard*

Convert the percentages to lifts:

Comp	***NL 75***			
	Week 5	***Week 6***	***Week 7***	***Week 8***
1-3a	*35%*	*13%*	*29%*	*23%*
NL	*26*	*10*	*22*	*17*

Then do the same for the intensity table:

% 1RM	*73%*	*81%*	*91%*	*>91%*	***HARI***
Comp			*93%*		
500lbs.	*365*	*405*	*455*		
	50%	*39%*	*11%*		***78.3%***
NL 75	*38*	*29*	*8*		

Divide the NL 8 in the top zone between 455 (91%) and heavier lifts and then map out the dynamics. Given a very low number of lifts in the top zone, no need to be concerned about a variant; just make assignments based on your experience with heavy deadlifts.

Say, you decided to work up to ~90% two weeks out and 95% a week out:

Week 5	***Week 6***	***Week 7***	***Week 8***
>90% (NL 8)			
---	*3*	*2*	*3*
Out of that number >91% (NL 3)			
---	*---*	*1*	*2*

Assign 80% lifts by week. Your first consideration: you already have a very specific comp week in mind:

On the comp day, Saturday, NL 8:

73% x 1/3, 80% x 1/2; opener 91%; 2 attempts

On Thursday NL 3 after a thorough warm-up with 50-60%:

73% x 1/3

On Tuesday the remaining NL 6:

73% x 2/2, 80% x 1/2

Hence your 80% budget for week 4 is NL 4. 25 lifts remain to assign to the first three weeks. You choose variant 1—similar to total volume's 1-3a. You choose an aggressive breakdown between three weeks, 15-35-50%:

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Comp	NL 75			
	Week 5	Week 6	Week 7	Week 8
1-3a	35%	13%	29%	23%
NL	26	10	22	17
	>90% (NL 8)			
	---	3	2	3
	Out of that number >91% (NL 3)			
	---	---	1	2
	80% (NL 29)			
	50%	35%	15%	---
NL	12	9	4	4

It does not work: you run out of space in week two. Try a distribution with the lowest middle number—22, 28, 50%. Still, 28% of NL 25 is only NL 7. 30% of NL 25 is 7 Still a problem—no room even for a single rep with 73%.

You realize either you have to abandon variant I, or make up your own distribution and frontload week one. Try 60-25-15%:

Comp	NL 75			
	Week 5	Week 6	Week 7	Week 8
1-3a	35%	13%	29%	23%
NL	26	10	22	17
	>90% (NL 8)			
	---	3	2	3
	Out of that number >91% (NL 3)			
	---	---	1	2
	80% (NL 29)			
I	60%	25%	15%	---
NL	15	6	4	4

You are pleased: week one is still very manageable, there is room for one rep with 73% on week two, and week three is easy as it should be. You decide that you want at least a double with 70% on week two and shift one rep to week one. Done:

	80% (NL 29)			
I				
NL	16	5	4	4

Subtract the listed 80% and heavier reps from the total and arrive at the NL with 73%:

Comp	NL 75			
	Week 5	Week 6	Week 7	Week 8
NL	26	10	22	17
>90%	---	3	2	3
80%	16	5	4	4
	26-16=10	10-3-5=2	22-2-4=16	Already set
73%	10	2	16	10

Here you have it, a solid plan for a lifter thriving on low volume.

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Onto the weeks.

Train to Tuesdays, Saturdays, and, at times Thursdays. Maintain the taper dynamics of the comp week throughout the month.

Week 5

NL 26	73%	80%	91%	>91%
	365	**405**	**455**	
	10	16	---	---
Reps per set	3-5	2-4	1-2	1

	Tuesday	Thursday	Saturday
	365x3 405x2, 4, 3 365x4		365x3 405x3, 4
%	60%		40%
NL	**16**		**10**

Week 6

NL 10	73%	80%	91%	>91%
	365	**405**	**455**	
	2	5	3	---
Reps per set	3-5	2-4	1-2	1

Tuesday	Thursday	Saturday
405x3		365x2 405x2 455x1/3
3		**7**

405x3 looks lonely. You could move it to Saturday as a back-off set. Or just let it be and learn to take large weight jumps, as a powerlifter should be able to. You would have to go to 80% straight after 50-65% warm-ups.

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Week 7

This is a week for working up to around 95%, and greasing the groove with 73%. As an option, you may lower the reps with each weight to the levels used in the comp week.

NL 22	73%	80%	91%	>91%
	365	**405**	**455**	
	16	4	1	1
Reps per set	3-5	2-4	1-2	1
Or lower	1-2	1-2	1	1

Tuesday	Thursday	Saturday
365x3/2 405x2 365x4	365x2/2	365x2 405x1/2 455x1 475-495x1
Never mind dividing the total volume between two sessions according to a certain ratio; just write up the Saturday plan and place the rest of the lifts on Tuesday.		
12	**4**	**6**

Week 8

NL 17	73%	80%	91%	>91%
	365	**405**	**455**	
	10	4	1	2
Reps per set	1-2	1-2	1	1

Tuesday	Thursday	Saturday
365x2/2 405x1/2	365x1/3	365x1/3 405x1/2 Opener: 455 3 more attempts
All planned.		
6	**3**	**8**

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PLAN STRONG™ 50

FIVE INTENSITY ZONES

Planning using all five zones is more complex than planning for the high intensity zones only. It has an advantage though: more precise allocation of 50-70% weights that do have a training effect that cannot be ignored. When the athlete makes hypertrophy training a priority, five-zone planning is almost a must, as lighter lifts must predominate.

Most of the recommendations from high intensity zones planning apply, unless specified otherwise.

VOLUME

First, you need to up the monthly NL, now that you are including the 50-70% lifts into the total:

Monthly NL per class of lift in the preparatory period, ≥50% 1RM
(StrongFirst)

- Squats and/or[3] deadlifts—**200-400**
- Presses, pullups—**300-500** and higher

Your daily volume will also go up:

NL per class of exercise per day, ≥50% 1RM
(StrongFirst)

- **10-25**—minimal
- **25-50**—optimal
- **50-100**—maximal

Volume in all main exercises in a session
(Roman, 1986)

- ≤ **50**—small
- **51-100**—medium
- >**100**—large

[3] For competitive powerlifters these numbers apply to each lift: the squat and the deadlift.

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ARI AND ITS VARIABILITY

With the lighter lifts counted, the target average intensity will be lower.

Monthly ARI should be in the 67-77% range.

Generally, keep the ARI below 70% in the prep period and above 70% in the comp period.

As a rule, increase the ARI in the competition month by 1-5% 1RM compared to the last preparatory month.

If the ARI was high in the last preparatory month, it can stay the same or even decrease in the competition month—while the shares of the two top zones increase.

The ARI remains the same—while the share of 81-90% lifts increases by 5%:

50-60%	61-70%	71-80%	81-90%	91-100%	ARI
20	*25*	*35*	***15***	***5***	***71%***
15	*30*	*40*	***10***	***5***	***71%***

In the following example the ARI decreases. And even though the combined share of the two top zones remains the same, the share of the 91-100% zone skyrockets from 0 to 11% at the expense of 81-90% lifts.

50-60%	61-70%	71-80%	81-90%	91-100%	ARI
8	*21*	*40*	***31***	***---***	***74.4%***
18	*22*	*30*	***20***	***11***	***73.5%***

In the final example the ARI increases very slightly and stays on the wrong side of 70% (from 69.0% to 69.5%), but the share of the two top zones increases from 5 to 19% and lifts in the top zone are introduced.

50-60%	61-70%	71-80%	81-90%	91-100%	ARI
31	*30*	*34*	*5*	*---*	***69,0%***
20	*35*	*26*	*18*	*1*	***69.5%***

Such tactics deliver a type of intensity variability, even with a nearly static ARI.

Generally change the intensity from week to week and month to month by 1-5% 1RM.

Between weeks, changes as low as 0.2% 1RM and, for rare athletes able to handle such variability, as high as 10% 1RM are acceptable at times.

It is not unusual to have large swings between some weeks and minimal between others.

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Sample intensity and intensity dynamics in the squat in a five-week competition period
(Chernyak, 1978)

Week	ARI	Change, % 1RM
1	69.6%	
2	69.4%	-0.2
3	75%	+5.6
4	68.9%	-6.1
5	71.6%	+2.7%
Average	**70.9%**	

Intensity and intensity dynamics in a six-week preparatory period for CMS and MS level powerlifters
(Sheyko)

Week	SQ		BP	
	ARI	Change, % 1RM	ARI	Change, % 1RM
1	70.9%		66.8%	
2	66%	-4.9	67.5%	-1.3
3	75%	+9	67.2%	-0.3
4	70.4%	-3.6	69.5%	+2.3
5	68.6%	-1.8	65.7%	-3.8
6	69.7%	+1.1	67%	+1.3
Average	**70%**		**67.2%**	

Intensity and intensity dynamics by month in the snatch for high razryad weightlifters
(Chernyak, 1978b)

Month	ARI	Change
1	74%	
2	71%	+3
3	69.8%	-1.2
4	68.2%	-1.6
5	74.5%	+6.3
6	73.3%	-1.2
7	69.8%	-3.5
8	74%	+4.2
9	71%	-3
Average	**71.5%**	

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INTENSITY ZONES

You may use the tactic from the high intensity zones' planning section: select 1-3 weights (convenient combinations of plates or available kettlebells) per zone. See the table "How to distribute a monthly volume across five intensity zones".

The advantages of this approach include simplicity and customization. Its disadvantage is the inability to give the same plan to lifters of different strength.

Kettlebell training weights in five zones as % of 1RM

1RM ↓	8 kg	10 kg	12 kg	14 kg	16 kg	18 kg	20 kg	22 kg	24 kg	28 kg	32 kg	36 kg	40 kg	44 kg
48kg									50%	58%	67%	75%	83%	92%
44kg								50%	55%	64%	73%	82%	91%	
40kg							50%	55%	60%	70%	80%	90%		
36kg						50%	56%	61%	67%	78%	89%			
32kg					50%	56%	63%	69%	75%	88%				
28kg				50%	57%	64%	71%	79%	86%					
24kg			50%	58%	67%	75%	83%	92%						
22kg			55%	64%	73%	82%	91%							
20kg		50%	60%	70%	80%	90%								
18kg		56%	67%	78%	89%									
16kg	50%	63%	75%	88%										

When planning in the high intensity zones only, you arrived at the number of lifts in the 70-80% zone in each week after assigning the rest of the weights by week and then subtracting their numbers from the totals. When planning in all five zones, you will be doing the same.

First, distribute the 91-100% and 81-90% lifts according to your chosen variants, then distribute the 50-60% and 61-70% weights according to the main variant. Then arrive to the NL in the 71-80% zone by subtraction.

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Why not just assign 60-70% weights according to the main variant and leave the rest to 50-60%?—Because you might end up with too many warm-up weights in one week and not have enough in others. Consider this example:

	Week 1	**Week 2**	**Week 3**	**Week 4**	**Total**
4-2a	22%	28%	15%	35%	
NL	88	112	60	140	**400**
	≥89%				
3b	22%	28%	35%	15%	
NL	9	11	14	6	**40**
	Out of that number 100%				
	—	1	3	—	4
	78% *				
* The 78% weight is treated as an 80-90% weight and gets its own variant. Otherwise, the low number of 89% lifts, would not provide much variability.					
2-1	28%	35%	15%	22%	
NL	50	63	27	40	**180**
	67%				
4-2a	22%	28%	15%	35%	
NL	26	34	18	42	**120**
	Subtract for 56%				
	3	4	1	52	**40**

And this is what it looks like when we assign 56% according to the main variant while leaving the rest to 67%. There are enough warm-up weights in all weeks (hence matching the main variant) and 67% serves as warm-ups in the first three weeks while providing a shock hypertrophy stimulus in week four:

	Week 1	**Week 2**	**Week 3**	**Week 4**	**Total**
4-2a	22%	28%	15%	35%	
NL	88	112	60	140	**400**
	…				
	56%				
4-2a	22%	28%	15%	35%	
	13	17	9	21	**60**
	Subtract for 67%				
	16	21	10	73	**120**

The traditional approach uses the central numbers for each zone: 55, 65%, etc. It does not work with kettlebells and bodyweight plus exercises but for barbell training it has the benefit of writing standardized plans for various strength levels using % 1RM.

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How to distribute a monthly volume across five zones with central weights

<table>
<tr><th>50-60%</th><th>61-70%</th><th>71-80%</th><th>81-90%</th><th>91-100%</th></tr>
<tr><td colspan="5">First establish the monthly ARI in the 67-77% range.
Generally, stay below 70% in the prep period and above 70% in the comp period.</td></tr>
<tr><td colspan="5">Calculate the ARI with the central weights of each zone—55, 65%, etc. The top zone is an exception with a bias towards the bottom half of the zone; use 92.5% as the central weight.</td></tr>
<tr><th>55%</th><th>65%</th><th>75%</th><th>85%</th><th>92.5%</th></tr>
<tr><td colspan="2">Assign the shares of these zones last after establishing the contribution of the top three zones.

Roughly 30-60% (usually 40-50%) of your volume should be in these two zones. For heavyweights focusing on hypertrophy the number can be as high as 75% Adjust the ratio between the 50-60% and 61-70% zones depending on the target average intensity: more lifts in the first zone to lower it; more in the second to raise.

The share of 50-70% lifts usually goes down in the competition period.</td><td colspan="3">Increase a share of these zones in the competition period.</td></tr>
<tr><td rowspan="2"></td><td colspan="3">Once you have established your ARI, the next step is to choose the predominant zone: one of these three.</td><td rowspan="2">Do not plan lifts in this zone by % but decide on an appropriate for you NL—and then convert this number into percentage of the total.

Many lifters can do well without lifts in this zone in the preparatory period.</td></tr>
<tr><td>For beginners, heavyweights, and during hypertrophy emphasis, this is the primary zone—35-40% of the total.</td><td>For most lifters this is the dominant zone—35-50% of the total (and at times as high as 60%).

If you are focusing on hypertrophy, reduce to 25-35% and increase the share of the lighter zones.</td><td>Typically assign 5-10% of your NBL to this zone.

If you respond well to high volume heavy lifts, gradually up this number to 20-40%. Adjust the volume in the lower three zones to keep the ARI the same. Migrate some reps from the 61-70% and 71-80% zones down to the 50-60% zone.</td></tr>
<tr><td></td><td colspan="3">Aim for a smooth curve but at times flatten the peak (a relatively even distribution across zones 2-4) or make two peaks (decrease the volume in zone 3 and increase in zones 2 and 4).</td><td></td></tr>
</table>

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Below are some of many possible distributions to use as templates in planning. Note that no lifts have been assigned to the top zone. You will need assign them after converting the decided NL into a percentage, then subtract this percentage from one or more of the lower zones. Recalculate the ARI, as it will have increased.

Sample distributions of volume by intensity zone

50-60%	61-70%	71-80%	81-90%	91-100%	ARI
33	33	**34**	---	---	**65.1%**
25	35	**40**	---	---	**66.5%**
25	**35**	**35**	5	---	**67%**
25	**35**	30	10	---	**67.5%**
20	**40**	35	5	---	
20	**40**	30	10	---	**68%**
20	35	**40**	5	---	
15	**40**	**40**	5	---	**68.5%**
25	**30**	25	20	---	**69%**
20	**35**	30	15	---	
15	**40**	35	10		
20	30	**40**	10	---	
25	**30**	20	25	---	**69.5%**
20	**35**	25	20	---	
15	35	**40**	10	---	
20	**30**	**30**	20	---	**70%**
15	**35**	**35**	15	---	
25	**30**	15	**30**	---	
10	**40**	**40**	10	---	
15	30	**45**	10	---	
10	30	**60**	---	---	
15	**35**	30	20	---	**70.5%**
10	**40**	35	15		
20	25	**35**	20	---	
15	30	**40**	15	---	
20	20	**45**	15	---	
15	25	**50**	10	---	
20	**30**	20	**30**	---	**71%**
	15	**30**	**30**	---	**71.5%**
15	20	**50**	15	---	
10	25	**55**	10	---	
15	**35**	15	**35**	---	**72%**
15	15	**55**	15	---	
10	20	**60**	10	---	
15	20	**40**	25	---	**72.5%**
10	25	**45**	20	---	
10	20	**55**	15	---	
10	**30**	**30**	**30**	---	**73%**
10	25	**40**	25	---	
10	25	**35**	30	---	**73.5%**
15	20	30	**35**	---	
10	20	**40**	30	---	**74%**
10	15	**45**	30	---	**74.5%**

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When planning the lifts in the top zone do not try to exactly hit the 92.5% average but surgically select the weights using your judgment and experience. Use the following table as a guideline:

Successful distribution of lifts in the 90-100% zone
(Medvedev, Verkhoshansky et al., 1992)

% 1RM	90-92.5	95	97.5	100
Share	65%	20%	15%	0%

The most straightforward way to plan with percentages is, having selected the monthly NL and distribution by intensity zones, calculate the NL in each zone and assign them to the central numbers—55, 65%, etc. (except the top zone where the central weight is 92.5%).

MONTHLY INTENSITY DYNAMICS

When working on intensity dynamics, use the following guidelines:

Algorithm for arriving at the desired intensity dynamics within a month
(StrongFirst)

<table>
<tr><th>Zones</th><th>50-60%</th><th>61-70%</th><th>71-80%</th><th>81-90%</th><th>91-100%</th></tr>
<tr><td>Central weights</td><td>55%</td><td>65%</td><td>75%</td><td>85%</td><td>92.5%</td></tr>
<tr><td rowspan="2">Dynamics</td><td rowspan="2">Main variant</td><td rowspan="2">Main variant</td><td rowspan="2">By subtraction of all other zones' NL from the total volume</td><td>The variant chosen to define the month's intensity dynamics</td><td>Try to abide by the chosen monthly intensity variant—but let your ability to perform heavy lifts on each given week prevail. Do not automatically assign ≥92.5% lifts to low volume weeks; Monday, Tuesday, Wednesday of a high volume week are also options.</td></tr>
<tr><td colspan="2">These zones' variant(s) tend to determine the monthly intensity dynamics.</td></tr>
<tr><td colspan="6">After you have completed all the calculations you may need to move some lifts around to achieve the desired intensity dynamics and variability. Generally trade between the 81-90% zone (above average) and the 61-70% zone (below average). If the 81-90% zone has too few lifts, trade the 71-80% zone.</td></tr>
</table>

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WEEKLY PLANNING

Then, when making weekly planning, push up the intensity in some reps by 5% (or 2.5%, or any other percentage less than five) while decreasing it by as much in the same number of reps.

Limit the use of this tactic in the 81-90% zone as this would introduce unplanned 90% lifts. One option is to stick with 2.5% increases and decreases in this zone. You will still acquire 80% lifts from the 71-80% zone.

Remember to consult the optimal rep range for each weight.

Reps per set for lifters with average to high endurance with a given RM
(StrongFirst, based on many sources)

<table>
<tr><th>% 1RM</th><th>50%</th><th>55%</th><th>60%</th><th>65%</th><th>70%</th><th>75%</th><th>80%</th><th>85%</th><th>90%</th><th>95%</th></tr>
<tr><td>RM, ≥</td><td colspan="4"></td><td>12RM</td><td>10RM</td><td>8RM</td><td>6RM</td><td>4RM</td><td>2RM</td></tr>
<tr><td>Optimal</td><td colspan="4"></td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td></tr>
<tr><td>Range</td><td colspan="4">4-10</td><td>4-8</td><td>3-6</td><td>3-5</td><td>2-4</td><td>1-2</td><td>1</td></tr>
<tr><td>Warm-ups</td><td colspan="2">4-5</td><td colspan="3">2-3</td><td colspan="2">1-2</td><td colspan="3">1</td></tr>
<tr><td>Hypertrophy</td><td colspan="5">5-10 (preparatory period only)</td><td colspan="5"></td></tr>
</table>

Reps per set for lifters with low endurance with a given RM
(StrongFirst, based on many sources)

<table>
<tr><th>% 1RM</th><th>50%</th><th>55%</th><th>60%</th><th>65%</th><th>70%</th><th>75%</th><th>80%</th><th>85%</th><th>90%</th><th>95%</th></tr>
<tr><td>RM</td><td colspan="4"></td><td>10RM</td><td>8RM</td><td>6RM</td><td>5RM</td><td>3RM</td><td>2RM</td></tr>
<tr><td>Optimal</td><td colspan="4"></td><td>5</td><td>4</td><td>3</td><td>2</td><td>1-2</td><td>1</td></tr>
<tr><td>Range</td><td colspan="4">4-6</td><td>3-6</td><td>3-5</td><td>2-4</td><td>2-3</td><td>1-2</td><td>1</td></tr>
<tr><td>Warm-ups</td><td colspan="2">3-5</td><td colspan="3">1-3</td><td colspan="2">1-2</td><td colspan="3">1</td></tr>
<tr><td rowspan="2">Hypertrophy</td><td colspan="2">5-10*</td><td colspan="3">5-8*</td><td colspan="5" rowspan="2"></td></tr>
<tr><td colspan="5">* Preparatory period only</td></tr>
</table>

E.g., a given week calls for NL 40 with 75%. You decide to be biased towards the center weight at the expense of the bottom and top weights, say 70x10, 75x20, 80x10. You fill out the table with ladders using correct rep counts for each weight:

% 1RM	***70%***	***75%***	***80%***
Rep range	*3-6*	*3-5*	*2-4*
NL	***10***	***20***	***10***
Reps by set	*4, 6*	*3, 4, 5, 3, 5*	*2, 4, 2*

A tactic: the session before maxing or near-maxing reduce the volume and do singles and doubles with 60-70% 1RM. (Vorobyev, 1964)

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PLAN STRONG™ TRAINING PLANS

STRONGFIRST MILITARY PRESS PLAN 501

- **This plan is for a girevik with 40kg military press 1RM.**

Although in this plan the intensity remained constant in both periods and in every week, most subjects successfully pressed 44kg after its completion.

	60%	70%	80%	90%	**ARI**
	24kg	**28kg**	**32kg**	**36kg**	
Reps	4-6	3-6	2-4	1-2	
	20%	30%	40%	10%	**74%**

Weeks 1-4	**NL 400**			
	Week 1	**Week 2**	**Week 3**	**Week 4**
3c	15%	28%	35%	22%
NL	60	112	140	88
Weeks 5-8	**NL 300**			
	Week 5	**Week 6**	**Week 7**	**Week 8**
2c	22%	35%	28%	15%
NL	66	105	84	45

Week 1

NL 60	60%	70%	80%	90%
	24kg	**28kg**	**32kg**	**36kg**
	12	18	24	6

Monday	**Wednesday**	**Friday**
24kg x 6 28kg x 5 32kg x 4	24kg x 6 28kg x 3 32kg x 2, 3 36kg x 1/2 32kg x 2, 3 28kg x 4	28kg x 3/2 32kg x 2/2 36kg x 1/4 32kg x 3/2
15	**25**	**20**

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Week 2

NL 112	60%	70%	80%	90%
	24kg	**28kg**	**32kg**	**36kg**
	22	33	46	11

Monday	**Wednesday**	**Friday**
24kg x 6 28kg x 5 (32kg x 4, 36kg x 2) x 3 28kg x 5	24kg x 4 28kg x 3 32kg x 2 36kg x 1 32kg x 3 36kg x 1 32kg x 4 36kg x 1 32kg x 3 36kg x 1 32kg x 2 36kg x 1 32kg x 3/2 28kg x 6/2 24kg x 6	24kg x 6 28kg x 4 32kg x 3, 4, 4, 3 28kg x 4
34	**50**	**28**

Week 3

NL 140	60%	70%	80%	90%
	24kg	**28kg**	**32kg**	**36kg**
	28	42	56	14

Monday	**Wednesday**	**Friday**
24kg x 6 28kg x 6 32kg x 4 36kg x 1, 2, 1, 2, 1 32kg x 4/3	24kg x 6 28kg x 5 32kg x 2 36kg x 1/3 28kg x 5 32kg x 2 36kg x 1/2 28kg x 5 32kg x 2 36kg x 1/2 32kg x 4, 3, 2, 4, 3, 2, 2 28kg x 5 24kg x 4/4	28kg x 6 32kg x 2, 3, 4, 2, 3 28kg x 5/2
35	**75**	**30**

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Week 4

NL 88	60%	70%	80%	90%
	24kg	**28kg**	**32kg**	**36kg**
	18	26	35	9

Monday	Wednesday	Friday
24kg x 6 28kg x 5 32kg x 3/3	24kg x 6 28kg x 5 32kg x 3/2 36kg x 1/3	24kg x 6 28kg x 6 (32kg x 3, 36kg x 2) x 2 (32kg x 3, 36kg x 1) x 2 32kg x 4/2 28kg x 5/2
20	**20**	**43**

Week 5

NL 66	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
	13	20	26	5	2

Monday	Wednesday	Friday
24kg x 4 28kg x 5 32kg x 2 36kg x 2 40kg x 1/2 32kg x 3/2	24kg x 5 28kg x 5/2 32kg x 3 36kg x 1, 2 32kg x 4, 3	24kg x 4 28kg x 5 32kg x 4/2
21	**30**	**15**

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Week 6

NL 105	60%	70%	80%	90%
	24kg	**28kg**	**32kg**	**36kg**
	21	31	42	11

Monday	**Wednesday**	**Friday**
24kg x 5 28kg x 5/2 32kg x 3, 2, 3 36kg x 1/2	24kg x 6 28kg x 5 32kg x 2/2 36kg x 2/3 32kg x 4, 3, 4, 3, 4 28kg x 6	24kg x 5 28kg x 5 32kg x 3 36kg x 1 32kg x 3 36kg x 2 32kg x 3/2 28kg x 5 24kg x 5
25	**45**	**35**

Week 7

NL 84	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
	17	25	34	6	2

Monday	**Wednesday**	**Friday**
24kg x 6 28kg x 5 32kg x 3 36kg x 2 32kg x 3 28kg x 5	24kg x 6 28kg x 5 32kg x 3 36kg x 2 40kg x 1/2 36kg x 2 32kg x 3/5 28kg x 5	24kg x 5 28kg x 5 32kg x 3, 4, 3
24	**40**	**20**

Week 8

NL 45	60%	70%	80%	≥90%
	24kg	**28kg**	**32kg**	**≥36kg**
	9	13	18	5

Monday	**Wednesday**	**Friday**
		TEST
28kg x 6 32kg x 3, 2, 3 36kg x 1/2	24kg x 4 28kg x 4 32kg x 3/2	24kg x 5 28kg x 3 32kg x 2/2 36kg x 1 40kg x 1 **42-44kg x 1**
17	**10**	**17**

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STRONGFIRST MILITARY PRESS PLAN 503

- **This plan is for a lady with a 20kg military press 1RM.**

The plan very similar to plan men's plan 501, although it uses a more limited weight selection. On high volume days you may separate the prescribed volume into two series or sessions.

Week	Monday	Wednesday	Friday
1	12kg x 4, 6 16kg x 2, 3, 2, 3	12kg x 3, 4 16kg x 2, 4, 2	12kg x 4, 5, 4 16kg x 3, 4, 2, 3
2	12kg x 4, 6, 4, 6 16kg x (2, 3) x 5 12kg x 4, 6, 4, 5, 6	12kg x 3, 4, 6 16kg x 3, 4, 2, 3	12kg x 4, 5, 4, 5 16kg x 2, 3, 4, 3, 4, 3
3	12kg x 4, 6 16kg x (2, 3) x 2 12kg x 4, 6 16kg x (2, 3) x 4 12kg x 4, 6	12kg x 6 16kg x (2, 4) x 3 12kg x 5, 6	12kg x 4, 5 16kg x 2, 3, 4 12kg x 4 16kg x 3, 4, 2, 4 12kg x 4, 6
4	12kg x 4, 6 16kg x 2, 3, 4, 2, 4 12kg x 5	12kg x 3, 4, 6 16kg x 3, 4, 2, 3	12kg x 6 16kg x 2, 3 12kg x 5 16kg x 2, 4, 2, 3 12kg x 6
5	12kg x 5 16kg x 2, 3, 2, 4, 2, 3	12kg x 3, 4 16kg x 2, 3, 2, 3 20kg x 1/3 16kg x 2, 3, 2, 3	12kg x 5 16kg x 3, 4, 3
6	12kg x 4, 5 16kg x 2, 3 20kg x 1/2 16kg x 2, 3 20kg x 1 16kg x 3	12kg x 4 16kg x 2, 3, 2, 3 20kg x 1/2 16kg x 4, 3, 4, 3, 4, 3, 4 12kg x 4	12kg x 4, 6 16kg x (2, 3) x 5
7	12kg x 5 16kg x 2, 3, 2, 3, 4, 2, 3	12kg x 5, 6 16kg x 2, 3, 2, 3 20kg x 1/2 16kg x 2, 3, 2, 3 20kg x 1/2 16kg x 2, 3	12kg x 5 16kg x 3, 2, 3, 4, 3
8	12kg x 4, 5 16kg x 2, 3, 4	12kg x 5 16kg x 2, 3	12kg x 6 16kg x 1, 2, 3 20kg x 1 **22kg x 1**

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STRONGFIRST MILITARY PRESS PLAN 501G

- **This plan is for a girevik with a 40kg military press 1RM.**

% 1RM	60%	70%	80%	90%	100%	ARI
NL 400, 3c	**PREP**					
ARI 1-4a	30%	30%	30%	9%	1%	**72.1%**
NL 300, 2c	**COMP**					
ARI 1	20%	30%	35%	13%	2%	**74.7%**

PREP					
	Week 1	**Week 2**	**Week 3**	**Week 4**	**Total**
3c	15%	28%	35%	22%	
NL	60	112	140	88	**400**
ARI 1-4a	75% *	71.3% *	71.5% *	72.3% *	**72.1%**
	≥90%				
2b	28%	35%	22%	15%	
NL	11	14	9	6	**40**
	Out of that number 100%				
	---	2	2	---	4
	80%				
3-4	22%	15%	35%	28%	
NL	26	18	42	34	**120**
	60%				
3c	15%	28%	35%	22%	
NL	18	34	42	26	**120**
	Subtract for 70%				
	5	46	47	22	**120**

* Established once all the calculations have been completed.

PREP		**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**	**60**	**112**	**140**	**88**	**400**
100%	**4**	---	2	2	---	
90%	**36**	11	12	7	6	
80%	**120**	26	18	42	34	
70%	**120**	5	46	47	22	
60%	**120**	18	34	42	26	
ARI		75%	71.3%	71.5%	72.3%	**72.1%**

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PREP

Week 1

NL 60	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
	18	5	26	11	---

Use 20, 35, 45% volume breakdown between days in this and other weeks with three sessions. Increase the volume towards the end of the week. Assign highest intensity lifts to the second session.

Monday	Wednesday	Friday
28kg x 5 32kg x 3 36kg x 1, 2, 1	24kg x 4, 5 32kg x 2, 3 36kg x 1, 2, 1, 2, 1	24kg x 4, 5 32kg x 4, 3, 4, 3, 4
12	**21**	**27**

Week 2

NL 112	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
	34	46	18	12	2

Break down the volume into four days using the 15, 22, 28, 35%. Maintain the trend of increasing volume towards the end of the week but add some variability by using variant 4-2a: 22-28-15-35%.

Monday	Wednesday	Friday	Saturday
24kg x 5 28kg x 3 32kg x 2 36kg x 1/2 32kg x 3 28kg x 5 24kg x 5	24kg x 4 28kg x 3 32kg x 2 36kg x 1 40kg x 1/2 After 30min or later: 28kg x 3 (32kg x 2, 36kg x 1) x 3 32kg x 3 28kg x 4	24kg x 4 32kg x 2 36kg x 1, 2, 1, 2 28kg x 5	24kg x 5 28kg x 5, 4, 6, 3, 5 24kg x 6, 5
25	**31**	**17**	**39**

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Week 3

NL 140	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
	42	47	42	7	2

The same 22-28-15-35%.

Monday	Wednesday	Friday	Saturday
24kg x 4 28kg x 3 32kg x 2 28kg x 4 32kg x 3 28kg x 3 32kg x 2 28kg x 5 24kg x 5	24kg x 4 28kg x 3 32kg x 2 36kg x 1 40kg x 1/2 After 30min or later: 24kg x 4 28kg x 3 32kg x 2, 3, 2, 3 28kg x 5 24kg x 5	24kg x 4 28kg x 3 (32kg x 2, 36kg x 1/2) x 3 32kg x 2	24kg x 5 28kg x 3 32kg x 3, 2, 3, 4, 3 After 30min or later: 24kg x 5 28kg x 3, 4, 3, 5 24kg x 6
31	**39**	**21**	**49**

Week 4

NL 88	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
	26	22	34	6	---

Same breakdown of volume as in week one.

Monday	Wednesday	Friday
24kg x 5 28kg x 3 32kg x 2, 3, 2, 3	24kg x 5 28kg x 3 32kg x 2 36kg x 1, 2, 1, 2 32kg x 3, 4, 2 28kg x 5	24kg x 6 28kg x 5 32kg x 2, 4, 3, 4 28kg x 6 24kg x 4, 6
18	**30**	**40**

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COMP

COMP					
	Week 5	**Week 6**	**Week 7**	**Week 8**	**Total**
2c	22%	35%	28%	15%	
NL	66	105	84	45	**300**
ARI 1	77.6%	74.4%	73.7%	73.1%	**74.7%**
	≥90%				
1	45%	35%	20%	---	
	18	14	8	5*	**45**
	Out of that number ≥100%				
	3	---	1	2	6
	80%				
2c	25%	42%	33%	11**	**105**
NL	24	39	31		
	60%				
2c	22%	35%	28%	15%	**60**
	13	21	17	9	
	Subtract for 70%				
	11	31	28	20	**90**

* Set the number of ≥90% lifts in the comp week at 5, distributed the remaining NL 40 according to the 20, 35, 45% constants using variant 1.

** Assigned 10% of the total to the comp week; assigned the remaining NL 94 according to the 25, 33, 42% constants using variant 2c.

COMP		**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Month**
% 1RM	**NL**	**66**	**105**	**84**	**45**	**300**
100%	**6**	3	---	1	2	
90%	**39**	15	14	7	3	
80%	**105**	24	39	31	11	
70%	**90**	11	31	28	20	
60%	**60**	13	21	17	9	
ARI *		77.6%	74.4%	73.7%	73.1%	**74.7%**

* Note how ARI variability and tapering dynamics have been achieved with the chosen algorithms.

Week 5

	60%	70%	80%	90%	100%
NL 66	**24kg**	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
	13	11	24	15	3

In all weeks taper volume and intensity. Use a more aggressive than in the prep period volume breakdown between days, 20, 30, 50%, in this and other weeks with three sessions.

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Monday	Wednesday	Friday
24kg x 4 28kg x 4 32kg x 2, 3 36kg x 1, 2, 1, 2, 1 After 30min or later: 28kg x 4 32kg x 3 36kg x 1, 2, 1, 2	24kg x 5 32kg x 2, 3, 4, 2, 3	24kg x 4 28kg x 3 32kg x 2 36kg x 1/2 40kg x 1/3 LC*: 44kg x 1/2
33	**19**	**14**

* "LC" = loaded clean.

Week 6

	60%	70%	80%	90%	100%
NL 105	**24kg**	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
	21	31	39	14	---

Given the week's high volume add a fourth training day on Saturday. To maintain the load rhythm, keep tapering from Monday to Friday. Use the 15, 22, 28, 35% percentages and variant 1-4b: 35-22-15-28%. Moved one rep from Friday to Saturday.

Monday	Wednesday	Friday	Saturday
24kg x 4 32kg x 2, 3 36kg x 1, 2, 1, 2 After 30min or later: 24kg x 4 28kg x 4 32kg x 3, 4, 3, 4	24kg x 5 28kg x 4 32kg x 2, 4, 3 28kg x 5	24kg x 4 28kg x 3 32kg x 2 36kg x 2, 1, 2, 1, 2 LC: 44kg x 1/3	24kg x 4 28kg x 4 32kg x 2 28kg x 6 32kg x 3 28kg x 5 32kg x 4
37	**23**	**17**	**28**

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Week 7

NL 84	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
	17	28	31	7	1

Break up the volume into 15, 35, 50%, an aggressive breakdown fitting a taper.

Monday	**Wednesday**	**Friday**
24kg x 4 28kg x 3 32kg x 3 36kg x 2 32kg x 3 36kg x 2 LC: 44kg x 1/2 After 30min or later: 24kg x 4 28kg x 4 32kg x 3, 2, 4, 3 28kg x 5	24kg x 5 28kg x 4 32kg x 4, 3, 4 28kg x 4, 5	24kg x 4 28kg x 3 32kg x 2 36kg x 1, 2 40kg x 1 LC: 44kg x 1, 48kg x 1/2
42	**29**	**13**

Week 8

NL 45	60%	70%	80%	90%	≥100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg, 44kg**
Reps per set*	3-6	2-6	1-4	1-2	1
	9	20	11	3	2

* Note the option of reduced reps at given weights.

First write the testing day's load, then distribute the rest between the first two days with a sharp taper.

Monday	**Wednesday**	**Friday**
28kg x 3, 4 32kg x 2 36kg x 1 32kg x 2, 3 28kg x 5	24kg x 5 28kg x 3 32kg x 1/2	24kg x 4 28kg x 2, 3 32kg x 2 36kg x 1 40kg x 1 **44kg x 1***
20	**10**	**14**

* Or, if 40kg x 1does not feel exceptionally strong, go for a rep record with 40kg.

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STRONGFIRST MILITARY PRESS PLAN 503G

- **This plan is for a lady with a 20kg military press 1RM.**

The plan is identical to men's plan 501G.

Week	Monday	Wednesday	Friday	Saturday
1	14kg x 5 16kg x 3 18kg x 1, 2, 1	12kg x 4, 5 16kg x 2, 3 18kg x 1, 2, 1, 2, 1	12kg x 4, 5 16kg x 4, 3, 4, 3, 4	
2	12kg x 5 14kg x 3 16kg x 2 18kg x 1/2 16kg x 3 14kg x 5 12kg x 5	12kg x 4 14kg x 3 16kg x 2 18kg x 1 20kg x 1/2 After 30min or later: 14kg x 3 (16kg x 2, 18kg x 1) x 3 16kg x 3 14kg x 4	12kg x 4 16kg x 2 18kg x 1, 2, 1, 2 14kg x 5	12kg x 5 14kg x 5, 4, 6, 3, 5 12kg x 6, 5
3	12kg x 4 14kg x 3 16kg x 2 14kg x 4 16kg x 3 14kg x 3 16kg x 2 14kg x 5 12kg x 5	12kg x 4 14kg x 3 16kg x 2 18kg x 1 20kg x 1/2 After 30min or later: 12kg x 4 14kg x 3 16kg x 2, 3, 2, 3 14kg x 5 12kg x 5	12kg x 4 14kg x 3 (16kg x 2, 18kg x 1/2) x 3 16kg x 2	12kg x 5 14kg x 3 16kg x 3, 2, 3, 4, 3 After 30min or later: 12kg x 5 14kg x 3, 4, 3, 5 12kg x 6
4	12kg x 5 14kg x 3 16kg x 2, 3, 2, 3	12kg x 5 14kg x 3 16kg x 2 18kg x 1, 2, 1, 2 16kg x 3, 4, 2 14kg x 5	12kg x 6 14kg x 5 16kg x 2, 4, 3, 4 14kg x 6 12kg x 4, 6	

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Week	**Monday**	**Wednesday**	**Friday**	**Saturday**
5	12kg x 4 14kg x 4 16kg x 2, 3 18kg x 1, 2, 1, 2, 1 After 30min or later: 14kg x 4 16kg x 3 18kg x 1, 2, 1, 2	12kg x 5 16kg x 2, 3, 4, 2, 3	12kg x 4 14kg x 3 16kg x 2 18kg x 1/2 20kg x 1/3 LC*: 22kg x 1/2 * "LC" = loaded clean	
6	12kg x 4 16kg x 2, 3 18kg x 1, 2, 1, 2 After 30min or later: 12kg x 4 14kg x 4 16kg x 3, 4, 3, 4	12kg x 5 14kg x 4 16kg x 2, 4, 3 14kg x 5	12kg x 4 14kg x 3 16kg x 2 18kg x 2, 1, 2, 1, 2 LC: 22kg x 1/3	12kg x 4 14kg x 4 16kg x 2 14kg x 6 16kg x 3 14kg x 5 16kg x 4
7	12kg x 4 14kg x 3 16kg x 3 18kg x 2 16kg x 3 18kg x 2 LC: 22kg x 1/2 After 30min or later: 12kg x 4 14kg x 4 16kg x 3, 2, 4, 3 14kg x 5	12kg x 5 14kg x 4 16kg x 4, 3, 4 14kg x 4, 5	12kg x 4 14kg x 3 16kg x 2 18kg x 1, 2 20kg x 1 LC: 22kg x 1, 24kg x 1/2	
8	14kg x 3, 4 16kg x 2 18kg x 1 16kg x 2, 3 14kg x 5	12kg x 5 14kg x 3 16kg x 1/2	12kg x 4 14kg x 2, 3 16kg x 2 18kg x 1 20kg x 1 **22kg x 1***	

* Or, if 20kg x 1does not feel exceptionally strong, go for a rep record with 20kg.

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STRONGFIRST MILITARY PRESS PLAN 499

- **This plan is a hypertrophy and strength maintenance plan for a girevik with 40kg military press 1RM. A 501 plan is recommended right after.**

% 1RM	60%	70%	80%	90%	100%	ARI
NL 450, 4-2a	**24kg**	**28kg**	**32kg**	**36kg**	**40kg**	
ARI 3c	30%	45%	20%	4%	1%	**70.1%**

	Week 1	Week 2	Week 3	Week 4	Total
4-2a	23%	27%	19%	31%	
NL	103	122	85	140	**450**
ARI 3c	68.9%	70.2%	72.6%	69.3%	**70.1%**
	≥90%				
3c	15%	28%	35%	22%	
NL	3	6	8	5	**22**
	Out of that number 100%				
	---	1	1	2	4
	80%				
3c	15%	28%	35%	22%	
NL	14	25	31	20	**90**
	60%				
4-2a	23%	27%	19%	31%	
NL	31	36	26	42	**135**
	Subtract for 70%				
	55	55	20	73	**203**

PREP		Week 1	Week 2	Week 3	Week 4	Month
% 1RM	**NL**	**103**	**122**	**85**	**140**	**450**
100%	**4**	---	1	1	2	
90%	**18**	3	5	7	3	
80%	**90**	14	25	31	20	
70%	**203**	55	55	20	73	
60%	**135**	31	36	26	42	
ARI		68.9%	70.2%	72.6%	69.3%	**70.1%**

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Week 1

NL 103	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
NL	31	55	14	3	---

Used 25, 33, 42% constants. Increase the volume towards the end of the week.
Place the heaviest lifts on Wednesday.

Monday	Wednesday	Friday
24kg x 5, 6 28kg x 5, 6 32kg x 4	24kg x 5, 6 28kg x 4 32kg x 3 36kg x 1, 2 32kg x 3, 4 28kg x 6	24kg x 4, 5 28kg x 4, 5, 4, 6, 4, 6, 5
26	**34**	**43**

Week 2

NL 122	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
NL	36	55	25	5	1

Use the 3b variant and classic constants to divide the volume between days: 22-28-35-15%.

Monday	Wednesday	Friday	Saturday
24kg x 6 28kg x 6 32kg x 3 36kg x 2 32kg x 3, 4, 3	24kg x 6 28kg x 3 32kg x 3 36kg x 1, 2 40kg x 1 32kg x 3 28kg x 4, 5, 6	24kg x 4, 5 28kg x 3 32kg x 2, 4 28kg x 4, 5, 6 24kg x 4, 6	24kg x 5 28kg x 4, 5, 4
27	**34**	**43**	**18**

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Week 3

NL 85	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
NL	26	20	31	7	1

Used 25, 33, 42% constants.

Monday	**Wednesday**	**Friday**
24kg x 6 28kg x 4 32kg x 3 36kg x 1, 2, 1 32kg x 4	24kg x 6 28kg x 3 32kg x 2 36kg x 1 40kg x 1 36kg x 2 32kg x 3, 4 28kg x 6	24kg x 6 28kg x 3 32kg x 2, 4, 3, 4, 2 28kg x 4 24kg x 8
21	**28**	**36**

Week 4

NL 140	60%	70%	80%	90%	100%
	24kg	**28kg**	**32kg**	**36kg**	**40kg**
Reps per set	4-6	3-6	2-4	1-2	1
NL	**42**	73	**20**	**3**	**2**

Used another variant, 3c, to make Monday easier and prepare for Wednesday: 15-28-35-22%.

Monday	**Wednesday**	**Friday**	**Saturday**
24kg x 6 28kg x 2, 4 32kg x 2, 3, 4	24kg x 5 28kg x 3 32kg x 2 36kg x 1 40kg x 1, 1 36kg x 2 After 30min or later: 24kg x 5 28kg x 4, 6 32kg x 2, 3, 4	24kg x 5 28kg x 4, 5, 6, 4, 5, 6, 4, 6, 4	24kg x 4, 6 28kg x 4, 6 24kg x 5, 6
21	**39**	**49**	**31**

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STRONGFIRST MILITARY PRESS PLAN 333

- **The plan is for a lady with 24kg x 1RM and perfect technique with heavy weights.**

The seven-week plan aims to consolidate your gains, prepare you for the SFG II strength test, and make you more responsive to higher volume in the next cycle.

% 1RM	75%	83%	92%	100%		**ARI**
	18kg	**20kg**	**22kg**	**24kg**		
NL 200, 3b	**PREP**					
ARI 4-1b	60%	30%	7%	3%		**79.3%**
NL 100, 1-2-3	**COMP (3 weeks)**					
ARI 2-1-3	50%	35%	10%	5%		**80.8%**

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
3b	22%	28%	35%	15%	
NL	44	56	70	30	**200**
ARI 4-1b	80.4%	79.3%	77.4%	82.4%	**79.3%**
	100%				
NL	---	2	---	4	**6**
	92%				
2-1	28%	35%	15%	22%	
	4	5	2	3	**14**
	83%				
1-3b	35%	22%	28%	15%	
NL	21	13	17	9	**60**
	Subtract for **75%**				
	19	36	51	14	**120**

PREP		**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**	**44**	**56**	**70**	**30**	**200**
100%	**6**	---	2	---	4	
92%	**14**	4	5	2	3	
83%	**60**	21	13	17	9	
75%	**120**	19	36	51	14	
ARI		80.4%	79.3%	77.4%	82.4%	**79.3%**

COMP		**Week 5**	**Week 6**	**Week 7**		**"Month"**
% 1RM	**NL**	**48**	**28**	**24**		**100**
100%	**6**	3	2	1		
92%	**10**	5	3	2		
83%	**35**	10	17	8		
75%	**49**	30	6	13		
ARI		80%	83.5%	80.1%		**80.8%**

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COMP					
Variant	**Week 5**	**Week 6**	**Week 7**		**Total**
1-2-3	48%	28%	24%		
NL	48	28	24		**100**
ARI 2-3-1	80%	83.5%	80.1%		**80.8%**
	100%				
NL	3	2	1		**6**
	92%				
NL	5	3	2		**10**
	83%				
2b	28%	48%	24%		
NL	10	17	8		**35**
	Subtract for **75%**				
NL	30	6	13		**49**

Before the first working set of each day press 12kg x 5, 16kg x 3. Use only these repetitions for each weight:

18kg	**20kg**	**22kg**	**24kg**
3-5	2-4	1	1

If 22kg or 24kg does not go up on a given day or it takes 100% effort, do not be concerned. Get help from an experienced spotter and complete all the listed singles. The spotter needs to assist enough to make each single a 95% effort—no more and no less. Two to three times a week perform two to three singles of loaded cleans with 28kg.

The plan is very flexible. You are to total a specified number of reps with given kettlebell sizes each week:

Week	**18kg**	**20kg**	**22kg**	**24kg**
1	19	21	4	---
2	36	13	5	2
3	51	17	2	---
4	14	9	3	4
5	30	10	5	3
6	6	17	3	2
7	13	8	2	1

It is up to you whether to execute the plan in three traditional sessions a week, to "grease the groove" throughout the day every day, or do a combination of both. If you are not greasing the groove, arrange your workload in the following manner for the last two weeks:

Week	**Sunday**	**Wednesday**	**Friday**
6	18kg x 3, 20kg x 2, 3 22kg x 1/2	20kg x 2, 3, 2, 3	18kg x 3, 20kg x 2 22kg x 1, 24kg x 1/2
7	18kg x 3, 20kg x 2 22kg x 1, 20kg x 2, 3	18kg x 3, 2, 3	18kg x 2, 20kg x 1 22kg x 1, **24kg x 1**

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STRONGFIRST MILITARY PRESS PLAN 502

- **This plan is for a girevik with 44kg military press 1RM who has successfully completed any of the 501 series plans.**

The four-week plan aims to consolidate your gains and enable you to press 44kg for five singles. A reduction in volume will make you more responsive to a higher volume in the next phase.

Start each session with 24kg x 5/1-2. When pressing 44kg rest for 5min or longer before each single. If 44kg does not go up in a given session or it takes 100% effort, get help from an experienced spotter and complete all the listed singles. The spotter needs to assist enough to make each single a 95% effort—no more and no less.

% 1RM	73%	82%	91%	100%		**HARI**
	32kg	**36kg**	**40kg**	**44kg**		
NL 250, 2b						
ARI 4	40%	40%	16%	4%		**80.6%**

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
2b	28%	35%	22%	15%	
NL	70	87	55	38	**250**
ARI 4	79.8%	80.3%	81.2%	81.5%	**80.6%**
	≥**91%**				
2c	22%	35%	28%	15%	
NL	11	17	14	8	**50**
	Out of the above number **100%**				
NL	---	2	3	5	10
	73%%				
2b	28%	35%	22%	15%	
NL	28	35	22	15	**100**
	Subtract for **82%** *				
NL	31	35	19	15	**100**

* Typically lifts in the 81-90% zone are assigned according to a variant and those in the 70-80% zone are calculated by subtraction. This time the opposite tactic was used.

		Week 1	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**	**70**	**87**	**55**	**38**	**250**
100%	**10**	---	2	3	5	
91%	**40**	11	15	11	3	
82%	**100**	31	35	19	15	
73%	**100**	28	35	22	15	
ARI		79.8%	80.3%	81.2%	81.5%	**80.6%**

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Week	Monday	Thursday	Saturday
1	32kg x 3 (36kg x 2, 40kg x 1) x 4 36kg x 3 32kg x 4 36kg x 3 32kg x 3, 5	32kg x 3, 4, 3 36kg x 2, 3, 2, 3	32kg x 3 36kg x 2 40kg x 1/7 (reps/sets) 36kg x 2, 3
2	32kg x 4 36kg x 2 40kg x 1, 2, 1, 2, 1 36kg x 3, 2, 3, 2, 3 32kg x 4, 5, 3, 4	32kg x 3 36kg x 3 40kg x 1/2 36kg x 3 40kg x 1/2 36kg x 2, 3, 2 32kg x 4	32kg x 3 36kg x 2 40kg x 1 44kg x 1 40kg x 1 44kg x 1 40kg x 2 36kg x 2, 3 32kg x 5
3	32kg x 3 (36kg x 2; 40kg x 1, 2) x 3 32kg x 3, 5	32kg x 3 36kg x 3, 2, 3 32kg x 5	32kg x 3 36kg x 2 40kg x 1/2 44kg x 1/3 36kg x 3
4	32kg x 4 36kg x 2, 3 40kg x 1/2 36kg x 3, 2, 3	32kg x 3/3	32kg x 2 36kg x 1/2 40kg x 1 **44kg x 1/5**

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STRONGFIRST MILITARY PRESS PLAN 505

- **This plan is for a lady girevik with 24kg military press 1RM who has successfully completed any of the 503 series plans.**

Almost identical to 502 but slightly higher ARI:

% 1RM	75%	83%	92%	100%		**HARI**
	18kg	**20kg**	**22kg**	**24kg**		
NL 250, 2b						
ARI 4	40%	40%	16%	4%		**81.9%**

Start each session 12kg x 5, 16kg x 3. If 24kg does not go up in a given session or it takes 100% effort, get help from an experienced spotter and complete all the listed singles. The spotter needs to assist enough to make each single a 95% effort—no more and no less.

Week	Monday	Thursday	Saturday
1	18kg x 3 (20kg x 2, 22kg x 1) x 4 20kg x 3 18kg x 4 20kg x 3 18kg x 3, 5	18kg x 3, 4, 3 20kg x 2, 3, 2, 3	18kg x 3 20kg x 2 22kg x 1/7 (reps/sets) 20kg x 2, 3
2	18kg x 4 20kg x 2 22kg x 1, 2, 1, 2, 1 20kg x 3, 2, 3, 2, 3 18kg x 4, 5, 3, 4	18kg x 3 20kg x 3 22kg x 1/2 20kg x 3 22kg x 1/2 20kg x 2, 3, 2 18kg x 4	18kg x 3 20kg x 2 22kg x 1 24kg x 1 22kg x 1 24kg x 1 22kg x 2 20kg x 2, 3 18kg x 5
3	18kg x 3 (20kg x 2; 22kg x 1, 2) x 3 18kg x 3, 5	18kg x 3 20kg x 3, 2, 3 18kg x 5	18kg x 3 20kg x 2 22kg x 1/2 24kg x 1/3 20kg x 3
4	18kg x 4 20kg x 2, 3 22kg x 1/2 20kg x 3, 2, 3	18kg x 3/3	18kg x 2 20kg x 1/2 22kg x 1 **24kg x 1/5**

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STRONGFIRST MILITARY PRESS PLAN 611

- **This plan is for a girevik with 44kg military press 1RM who has successfully completed any of the 501 series plans—and thrives on very high volume and intensity.**

% 1RM	64%	73%	82%	91%	100%	ARI
	28kg	**32kg**	**36kg**	**40kg**	**44kg**	
NL 500, 4-2b	**PREP**					
ARI 1-3b	15%	63%	20%	2%	<1%	**73.9%**
NL 250, 1	**COMP**					
ARI 2a	23%	30%	35%	10%	2%	**76.4%**

Note a sharp, 50% volume reduction from the very high volume prep period.

Although ARI is high, the shares of the two top zones are moderate.

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
4-2b	15%	28%	22%	35%	
NL	75	140	110	175	**500**
ARI 1-3b	76%	73%	75.5%	72.4%	**73.9%**
	91%				
NL	4	---	6	---	**10**
	82%				
3-1b	28%	22%	35%	15%	
NL	28	22	35	15	**100**

PREP		**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**	**75**	**140**	**110**	**175**	**500**
100%	**---**	---	---	---	---	
91%	**10**	4	---	6	---	
82%	**100**	28	22	35	15	
73%	**315**	32	97	52	134	
64%	**75**	11	21	17	26	
ARI		76%	73%	75.5%	72.4%	**73.9%**

COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
1	35%	28%	22%	15%	
NL	87	70	55	38	**250**
2a	74.7%	78.5%	76.9%	76%	**76.4%**
	≥ 100%				
	---	2	1	2	**5**
	91%				
2c	22%	38%	29%	11%	
NL	5	10	7	3	**25**
	82%				
2b	29%	38%	22%	11%	
NL	26	33	19	10	**88**

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COMP		Week 5	Week 6	Week 7	Week 8	Month
% 1RM	**NL**	**87**	**70**	**55**	**38**	**250**
100%	**5**	---	2	1	2	
91%	**25**	5	10	7	3	
82%	**88**	26	33	19	10	
73%	**75**	36	9	16	14	
64%	**57**	20	16	12	9	
ARI		74.7%	78.5%	76.9%	76%	**76.4%**

Your sessions are recorded in the following manner:

28kg x 4
32kg Σ 12
36kg Σ 10

The "Σ" symbol or "sigma" stands for the total of reps at the given weight. You are to fulfill the listed volume for each arm using any combination of the listed reps:

32kg	36kg	40kg
3, 4, 5	2, 3	1, 2

Ladder whenever possible. Avoid doing sets with the same rep count back to back. E.g., do 3, 5 rather than 4, 4. A series of heavy singles is an exception. Following are three of many options for performing the above workout:

A	B	C
28kg x 4 32kg x 4 36kg x 2, 3, 2, 3 32kg x 3, 5	28kg x 4 32kg x 4 36kg x 3 32kg x 4 36kg x 4 32kg x 4 36kg x 3	28kg x 4 32kg x 3, 5, 4 36kg x 2, 3, 2, 3

Typically do your heavy lifts early in the session, as in the example A. Occasionally, on easier days and weeks, you may sandwich your heavy sets between lighter sets as in example B and even do them last as in example C.

Do one clean for every set of presses. Train both arms equally. The rest periods are self-paced; this is not a density protocol. Rest actively: do "fast and loose" drills, walk around.

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Week	Monday	Tuesday	Thursday	Saturday
1	28kg x 4 32kg Σ 14 36kg Σ 18		28kg x 3 32kg Σ 14 36kg Σ 4	28kg x 4 32kg Σ 4 36kg Σ 6 40kg Σ 4
2	28kg x 4 32kg Σ 19 36kg Σ 8	28kg x 5 32kg Σ 15 After 30min or later: 28kg x 3 32kg Σ 26	28kg x 5 32kg Σ 34	28kg x 4 32kg x 3 36kg Σ 14
3	28kg x 4 36kg Σ 20	28kg x 4 32kg Σ 34	28kg x 5 32kg Σ 15 36kg Σ 11	28kg x 4 32kg x 3 36kg x 2, 2 40kg Σ 6
4	28kg x 5 32kg Σ 29 36kg Σ 5	28kg x 4 32kg Σ 20 After 30min or later: 28kg x 4 32kg Σ 33	28kg x 5 32kg Σ 15 After 30min or later: 28kg x 4 32kg Σ 25	28kg x 4 32kg Σ 12 36kg Σ 10
5	28kg x 5, 5 32kg x 4 36kg Σ 28 LC 44kg x 1/3 (reps/sets)		28kg x 5 32kg Σ 19	28kg x 5 32kg x 3 36kg Σ 8 40kg Σ 5 LC 48kg x 1/2
6	28kg x 6 32kg x 3 36kg Σ 20 40kg Σ 4 LC 48kg x 1/3		28kg x 6 32kg x 3 36kg Σ 11	28kg x 4 32kg x 3 36kg x 2 40kg Σ 6 44kg x 1/2 LC 48kg x 1/2
7	28kg x 4 32kg x 2 36kg Σ 17 40kg Σ 3 LC 48kg x 1/3		28kg x 4 32kg Σ 11	28kg x 4 32kg x 3 36kg x 2 40kg Σ 4 44kg x 1 LC 48kg x 1/2
8	28kg x 3 32kg x 4 36kg x 2 40kg x 1/2 LC 48kg x 1/2 36kg x 3/2		28kg x 3 32kg x 2/3	28kg x 3 32kg x 2/2 36kg x 1/2 40kg x 1 44kg x 1 **48kg x 1**

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STRONGFIRST MILITARY PRESS PLAN 106

- **This plan is for a girevik with 36kg x 2RM and 2*32kg x 2RM.**

2*32kg x 2RM (Est. 1RM 2*33.5kg)
36kg x 2RM (Est. 1RM 38kg)
28kg x 10RM (74% 1RM)

Sample calculations

Est. 38kgx1RM	***53%***	***63%***	***74%***	***84%***	***95%***	***ARI***
	20kg	***24kg***	***28kg***	***32kg***	***36kg***	
	10%	*15%*	*55%*	*19%*	*1%*	**72.4%**
	530	*945*	*4070*	*1596*	*95*	
	10%	*10%*	*55%*	*24%*	*1%*	**73.4%**
	530	*630*	*4070*	*2016*	*95*	
	10%	*10%*	*50%*	*29%*	*1%*	**73.9%**
	530	*630*	*3700*	*2436*	*95*	

Phase I (Prep): Weeks 1-4

- Do a complex with two kettlebells: C&P+SQ. Do all the cleans, then all the presses, and finally all the squats (remember to narrow you stance). The rep numbers given refer to the cleans and presses; you are to do 3 squats after ever set of C&Ps, regardless of the weight and C&P reps. E.g., "24kg x 4" means 4 C + 4P + 3SQ with a pair of 24kg kettlebells. "32kg x 1" means 1C + 1P + 3SQ.

- Keep the rest periods between sets brief; just long enough to assure high quality of reps. Rest actively: walk around, do "fast and loose" drills.

- Do no swings, snatches, jerks, or pullups. You may do SFB type hanging leg raises on alternate days to maintain your pullups.

- On Saturdays do easy windmills, get-ups, and other restorative exercises.

Doubles, est. 1RM 2*33.5kg	48%	60%	72%	84%	96%	**ARI**
	2*16kg	**2*20kg**	**2*24kg**	**2*28kg**	**2*32kg**	
	30%	30%	30%	9%	1%	**62.5%**
Reps	4-8	4-6	3-6	1-2	1	

The ARI is very low, 62.5%, to compensate for the greater difficulty imposed by "complexes".

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
3b	22%	28%	35%	15%	
NL	88	112	140	60	**400**
	≥ **84%**				
2b	28%	35%	22%	15%	
NL	11	14	9	6	**40**
	Out of that number 96%				
	---	3	1	---	4

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Week	Monday	Wednesday	Friday	Saturday
	Doubles' "Complexes"			
1	16kg x 5 20kg x 5 20kg x 5 24kg x 4, 6, 4	16kg x 5 20kg x 4 24kg x 3 28kg x 1, 2, 3, 2, 1	16kg x 7 20kg x 6 24kg x 5 20kg x 6 24kg x 4 16kg x 8	
2	16kg x 5 20kg x 5 24kg x 4 28kg x 1 24kg x 5 20kg x 5 24kg x 5	16kg x 5 20kg x 4 24kg x 4 (28kg x 2, 32kg x 1) x 3 28kg x 2/2	16kg x 6 20kg x 5 24kg x 4 20kg x 4 24kg x 6 20kg x 4 24kg x 6 20kg x 6 16kg x 8, 7	
3	16kg x 5 20kg x 5 24kg x 4 28kg x 2 24kg x 4 28kg x 2 24kg x 5 20kg x 5	16kg x 5 20kg x 4 24kg x 3 28kg x 2 32kg x 1 28kg x 2 24kg x 5 20kg x 5 16kg x 5 20kg x 5	16kg x 6 20kg x 5 24kg x 6 20kg x 6	16kg x 5 20kg x 4 24kg x 6, 3, 6 20kg x 5 16kg x 7, 5, 7
4	16kg x 6 20kg x 5 24kg x 4	16kg x 5 20kg x 4 24kg x 3 28kg x 2/3	16kg x 7 20kg x 4 24kg x 6 20kg x 5 24kg x 5	

Phase II (Comp): Weeks 5-8

- No more complexes. You are training the single kettlebell press; perform the specified number of reps for both arms. Reclean between the reps. Take plenty of rest between sets.

- Do double kettlebell front squat after you have finished all the C&Ps in an "easy strength" format: 10-15 total reps in sets of 2-5, fairly heavy and far from failure.

- You may do a non-demanding pullup program on alternate days. A low volume of swings and/or snatches is also a possibility. Use your judgment. On Saturdays do easy windmills, get-ups, and other restorative exercises.

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Est. 38kg x 1RM	53%	63%	74%	84%	95%	**ARI**
	20kg	**24kg**	**28kg**	**32kg**	**36kg**	
	10%	10%	50%	28%	2%	**74%**
Reps	4-8	4-6	3-6	1-2	1	

COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
1	35%	28%	22%	15%	
NL	105	84	66	45	**300**
	≥84%				
2c	22%	35%	28%	15%	
NL	20	32	25	13	**90**
	Out of that number ≥95%				
	---	2	1	2	**3**

Week	**Monday**	**Wednesday**	**Friday**
5	20kg x 6 24kg x 6 28kg x 3 32kg x 2 28kg x 4 32kg x 2 28kg x 4 32kg x 1 28kg x 5	20kg x 5 24kg x 4 28kg x 3 (32kg x 1, 2) x 5 LC: 36kg x 1/2 LC: 40kg x 1/2	20kg x 6 24kg x 6 (28kg x 5, 6) x 3
6	20kg x 4 28kg x 4 32kg x 2, 1, 2 28kg x 3 32kg x 2, 1, 2 28kg x 5 32kg x 1	20kg x 4 24kg x 4 28kg x 3 32kg x 1/2 LC: 40kg x 1 36kg x 1 32kg x 1 LC: 40kg x 1 36kg x 1 32kg x 2/2 LC: 40kg x 1	24kg x 4 28kg x 4 (32kg x 2, 1) x 4 28kg x 6, 5, 6
7	20kg x 4 28kg x 4 32kg x 2/5 28kg x 4	20kg x 3 24kg x 3 28kg x 3 32kg x 2/2 LC: 40kg x 1/2 36kg x 1 LC: 40kg x 1/2 32kg x 2	24kg x 4 (28kg x 4, 32kg x 2) x 4
8	20kg x 5 24kg x 5 28kg x 4 (32kg x 1, LC 40kg x 1) x 3	20kg x 5 24kg x 4 28kg x 2/2	20kg x 5 24kg x 3/2 28kg x 2 32kg x 1 36kg x 1 **40kg x 1**

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STRONGFIRST MILITARY PRESS PLAN 119

- **This plan is for a lady girevik with 18kg military press 1RM.**

% 1RM	67%	78%	89%	100%	HARI
	12kg	**14kg**	**16kg**	**18kg**	
NL 300, 3-1a	**PREP**				
ARI 2-4a	35%	45%	19%	1%	**76.5%**
NL 250	**COMP**				
ARI 1	20%	50%	28%	2%	**79.3%**

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
3-1a	28%	15%	35%	22%	
NL	84	45	105	66	**300**
2-4a					
ARI	74.3%	82.6%	74.6%	77.8%	
	≥89%				
2-4a	15%	35%	22%	28%	
NL	9	21	13	17	**60**
	Out of the above number 100%				
NL	---	2	---	1	3
	78%				
3-1a	28%	15%	35%	22%	
NL	38	20	47	30	**135**
	Subtract for **67%**				
	37	4	45	19	**105**

PREP		**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**	**84**	**45**	**105**	**66**	**300**
100%	**3**	---	2	---	1	
89%	**57**	9	19	13	16	
78%	**135**	38	20	47	30	
67%	**105**	37	4	45	19	
HARI		74.3%	82.6%	74.6%	77.8%	**76.5%**

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COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
I	35%	28%	22%	15%	
NL I	87	70	55	38	**250**
I					
HARI	80.9%	80.5%	79%	74.2%	
	≥89%				
I	42%	33%	25%	---	
NL	29	23	18	5*	**75**
	Out of that number ≥100%				
NL	2	1	1	2	6
	78%				
I	45%	35%	20%	---	
NL	50	39	23	13**	**125**
	Subtract for **67%**				
NL	8	8	14	20	**50**

* Set the number of ≥89% lifts in the comp week at 5, distributed the remaining NL 70 according to the 25, 33, 42% constants using variant 1.

** Assigned 10% of the total to the comp week; assigned the remaining NL 112 according to-the 25, 33, 42% constants using variant 1. Turned out that IARI was the same in weeks 6 and 7, 80%. Used a different distribution to lower the ratio of 78% weights in week 5: 20, 35, 45%.

COMP		**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Month**
% 1RM	**NL**	**87**	**70**	**55**	**38**	**250**
≥100%	**6**	2	1	1	2	
89%	**69**	27	22	17	3	
78%	**125**	50	39	23	13	
67%	**50**	8	8	14	20	
ARI		80.9%	80.5%	79%	74.2%	**79.3%**

Week	**Monday**	**Wednesday**	**Friday**	**Saturday**
1	12kg x 3, 4 14kg x 3, 4 16kg x 1, 2	12kg x 3, 4 14kg x 3 16kg x 1, 2 14kg x 3 16kg x 1, 2 14kg x 4 12kg x 6	12kg x 3, 5 14kg x 4, 5, 3, 5, 4 12kg x 4, 5	
2	14kg x 2, 3 16kg x 1, 2, 1, 2 14kg x 4	12kg x 4 14kg x 2 16kg x 1, 2 18kg x 1/2	14kg x 2, 3 16kg x 1, 2, 1, 2, 1, 2, 1 14kg x 5	

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Week	Monday	Wednesday	Friday	Saturday
3	12kg x 5 14kg x 4 16kg x 1, 2, 1 14kg x 5 12kg x 5	12kg x 3 14kg x 2 16kg x 1, 2 14kg x 3 16kg x 1, 2 14kg x 3	12kg x 4, 5 14kg x 3, 4 16kg x 1, 2 14kg x 4 12kg x 5	12kg x 3, 5 14kg x 3, 5, 3, 5, 3 12kg x 4, 6
4	12kg x 4 14kg x 3 16kg x 1, 2, 1, 2, 1, 2 14kg x 4	12kg x 4 14kg x 2, 3 16kg x 1, 2 18kg x 1	12kg x 5 14kg x 4 16kg x 1, 2, 1 14kg x 5, 4, 5 12kg x 6	
5	14kg x 2, 3 16kg x 1, 2, 1, 2, 1 After 30min or later: 14kg x 2, 3 16kg x 2, 1, 2, 1, 2 14kg x 2, 4	12kg x 4 14kg x 4 16kg x 1, 2, 1, 2, 1 14kg x 4	12kg x 4 14kg x 2 16kg x 1, 2 18kg x 1 16kg x 2 18kg x 1 LC: 20kg x 1/2	14kg x 3, 4, 5, 3, 4, 5
6	14kg x 2, 3 16kg x 1, 2, 1, 2 After 30min or later: 14kg x 2, 3 16kg x 1, 2, 1, 2 14kg x 3, 4, 2, 4	12kg x 4 14kg x 3 16kg x 1, 2, 1, 2, 1 LC: 20kg x 1/2 14kg x 3, 4	12kg x 4 14kg x 3 16kg x 1, 2 18kg x 1 LC: 20kg x 1/2 14kg x 3	
7	12kg x 6 14kg x 3 16kg x 2, 1, 2, 1, 2 14kg x 4, 2, 4	12kg x 4 14kg x 3 16kg x 1, 2, 1, 2 LC: 20kg x 1/2 14kg x 4	12kg x 4 14kg x 3 16kg x 1, 2 18kg x 1 LC: 20kg x 1/2	
8	12kg x 3, 4 14kg x 2, 3, 4 16kg x 1/2	12kg x 3, 4 14kg x 2	12kg x 3/2 14kg x 2 16kg x 1 18kg x 1 **20kg x 1**	

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STRONGFIRST MILITARY PRESS PLAN 220

- **The program is for a lady with 22kg x 1RM.**

% 1RM	72%	82%	91%	100%	HARI
	16kg	**18kg**	**20kg**	**22kg**	
NL 250, 3b	**PREP**				
ARI 2-4b	50%	40%	10%		**78%**
NL 200, 1	**COMP**				
ARI 2b	35%	50%	15%		**80%**

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
3b	22%	28%	35%	15%	
NL	55	70	87	38	**250**
	≥**91%**				
2-4b	22%	35%	15%	28%	
	5	9	4	7	**25**
	Out of that # 100%				
NL	---	2	---	3	5

COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
1	35%	28%	22%	15%	
NL	70	56	44	30	**200**
	≥**91%**				
2b	28%	35%	22%	15%	
	8	10	7	5	**30**
	Out of that # 100%				
NL	6	---	2	2	10

Your sessions are recorded in the following manner:

16kg Σ 16
18kg Σ 10

The "Σ" symbol or "sigma" stands for the total of reps at the given weight. You are to fulfill the listed volume for each arm using any combination of the listed reps:

16kg	18kg	20kg	22kg
3-5	2-3	1-2	1

Use the session planning guidelines from the manual.

Start each session with 8kg x 5.

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Week	Tuesday	Thursday	Saturday
1	16kg Σ 16 18kg Σ 10	16kg Σ 8 18kg Σ 8	16kg x 3 18kg Σ 5 20kg Σ 5
2	16kg Σ 15 18kg Σ 13 20kg Σ 5	16kg Σ 12 18kg Σ 8	16kg x 3 18kg x 2 20kg x 1/2 (reps/sets) 22kg x 1/2 18kg x 3 16kg x 5
3	16kg Σ 32 18kg Σ 10 You may separate this session into two divided by 30min or longer	16kg Σ 8 18kg Σ 16	16kg x 3 18kg Σ 14 20kg Σ 4
4	16kg Σ 9 18kg Σ 6 20kg Σ 3	16kg Σ 7 18kg Σ 4	16kg x 3 18kg x 2 20kg x 1 22kg x 1/3
5	16kg Σ 16 18kg Σ 17	16kg Σ 12 18kg Σ 8	16kg x 3 18kg x 2 20kg x 1/2 22kg x 1/6 16kg x 4
6	16kg Σ 22 18kg x 2 20kg Σ 3	16kg x 3 18kg Σ 11 20kg Σ 2	16kg x 3 18kg Σ 5 20kg Σ 5
7	16kg Σ 10 18kg Σ 8 20kg Σ 3	16kg Σ 9 18kg x 3	16kg x 3 18kg x 2 20kg x 1/2 22kg x 1/2 18kg x 2
8	16kg x 3 18kg x 3 20kg x 1 16kg x 3 18kg x 3 20kg x 1	16kg x 3/2 18kg x 2	16kg x 3 18kg x 1/2 20kg x 1 22kg x 1 **24kg x 1**

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STRONGFIRST CUSTOM DEADLIFT PLAN FOR 456X1TM

456x1 was a training max. Estimated 465x1RM. The plan was intentionally written using low volume, low volume and intensity variability, considering that the athlete, a strong young man, is far from reaching his potential and does not need "strong medicine" to progress.

% 1RM	59%	68%	78%	87%	≥91%	ARI
	275	**315**	**365**	**405**	**≥425**	
Reps	4-6	3-6	2-4	1-3	1	
NL 200, 2c	**PREP**					
ARI 2-4a	20%	60%	15%	5%	---	**68.7%**
NL 150, 1	**COMP**					
ARI 2b	24%	35%	26%	11%	4%	**71.5%**

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
2c	23%	27%	31%	19%	
NL	46	54	62	38	**200**
2-4a	67.8%	69.1%	68.2%	68.9%	**68.7%**
	87%				
	---	7	2	1	**10**

PREP		Week 1	Week 2	Week 3	Week 4	Month
% 1RM	**NL**	**46**	**54**	**62**	**38**	**200**
93%	**---**	---	---	---	---	
87%	**9**	---	4	2	3	
78%	**29**	7	8	9	5	
68%	**121**	30	31	38	22	
59%	**41**	9	11	13	8	
ARI		67.8%	69.1%	68.2%	68.9%	**68.7%**

COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
1	31%	27%	23%	19%	
NL	46	41	35	28	**150**
ARI 2-b	72.6%	72.7%	72%	67.9%	**71.5%**
	≥91%				
	---	2	1	3	**6**
	87%				
1	48%	28%	24%	---	
	8	5	3	1	**17**

COMP		Week 5	Week 6	Week 7	Week 8	Month
% 1RM	**NL**	**46**	**41**	**35**	**28**	**150**
93%	**6**	---	2	1	3	
87%	**17**	8	5	3	1	
78%	**39**	14	12	12	1	
68%	**52**	15	14	12	11	
59%	**36**	9	8	7	12	
ARI		72.6%	72.7%	72%	67.9%	**72.3%**

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Week	Monday	Thursday	Saturday
1	275x3 315x3, 5, 3, 6	275x3 315x3, 4	275x3 315x3, 3 365x2, 3, 2
2	275x4 315x4 365x2, 3 315x5, 6	275x4 315x3, 4, 3	275x3 315x3 365x3 405x1, 2, 1 315x3
3	275x5 315x3, 5 365x2, 3 315x5, 3, 5	275x4 315x3, 4, 3, 4	275x4 315x3 365x2 405x1/2 365x2
4	275x3 315x3 365x2, 2 315x5, 3, 5	275x2 315x3	275x3 315x3 365x1 405x1 435x1 455x1

Week	Tuesday	Thursday	Saturday
5	275x5 315x3 365x4, 3, 4 315x4, 5		275x4 315x3 365x3 405x3, 2, 3
6	275x4 315x4 365x3 405x2/2 365x3, 2, 3 315x3, 4		275x4 315x3 365x1 405x1 445x1 465x1
7	275x4 315x3 365x3 405x2 365x3, 2, 3 315x3/2		275x3 315x3 365x1 405x1 445x1
8	275x3/2 315x3/3		275x3/2 315x2 365x1 405x1 3 attempts at TSC. **Pulled 500.**

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STRONGFIRST CUSTOM BENCH PRESS COMPETITION PLAN

% 1RM	55%	65%	75%	85%	92.5%	ARI
NL 350, 2a	**PREP I**					
ARI 1-3a	10%	20%	37%	30%	3%	**74.5%**
NL 300, 3-1-2	**PREP II (3 weeks)**					
ARI 2-3-1	10%	40%	40%	10%	---	**70%**
NL 250, 2b	**COMP**					
ARI 2a	20%	15%	35%	25%	5%	**72.9%**

Week	50-60%	61-70%	71-80%	81-90%	91-100%
1	3	4	29	16	1
2	17	35	45	23	2
3	7	15	36	37	3
4	8	16	20	29	4
5	14	54	30	7	---
6	4	15	42	14	---
7	13	50	48	9	---
8	19	14	19	16	2
9	13	10	31	28	5
10	8	6	25	14	2
11	10	7	13	5	3

Intensity was varied in a manner different from the one described in the manual. The top three zones' lifts were assigned according to different variants, then the NL in the two bottom zones was arrived at by subtraction from the total. The 50-70% lifts were divided between 55% and 65% according to the 1:2 ratio determined by their contribution to the total in the given phase.

PREP I									
Variant	**Week 1**		**Week 2**		**Week 3**		**Week 4**		**Total**
2a	15%		35%		28%		22%		
NL	53		122		98		77		**350**
ARI 1-3a	76.5%		71.5%		76.4%		75.5%		**74.5%**
	92.5%								
NL	1		2		3		4		**10**
	Out of the above number ≥**95%**								
NL	---		---		1		2		
	85%								
3a	15%		22%		35%		28%		
NL	16		23		37		29		**105**
	75%								
2c	22%		35%		28%		15%		
NL	29		45		36		20		**130**
	Subtract for **55, 65%**								
NL	7		52		22		24		**105**
	55	**65**	**55**	**65**	**55**	**65**	**55**	**65**	
	1:2		1:2		1:2		1:2		
	3	4	17	35	7	15	8	16	

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PREP II					
Variant	**Week 5**	**Week 6**	**Week 7**		**Total**
3-1-2	35%	25%	40%		
NL	105	75	120		**300**
ARI 2-3-1	67.9%	73.8%	69.4%		
	92.5%				
NL	---	---	---	---	
	Out of the above number ≥**95%**				
NL	---	---	---	---	**---**
	85%				
2-3-1	24%	48%	28%		
NL	7	14	9		**30**
	75%				
3	25%	35%	40%		
NL	30	42	48		**120**
	Subtract for **55, 65%**				
NL	68	19	63		**150**
	55 65	**55 65**	**55 65**		
	1:4	1:4	1:4		
	14 54	4 15	13 50		

COMP					
Variant	**Week 8**	**Week 9**	**Week 10**	**Week 11**	**Total**
2b	28%	35%	22%	15%	
NL	70	87	55	38	**250**
ARI 2a	70.4%	75.1%	74.2%	70.6%	
	92.5%				
NL	2	5	2	3	**12**
	Out of the above number ≥**95%**				
NL	---	---	1	3	
	85%				
2b	28%	48%	24%	---	
NL	16	28	14	5	**63**
	75%				
2c	22%	35%	28%	15%	
NL	19	31	25	13	**88**
	Subtract for **55, 65%**				**87**
NL	33	23	14	17	
	55 65	**55 65**	**55 65**	**55 65**	
	4:3	4:3	4:3	4:3	
	19 14	13 10	8 6	10 7	

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STRONGFIRST CUSTOM PULLUP PLAN

32kg x 1RM

Bodyweight (240) = 77%
12kg (266) = 86%
16kg (275) = 89%
24kg (293) = 95%
32kg (310) = 100%

Although this plan was designed a prep cycle, with no intention of peaking, the athlete set a 40kg x 1 PR in week four.

% 1RM	77%	89%	95%	100%	**HARI**
	Bodyweight	**16kg**	**24kg**	**32kg**	
NL 400, 3c	**PREP**				
ARI 4-2a	65%	33%	2%		**81.3%**

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
3c	15%	28%	35%	22%	
NL	60	112	140	88	400
ARI 4-2a	81.2%	82.6%	79.7%	82.6%	**81.3%**
	≥**89%**				
2-4a	15%	35%	22%	28%	
NL	21	49	31	39	**140**
	Out of the above number ≥**95%**				
NL	---	5	1	4	8

77% lifts were assigned by subtraction, then some were traded for 89% lifts between weeks.

PREP		**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**	**60**	**112**	**140**	**88**	**400**
100%	**3**	---	1	---	2	
95%	**6**	---	4	1	1	
89%	**131**	21	44	30	36	
77%	**260**	39	63	109	49	
ARI		81.2%	82.6%	79.7%	82.6%	**81.3%**

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Week	Monday	Wednesday	Friday	Saturday
1	B (bodyweight) x 5 (16kg x 1, 2) x 5 B x 5		B x 5 (16kg x 1, 2) x 2 (B x 3, 5) x 3	
2	B x 4/2 16kg x 1, 2 24kg x 1/2 (16kg x 1, 2) x 7	B x 3/3 16kg x 2 24kg x 1 32kg x 1 24kg x 1 16kg x 2, 1	B x 5/2 (16kg x 1, 2) x 4 After 30min or later: (B x 3, 4, 5) x 3	
3	B x 4 (16kg x 1, 2) x 3 (B x 3, 5) x 3	B x 4 16kg x 1, 2 24kg x 1 (16kg x 1, 2) x 3 B x 3, 4, 5	B x 5 (16kg x 1, 2) x 3 B x 3, 4, 5	(B x 3, 5) x 6
4	B x 3, 4 (16kg x 1, 2) x 9 30min to 6 hours later: (B x 3, 4) x 4	B x 3, 4 (16kg x 1, 2) x 2	B x 3, 4 16kg x 1/3 24kg x 1 32kg x 1 **36kg x 1** **(Did 40kg x 1)**	

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STRONGFIRST CUSTOM PISTOL PLAN

36kg x 1RM
32kg x 5RM

36kg (128) = 100%
32kg (124) = 97%
24kg (116) = 91%
16kg (108) = 84%
Bodyweight (240) = 77%

% 1RM	84%	91%	97%	100%	HARI
	16kg	**24kg**	**32kg**	**36kg**	
NL 400, 3c	**PREP**				
HARI 1-3	50%	30%	20%		**88.8%**
NL 300, 2c	**COMP**				
HARI 1	40%	30%	30%		**90.3%**

We selected extremely high HARI, 88.8% in the prep period and 90.3% in the competition. In our experience, athletes who can do multiple reps with near maximal weights (as this gentleman whose 5RM=97% 1RM) are able to tolerate very high average intensity and many lifts in the top zone due to a lower nervous excitation required to lift a given weight when one knows he has several reps in the tank—especially in exercises like pistols and pullups that are much less stressful psychologically than squats or deadlifts.

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
3c	15%	28%	35%	22%	**400**
NL	60	112	140	88	
HARI 1-3	90%	89.4%	87.3%	89.4%	**88.8%**
	≥**97%**				
2-4b	22%	35%	15%	28%	
NL	18	28	12	22	**80**
	Out of the above number **100%**				
2-4a	15%	35%	22%	28%	
	1	4	2	3	10

PREP		**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**	**60**	**112**	**140**	**88**	**400**
100%	**10**	1	4	2	3	
97%	**70**	17	24	10	19	
91%	**119**	18	33	42	26	
84%	**201**	24	51	86	40	
HARI		90%	89.4%	87.3%	89.4%	**88.8%**

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COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
2c	22%	35%	28%	15%	
NL	66	105	84	45	**300**
HARI 1	91.5%	90.4%	89.3%	89.2%	**90.3%**
	≥**97%**				
2	29%	38%	22%	11%	
	26	34	20	10	**90**
	Out of the above number **100%**				
1-3	38%	20%	28%	14%	
	6	3	4	2	15

COMP		Week 5	Week 6	Week 7	Week 8	Month
% 1RM	**NL**	**66**	**105**	**84**	**45**	**300**
≥**100%**	**15**	6	3	4	2	
97%	**75**	20	31	16	8	
91%	**91**	20	32	25	14	
84%	**119**	20	39	39	21	
HARI		91.5%	90.4%	89.3%	89.2%	**90.3%**

Week	**16kg**	**24kg**	**32kg**	**36kg**	**40kg**
Reps	**2-4**	**2-3**	**1-2**	**1**	**1**
Week					
1	24	18	17	1	---
2	51	33	24	4	---
3	86	42	10	2	---
4	40	26	19	3	---
5	20	20	20	6	---
6	39	32	31	3	---
7	39	25	16	4	---
8	21	14	8	1	**1**

The athlete was instructed to fulfill the above plan in a flexible manner using ladders and a specified rep range for each weight. He trained five times a week and after week five the athlete set a PR of 48kg—up from 36kg.

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STRONGFIRST SQ AND/OR DL PLAN 787A

All plans of the 787 series feature the same moderately high volume (NL 225 in the preparatory period) but vary in intensity, its dynamics, and the share of top zone lifts. Note that weekly volume in the complete cycle averages out to NL 50—that of *Power to the People!*

Plan 787A has moderate intensity (67.5% and 70.9% in the preparatory and competition period respectively) but great intensity variability and a high number of lifts in the top two zones, especially in the competition period.

% 1RM	55%	65%	75%	85%	92.5%	ARI
NL 225, 4-2a	**PREP**					
3c	25%	**35%**	30%	10%	---	**67.5%**
NL 175, 1-3b	**COMP**					
2b	20%	25%	**35%**	15%	5%	**70.9%**

PREP					
	Week 1	**Week 2**	**Week 3**	**Week 4**	**Total**
4-2a	22%	28%	15%	35%	
NL	49	63	34	79	**225**
COMP					
	Week 5	**Week 6**	**Week 7**	**Week 8**	
1-3b	35%	22%	28%	15%	
NL	61	39	49	26	**175**

First draft of the prep cycle:

PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
4-2a	22%	28%	15%	35%	
NL	49	63	34	79	**225**
ARI ~~3c~~ 3b	67.2%	67.4%	69.1%	67%	**67.5%**
	85%				
3c	15%	28%	35%	22%	
NL	3	6	8	5	**22**
	65%				
4-2a	22%	28%	15%	35%	
NL	17	22	12	28	**79**
	55%				
4-2a	22%	28%	15%	35%	
NL	12	16	8	20	**56**
	Subtract for 75%				
NL	17	19	6	26	**68**

Since the share of 85% is low (10%), its variant does not produce high intensity variability:

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PREP		Week 1	Week 2	Week 3	Week 4	Month
% 1RM	**NL**	**49**	**63**	**34**	**79**	**225**
92.5%	**---**	---	---	---	---	
85%	**22**	3	6	8	5	
75%	**68**	17	19	6	26	
65%	**79**	17	22	12	28	
55%	**56**	12	16	8	20	
ARI 3b		67.2%	67.4%	69.1%	67%	**67%**

In addition, the intended variant 3c changed into 3b. Still, the above plan is acceptable. But it can be improved. Up the intensity in week three and lower in week one. Move all 85% lifts, NL 3, from week one to week three and trade for 65% lifts. Given the plan's low volume, trading just these three lifts should have enough impact:

PREP		Week 1	Week 2	Week 3	Week 4	Month
% 1RM	**NL**	**49**	**63**	**34**	**79**	**225**
92.5%	**---**	---	---	---	---	**Final version**
85%	**22**	~~3~~ ---	6	~~8~~ 11	5	
75%	**68**	17	19	6	26	
65%	**79**	~~17~~ 20	22	~~12~~ 9	28	
55%	**56**	12	16	8	20	
ARI 3c		66%	67.4%	70.9%	67%	**67.5%**

Indeed it has. The changes in ARI between weeks are now +1.4, +1.5, -3.9% and the planned variant 3c has been restored.

PREP

Week 1

	55%	65%	75%	85%	92.5%	ARI
NL 49	12	20	17	---	---	66%

25, 33, 42% constants, then moved one rep.

Monday	Wednesday	Friday
55x4 65x4 60x4	60x3 70x2 80x2, 3 75x3, 4	50x4 60x5 70x5, 6
25% *	33% *	42% *
12	**17**	**20**

* Moved one rep.

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Week 2

	55%	65%	75%	85%	92.5%	ARI
NL 63	16	22	19	6	---	67.4%

Monday	Wednesday	Friday
50x5 60x4 70x4	55x6 65x3 75x2 (85x2, 80x2) x 3	60x4 70x5 75x5 65x5 70x4 60x4
25% *	33% *	42% *
13	**23**	**27**

* Adjusted slightly.

Week 3

	55%	65%	75%	85%	92.5%	ARI
NL 34	8	9	6	11	---	70.9%

Monday	Wednesday	Friday
55x4 70x3 80x2	60x3 75x2 (85x2, 90x1) x 2	55x4 65x3 75x4 85x3
25%	33%	42%
9	**11**	**14**

Week 4

	55%	65%	75%	85%	92.5%	ARI
NL 79	20	28	26	5	---	67%

Monday	Wednesday	Friday
50x5 60x5 70x3, 4, 3	55x4 60x4 65x4 75x2 85x2, 3 80x3, 4	50x3 60x3 70x3, 4 75x3, 4, 3 65x4 60x6
25%	33%	42%
20	**26**	**33**

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PREP, 5 days a week version

Although less practical, dividing the volume into five days is more effective. All series 787 plans' prep periods can be converted in this manner.

Week 1

	55%	65%	75%	85%	92.5%	ARI
NL 49	12	20	17	---	---	66%

Monday	Tuesday	Wednesday	Thursday	Friday
50x4 65x3	55x4 60x4 70x4	60x3 70x2 80x2, 3	65x5	60x3 70x5 75x3, 4
7	**12**	**10**	**5**	**15**
15%	25%	20%	10%	30%

Week 2

	55%	65%	75%	85%	92.5%	ARI
NL 63	16	22	19	6	---	67.4%

Monday	Tuesday	Wednesday	Thursday	Friday
55x3 60x3 65x3 70x3 80x2 85x2	50x5 65x4	55x3 65x3 75x2 85x2, 2	60x3 70x4	60x5 70x5 75x5 80x4
16	**9**	**12**	**7**	**19**
25%	15%	20%	10%	30%

Week 3

	55%	65%	75%	85%	92.5%	ARI
NL 34	8	9	6	11	---	70.9%

Monday	Tuesday	Wednesday	Thursday	Friday
75x3 85x2, 3	65x3 80x2	70x3 85x2 90x1/2	55x4	55x4 60x3 75x3
8	**5**	**7**	**4**	**10**
25%	15%	20%	10%	30%

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Week 4

	55%	65%	75%	85%	92.5%	ARI
NL 79	20	28	26	5	---	67%

Monday	Tuesday	Wednesday	Thursday	Friday
50x5 60x5 70x3 80x2, 3, 2	55x5 65x4 70x3	55x5 65x4 75x2 85x2, 3	60x4 70x4	60x6 70x3, 4 75x3, 4, 3
20	**12**	**16**	**8**	**23**
25%	15%	20%	10%	30%

COMP

First write up the comp week, then assign the rest of the volume between the first three weeks.

If the plan is used for a powerlifting competition, shift the days one forward.

Week 8

	55%	65%	75%	85%	92.5%	ARI
NL 26	15	2	5	1	3	65% *

* Determined after later calculations.

Monday	Wednesday	Thursday	Friday
50x3 60x3 70x2/2	50x3 60x3	Optional: 50x3/2-3	55x3 65x2 75x1 85x1 ~92.5x1 ~97.5x1 ~102.5-105x1
10	**6**		**10**

Note that not all of the weights on the list could be balanced out against the central weights: 70x2/2 and 75x1 all ended up listed as 75%. It is not significant.

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COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
1-3b	35%	22%	28%	15%	
NL	61	39	49	26	**175**
ARI 2b	71.8%	75.8%	68.9%	65%	**70.9%**
	92.5%				
2-4b					
NL	2	4	---	3	**9**
	Out of the above number ≥**95%**				
	---	1	---	2	3
	85%				
2b	30%	50%	20%	---	
NL	8	12	5	1	**26**
	65%				
1-3	50%	20%	30%	---	
NL	21	8	13	2	**44**
	55%				
1-3	50%	20%	30%	---	
NL	10	4	6	15	**35**
	Subtract for **75%** in first three weeks				
NL	20	11	25	5	**61**

Week seven ends up with NL 25 in the 71-80% zone—too many so close to maxing. These are strength building weights, not tapering ones. Trade NL 10 with week five for 61-70%. Calculate the ARI and identify the dynamics:

COMP		**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Month**
% 1RM	**NL**	**61**	**39**	**49**	**26**	**175**
92.5%	**9**	2	4	---	3	**Final version**
Out of that ≥95%:	3	---	1	---	2	
85%	**26**	8	12	5	1	
75%	**61**	~~20~~ 30	11	~~25~~ 15	5	
65%	**44**	~~21~~ 11	8	~~13~~ 23	2	
55%	**35**	10	4	6	15	
ARI 2b		71.8%	75.8%	68.9%	65%	**70.9%**

The resulting intensity variability is very high and the variant is the planned 2b.

If the intensity variability is too high for the given athlete, use less variable three constants when dividing volume between the first three weeks, e.g. 25, 33, 42%.

Onto the rest of the weeks:

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Week 5

	55%	65%	75%	85%	92.5%	ARI
NL 61	10	11	30	8	2	71.8%

Monday	Wednesday	Friday
60x4 70x4 80x2 85x2 80x2 85x2 80x3 75x2 80x2 75x3	55x4 65x3 70x3 75x3 70x3 75x4	50x3 60x3 70x3 80x2 90x1/4 (reps/sets)
42%	33%	25%
26	**20**	**15**

Week 6

	55%	65%	75%	85%	92.5%	ARI
NL 39	4	8	11	12	4	75.8%

Monday	Wednesday	Friday
65x4 75x3 80x2 85x1 90x1 85x2, 3	65x4 75x2 85x1, 2, 1, 2	55x4 70x3 80x1 90x1 92.5x1 92.5-95x1
42% *	33% *	25% *
16	**12**	**11**

* Moved one inconvenient rep.

Week 7

	55%	65%	75%	85%	92.5%	ARI
NL 49	6	23	15	5	---	68.9%

Monday	Wednesday	Friday
55x6 65x3 70x3 75x3 80x3 75x3	60x4 65x4 70x4 65x4	65x4 75x3 85x2, 1, 2
42%	33%	25%
21	**16**	**12**

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Week	Monday	Wednesday	Friday
1	55x4 65x4 60x4	60x3 70x2 80x2, 3 75x3, 4	50x4 60x5 70x5, 6
2	50x5 60x4 70x4	55x6 65x3 75x2 (85x2, 80x2) x 3	60x4 70x5 75x5 65x5 70x4 60x4
3	55x4 70x3 80x2	60x3 75x2 (85x2, 90x1) x 2	55x4 65x3 75x4 85x3
4	50x5 60x5 70x3, 4, 3	55x4 60x4 65x4 75x2 85x2, 3 80x3, 4	50x3 60x3 70x3, 4 75x3, 4, 3 65x4 60x6
5	60x4 70x4 (80x2, 85x2) x 2 80x3 75x2 80x2 75x3	55x4 65x3 70x3 75x3 70x3 75x4	50x3 60x3 70x3 80x2 90x1/4 (reps/sets)
6	65x4 75x3 80x2 85x1 90x1 85x2, 3	65x4 75x2 85x1, 2, 1, 2	55x4 70x3 80x1 90x1 92.5x1 92.5-95x1
7	55x6 65x3 70x3 75x3 80x3 75x3	60x4 65x4 70x4 65x4	65x4 75x3 85x2, 1, 2
8	50x3 60x3 70x2/2	**Wednesday**: 50x3 60x3 **Thursday**: Optional: 50x3/2-3	55x3, 65x2, 75x1, 85x1 ~92.5x1 ~97.5x1 ~102.5-105x1

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STRONGFIRST SQ AND/OR DL PLAN 787B

Plan 787B has high intensity, great intensity variability, and a high number of lifts in the top two zones in the competition period. The competition phase is too intense for most conventional pullers for DL only training.

The plan may also be used for a press for lifters who progress on low volume.

% 1RM	55%	65%	75%	85%	92.5%	ARI
NL 250, 1-3c	**PREP**					
ARI 3c	15%	**40%**	35%	10%	---	**69%**
NL 175, 1-3b	**COMP**					
ARI ~~2b~~ 2-4a	10%	25%	**35%**	25%	5%	**73.9%**

PREP					
	Week 1	**Week 2**	**Week 3**	**Week 4**	**Total**
4-2a	22%	28%	15%	35%	
NL	55	70	38	87	**250**
COMP					
	Week 5	**Week 6**	**Week 7**	**Week 8**	
1-3b	35%	22%	28%	15%	
NL	61	39	49	26	**175**

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PREP					
Variant	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Total**
4-2a	22%	28%	15%	35%	
NL	**55**	**70**	**38**	**87**	**250**
ARI 3c	69%	69.1%	70.3%	68.4%	**69%**
3b	22%	28%	35%	15%	
	85%				
3b	22%	28%	35%	15%	
NL	5	7	9	4	**25**
	65%				
4-2a	22%	28%	15%	35%	
NL	22	28	15	35	**100**
	55%				
4-2a	22%	28%	15%	35%	
NL	8	10	6	13	**37**
	Subtract for **75%**				
NL	20	25	8	35	**88**

PREP		**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**	**55**	**70**	**38**	**87**	**250**
92.5%	**---**	---	---	---	---	
85%	**25**	5	7	9	4	
75%	**88**	20	25	8	35	
65%	**100**	22	28	15	35	
55%	**37**	8	10	6	13	
ARI 3c		69%	69.1%	70.3%	68.4%	**69%**

The result matches the intensity variant but the change between the first two weeks is too low. There are multiple ways to address this. One is to increase the intensity in week three at the expense of week one. Try trading all 85% lifts for 65% lifts:

PREP		**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Month**
% 1RM	**NL**	**55**	**70**	**38**	**87**	**250**
92.5%	**---**	---	---	---	---	**Final version**
85%	**25**	~~5~~ 0	7	~~9~~ 14	~~5~~ 0	
75%	**88**	20	25	8	20	
65%	**100**	~~22~~ 27	28	~~15~~ 10	~~22~~ 27	
55%	**37**	8	10	6	8	
ARI 3c		~~69%~~ 67.2%	69.1%	~~70.3%~~ 72.9%	68.4%	**69%**

Higher variability has been achieved—but now the intensity variant has changed from 3b to 3c. Totally acceptable. In retrospect, a similar result could have been achieved by spreading 85% lifts between three weeks only, e.g. 0-15-30-55%.

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COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
1-3b	35%	22%	28%	15%	
NL	61	39	49	26	**175**
ARI ~~**2b**~~ **2-4a***	73.4%	76.5%	72.6%	73.9%	
	92.5%				
2-4b					
NL	2	4	---	3	**9**
	Out of the above number ≥**95%**				
	---	1	---	2	3
	85%				
2b	33%	42%	25%	---	
NL	14	17	10	3	**44**
	65%				
1-3b	35%	22%	28%	15%	
NL	15	10	12	7	**44**
	55%				
1-3b	35%	22%	28%	15%	
NL	6	4	5	2	**17**
	Subtract for **75%**				
NL	24	4	22	11	**61**

* Since we count the competition lifts as part of the total, a slight intensity spike on week four that changes the variant is not surprising.

COMP		**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Month**
% 1RM	**NL**	**61**	**39**	**49**	**26**	**175**
92.5%	**9**	2	4	---	3	
Out of that ≥95%:	3	---	1	---	2	
85%	**44**	14	17	10	3	
75%	**61**	24	4	22	11	
65%	**44**	15	10	12	7	
55%	**17**	6	4	5	2	
ARI		73.4%	76.5%	72.6%	73.9%	**73.9%**

Having mapped out the competition week you realize that you have too many lifts in the 81-90% zone; one would suffice. You trade NL 2 in this zone for the same number of lifts in the bottom zone with week one. You have chosen week one to bump up its intensity and attempt to meet the original goal of the 2b intensity variant. Mission accomplished:

COMP		**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Month**
% 1RM	**NL**	**61**	**39**	**49**	**26**	**175**
92.5%	**9**	2	4	---	3	**Final version**
Out of that ≥95%:	3	---	1	---	2	
85%	**44**	~~14~~ 16	17	10	~~3~~ 1	
75%	**61**	24	4	22	11	
65%	**44**	15	10	12	7	
55%	**17**	~~6~~ 4	4	5	~~2~~ 4	
ARI 2-4a		~~73.4%~~ 74.4%	76.5%	72.6%	~~73.9%~~ 71.6%	**73.9%**

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Week	Monday	Wednesday	Friday	Saturday
1	55x4 60x5 65x5	60x4 70x3, 4 80x2, 3, 2	55x4 65x4 75x3, 70x4 75x3, 70x5	
2	55x5 70x4 80x2	60x5 75x3 82.5x2 87.5x2 85x3	60x4 70x4, 5 80x3, 4 75x4	55x5 65x5 70x5 65x5
3	55x6 75x3	60x3 70x3 75x3 82.5x2 87.5x2	65x4 75x2 85x2, 3, 2, 3	
4	55x4 65x4 60x5	55x5 65x5 75x3 80x2 85x2, 2	60x6 70x4 80x3, 4 75x3, 4, 3, 4	55x4 65x4 70x5, 6, 5
5	65x5 75x3 85x1, 2, 3, 2, 3 75x4, 3	60x5 70x3, 80x3 75x4, 70x5	55x4 70x2 80x2 85x1, 2 90x1/2 85x2	
6	65x4 75x2 85x2, 3, 2, 3	60x3 70x3 85x1, 2, 1, 2	55x4 75x2 85x1, 90x1 95x1, 90x1/2	
7	65x4 75x3 85x2, 3, 2 70x3, 4	60x3 70x3 80x2, 3, 2 75x3	55x5 65x2 75x2 85x1, 2	
8	65x3 75x1 80x1 75x2 80x1 75x1	60x2 70x3	55x4 70x2 75x1 80x1 85x1 ~92.5x1 ~97.5x1 **~102.5-105x1**	If the plan is used for a powerlifting competition, shift the days one forward in the competition period or both periods.

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STRONGFIRST SQ AND/OR DL PLAN 787C

Plan 787C has a prep period identical to that of 787B. The competition period is lower in ARI and has a much lower number of lifts in the 81-90% zone.

% 1RM	55%	65%	75%	85%	92.5%	ARI
NL 250, 1-3c	**PREP**					
ARI 3c	15%	**40%**	35%	10%	---	**69%**
NL 175, 1-3b	**COMP**					
ARI ~~2b~~ 2-4b	15%	30%	**40%**	10%	5%	**70.9%**

COMP					
Variant	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Total**
1-3b	35%	22%	28%	15%	
NL	61	39	49	26	**175**
ARI 2-4b	70.5%	74%	69.3%	70.9%	**70.9%**
	92.5%				
2-4b					
NL	2	4	---	3	**9**
	Out of the above number ≥**95%**				
	---	1	---	2	3
	85%				
2b	30%	55%	15%	---	
NL	5	10	3	---	**18**
	65%				
1-3b	35%	22%	28%	15%	
NL	18	11	15	8	**52**
	55%				
1-3b	35%	22%	28%	15%	
NL	9	5	8	4	**26**
	Subtract for **75%**				
NL	27	9	23	11	**70**

COMP		**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Month**
% 1RM	**NL**	**61**	**39**	**49**	**26**	**175**
92.5%	**9**	2	4	---	3	
Out of that ≥95%:	3	---	1	---	2	
85%	**18**	5	10	3	---	
75%	**52**	27	9	23	11	
65%	**26**	18	11	15	8	
55%	**70**	9	5	8	4	
ARI 2-4b		70.5%	74%	69.3%	70.9%	**70.9%**

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Week	Monday	Wednesday		Friday
5	55x3 65x3 75x4 85x2, 3 80x4 70x4, 5	60x5 65x5 75x4, 5		50x3 60x3 70x3 80x3 90x1, 1
6	65x5 75x2 85x2, 3, 2, 3	55x5 70x3 75x3		60x3 70x2 80x2 90x1 95x1 90x2
7	55x5 70x3 80x2, 3, 2 65x6	60x3 70x3 80x3 70x4, 3		55x3 65x3 75x3 85x1, 2
8	65x3 75x2 80x2 75x3	**Wednesday**	**Thursday**	50x2 60x2 70x2 80x1 ~92.5x1 ~97.5x1 ~102.5-105x1
		60x3 70x2, 2	Optional: 50x3/2-3	

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Thank you for attending Plan Strong™

As you have now discovered, this Manual contains a great deal of dense information—do not expect to be able to absorb and apply all of it immediately. Take your time, review a bit each week, and continue to practice designing more plans. The strategy behind this system and the process of programming will "click" and improve with practice—do not simply rely on the plans provided here.

Also be sure to request admission to the Plan Strong™ discussion group on Facebook: www.facebook.com/groups/PlanStrong/

A photo of your Certificate of Attendance may be required to be granted membership.

To experience even more advanced strength education offered by StrongFirst, please see our **SPECIAL EVENTS** page to see what's upcoming. New events are added regularly. www.strongfirst.com/special-events/

Power to you!

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ENDNOTES

[1] According to the degree of uncertainty, systems are divided into *deterministic* and *probabilistic*. The former are characterized by a totally predictable environment and all data known in advance. A frequently used example of operating a deterministic system is calculating the return on your deposit into a savings account. The interest rate is fixed and you can precisely predict how much money you are going to make. A probabilistic system, on the other hand, is characterized by incomplete data and an element of chance. The stock market is a probabilistic system.

[2] Not that it stopped them. Rigert, for instance, clean pulled 881 pounds from thin blocks. He deemed it unsafe to pull that much from the platform. This was before the days of powerlifting plates and bars designed to hold that much weight. Assistants had to hold the plates in place.

[3] **Understanding the Soviet weightlifting ranking system**

<table>
<tr><th>Rank</th><th>Olympic press for an 82.5kg /181lb. lifter (1965-1968 ranking table)</th><th>Squat
(Roman, 1986)</th><th>Comments</th></tr>
<tr><td>III</td><td>72.5kg / 160 lbs.</td><td rowspan="4">A double bodyweight squat was reached by all but heavyweights: at Level II by lightweights and Level I for 82.5 and 90kg lifters.</td><td rowspan="2">“Low ranks”—“razryadniki”</td></tr>
<tr><td>II</td><td>90kg / 198 lbs.</td></tr>
<tr><td>I</td><td>Seated bodyweight press (Evdokimov, 1971)</td><td rowspan="4">“Qualified athletes”

1,602 new MSs in 1979-1980 (Sandalov et al., 1982)</td></tr>
<tr><td>CMS</td><td></td></tr>
<tr><td>MS</td><td>120kg / 265 lbs.</td><td>MS lifters could squat slightly above 2 times bodyweight for heavyweights, up to 3 times for lightweights, 82.5 and 90kg classes in the middle with 2.5 times.</td></tr>
<tr><td>MSIC</td><td></td><td>Some MSICs were squatting 4 times their bodyweight.</td><td></td></tr>
<tr><td colspan="4">Future elite who start training at 16 typically make Level III by the end of the first year; Level II in 1.75 years; Level I in 2.5 years; CMS in 3.5 years; MS in 4 years; MSIC in 8 years. (Roman, 1986)

“Anyone can be trained to the level of a Master of Sport.” (Vorobyev, 2001)</td></tr>
</table>

[4] The NBL was not typically recorded at that time; here and in a number of following sources the given numbers are estimated using three reps as an average rep count, typical in that era.

[5] It must be noted that there is a great overlap between different classes of weightlifting exercises, especially the back. We are not talking bodybuilding chest versus biceps. Consider sample five sessions of a Level I lifter Chernyak used when suggesting the above numbers:

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Session #	NBL per exercise group							Session NBL
	SN	C&J	SQ	SN Pull	CL Pull	GM	P	
1	31	21	---	21	---	---	30	103
2	23	---	18	---	18	---	---	59
3	---	18	15	18	---	---	---	51
4	13	15	24	---	---	25	---	77
5	15	15	---	---	15	---	---	45

[6] Note the squat average annual intensity goes down as the athlete's level goes up: Level II—71%, Level I—70%, CMS—68%, MS-MSIC—66%. (Medvedev, 1986)

[7] Note that it averages out to 93%. This is the reason we use 92.5% rather than 95% as the "central weight" for the top zone.

[8] Use by some specialists of the "71-80%" nomenclature and by others (Roman et al.) of "70-79%" created confusion, especially when it came to tracking the lifts in the top zone (including 90% lifts or not). StrongFirst recommends a more ambiguous "70+" (Matveev), along with inclusion of 90% lifts into the top zones and planning using the central numbers.

[9] The Gaussian normal distribution applies exercise in general, and not just in weightlifting: fairly small volume falls in the low and very high intensity zones; most work is done with "relatively high intensity". (Matveev, 1991) "In physiology there is the law of medium loads. According to this law, the largest amount of work can be performed only with medium weights… It is in compliance with the law of medium loads that dozens of tons are lifted in various sports." (Vorobyev, 1972)

[10] Medvedev (1986): "3-4 repetition sets with 75-80% weights in main exercises are the core of the year round learning and training process.": 51% for razryadniki and CMS, 57% for elite regardless of the period.

[11] Note that over half of the volume it the ETK ROP comes from sets of 3 and 4 and higher and lower reps follow a normal distribution:

1	2	3	4	5
10%	19%	29%	26%	16%

[12] E.g., if for 75% 1RM 5 reps are optimal and the range is 3-6, assign 40% of the NL to sets of 5 and the remaining NL to sets of 3, 4, and 6 using a normal distribution. In other words, the NL made up with sets of 4 would be second to that of fives; threes and sixes would be represented the least.

[13] Based on what we know today about the hazards of lumbar flexion consider other ways of stretching the lower back.

[14] The NBL by zone percentages were rounded up for ease of reading.

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[15] Today this phenomenon is successfully used by many powerlifters, especially American (Louie Simmons' influence undoubtedly). A lifter might hit a heavy deadlift double two weeks out, then do some light speed pulls a week before the meet. Deload weeks with speed work are also practiced in the body of the main training period. Russian powerlifters used this phenomenon the Soviet times. In the iconic squat cycle by Smolov (1990) a two-week "switching cycle" made up of free style explosive work is sandwiched between a brutal loading cycle and the peaking cycle. That was done even though many Russian powerlifters do not squat explosively in competition and regular training.

[16] It is likely that Vorobyev's disagreements with Matveev were related to terminology more than substance and the fact that the former advocated greater variability—again, a quantitative rather than qualitative disagreement.

[17] There have been outliers in the 30s.